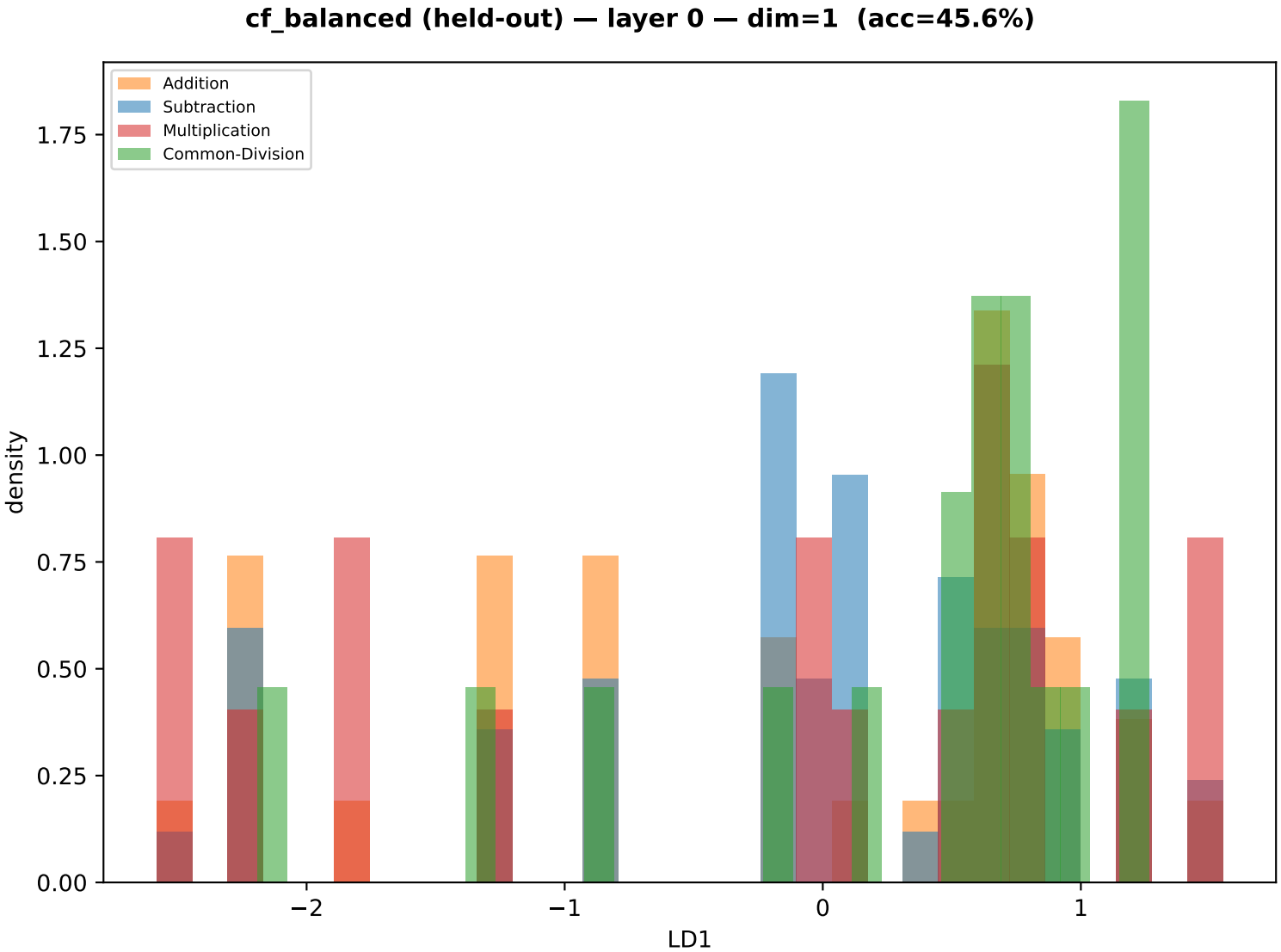
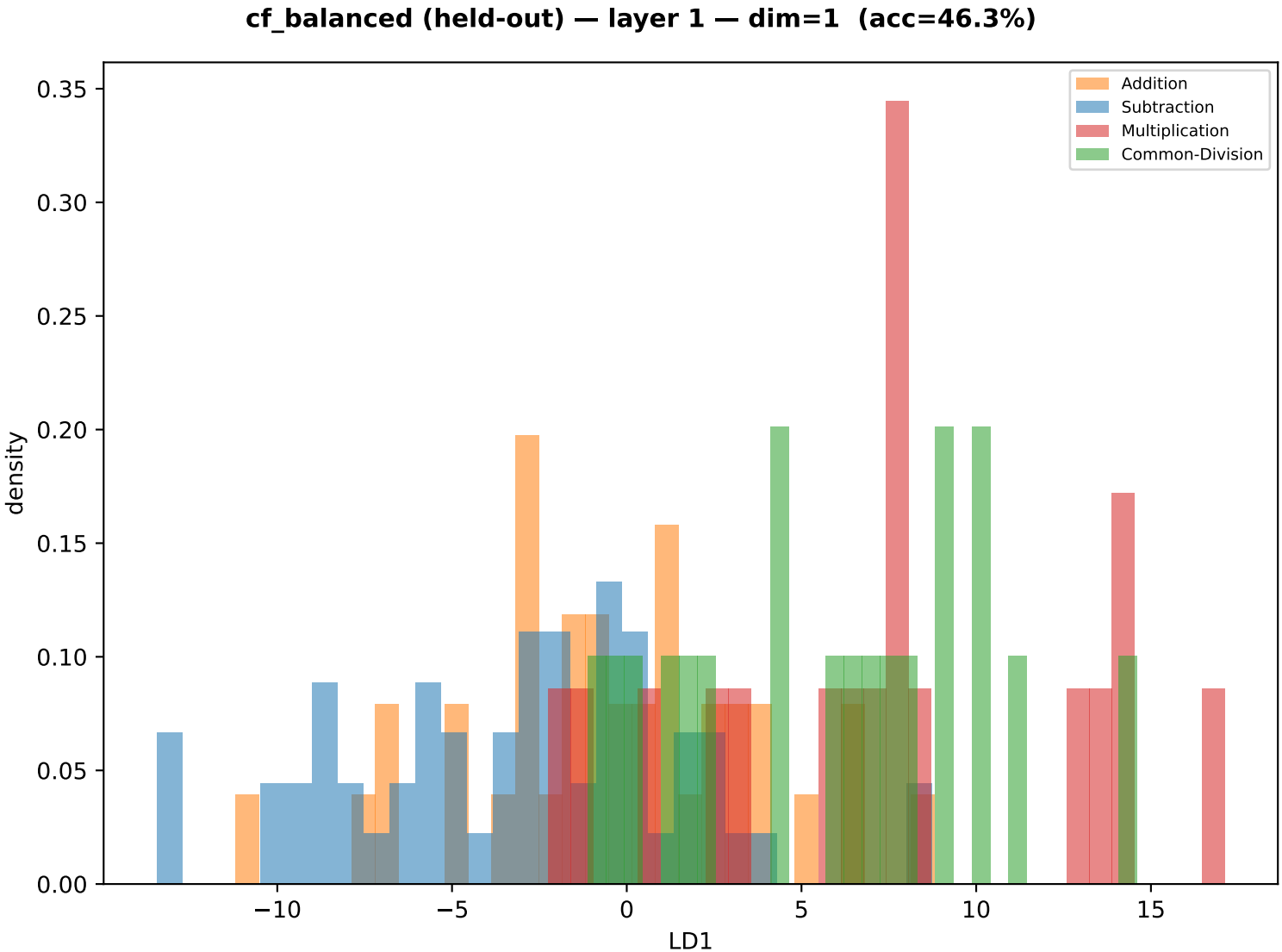
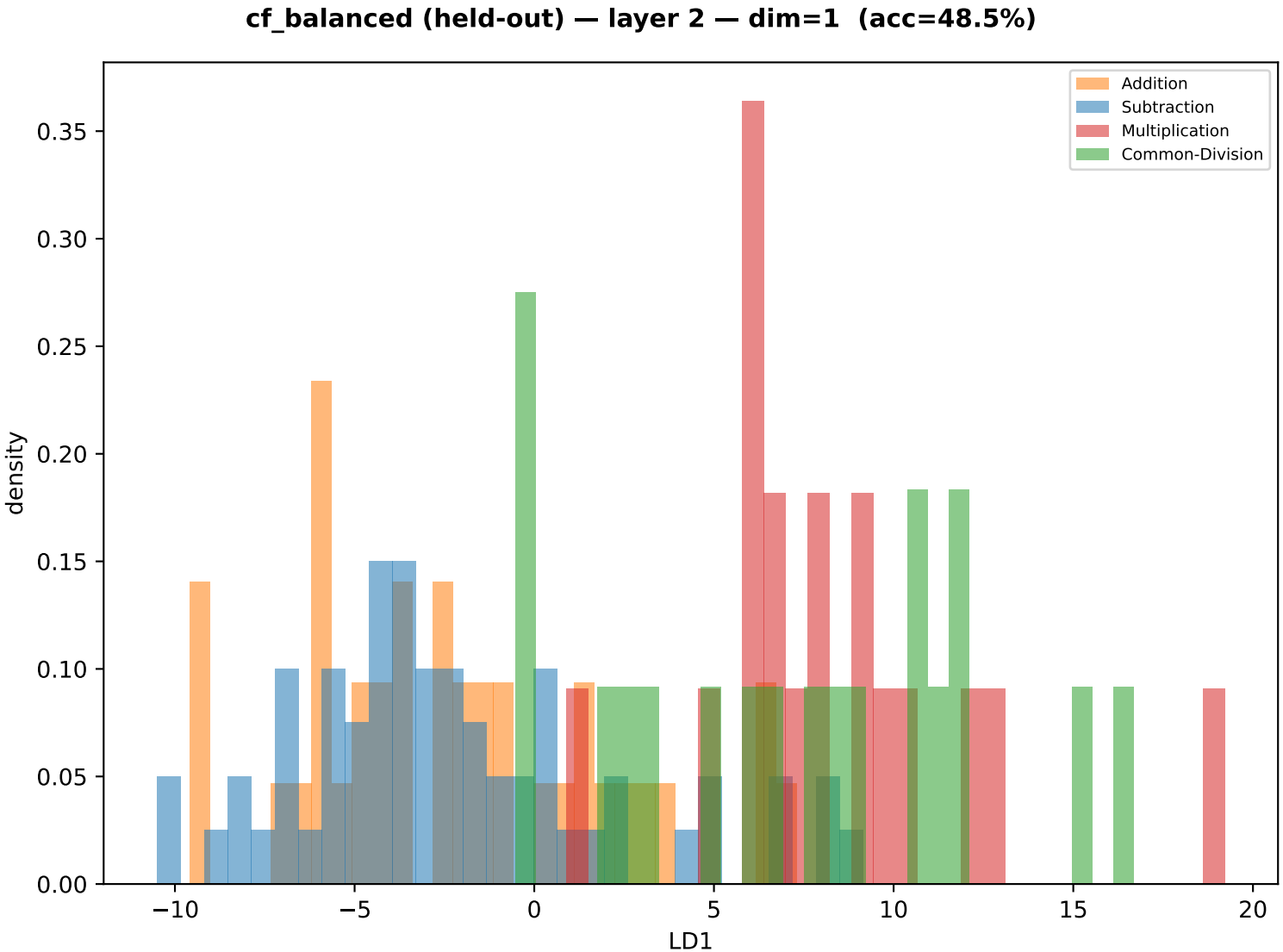
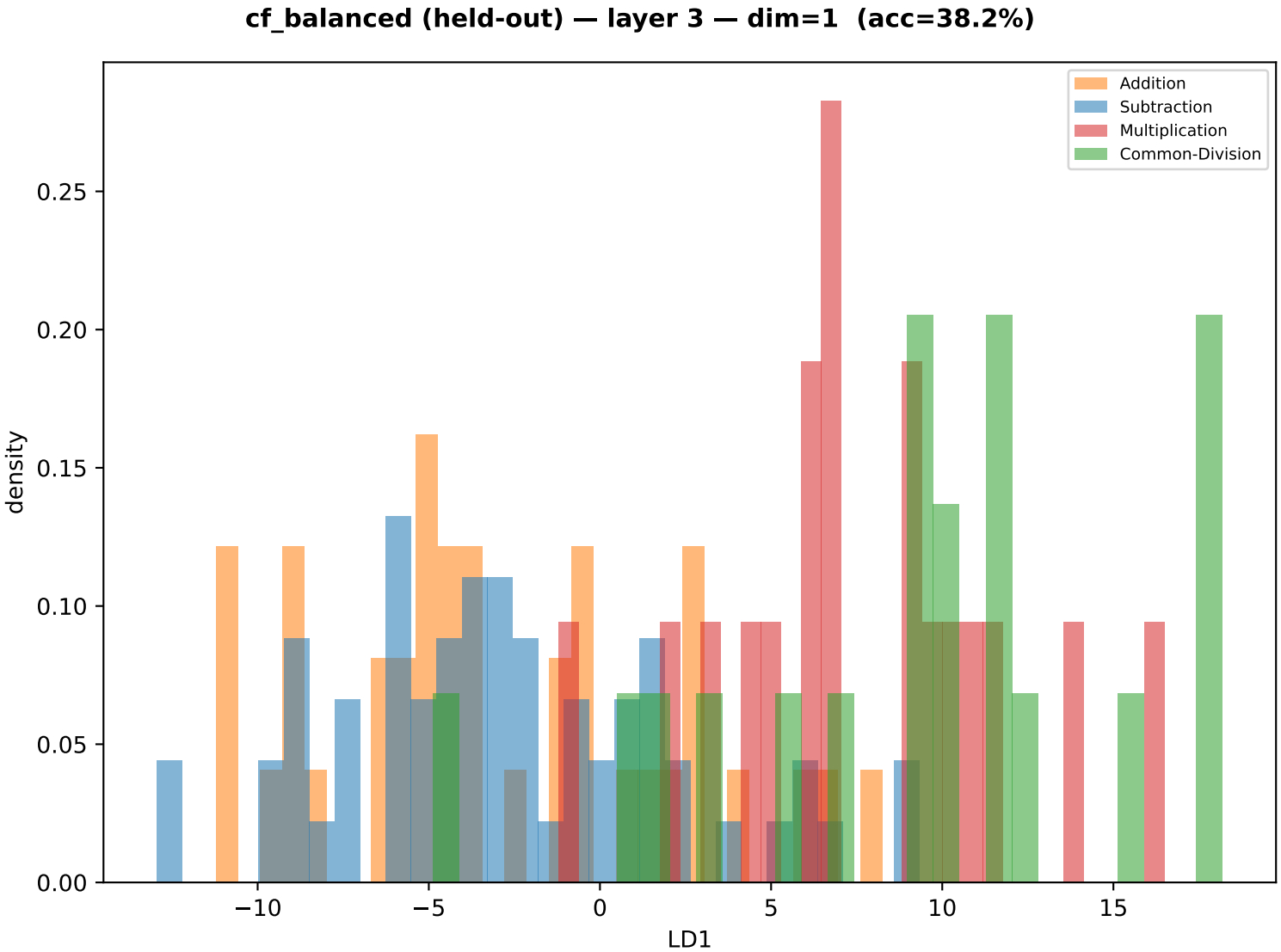
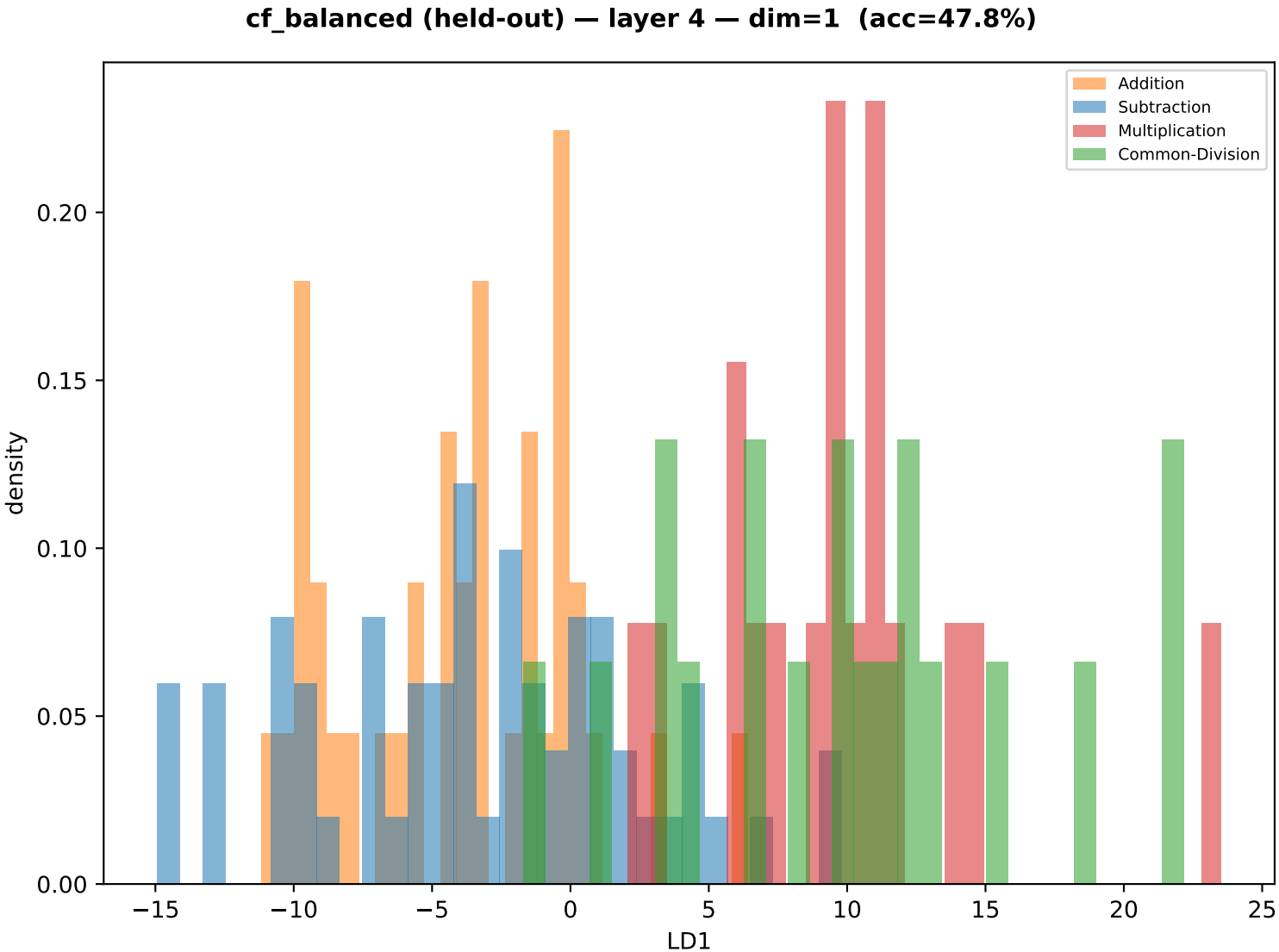
LDA(operator) trained on cf_balanced ':' residuals, evaluated at k-dim projection
cf_balanced held-out 20%
SVAMP (cross-dataset transfer)
k-dim LDA acc (%)
layer
k=1
k=2
k=3
100
90
80
70
60
50
40
30
20
0
2
4
6
8
10
12

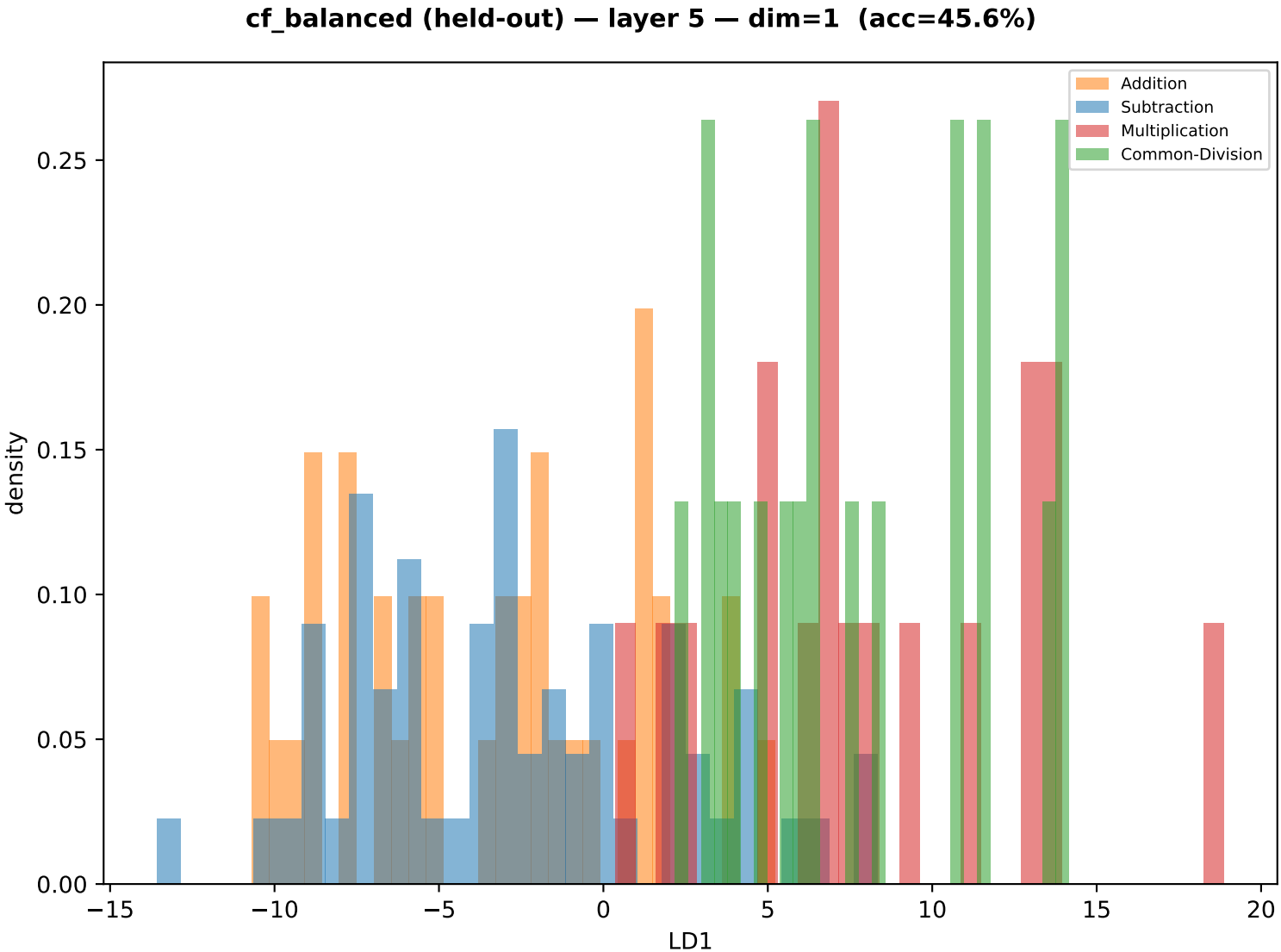

cf_balanced (held-out) — layer 0 — dim=1 (acc=45.6%)
Addition
Subtraction
Multiplication
Common-Division
density
LD1
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
−2
−1
0
1

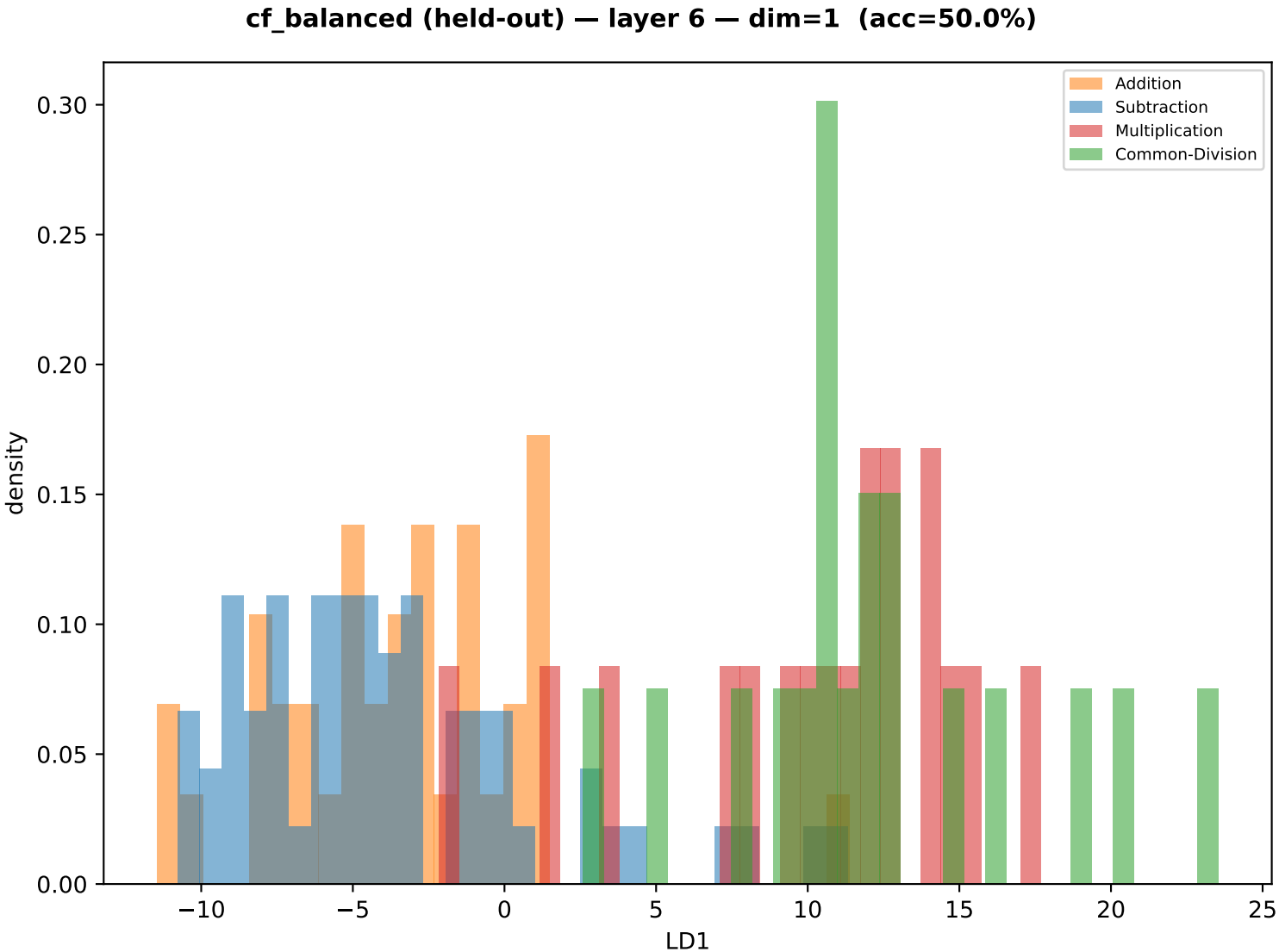

cf_balanced (held-out) — layer 1 — dim=1 (acc=46.3%)
Addition
Subtraction
Multiplication
Common-Division
density
LD1
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
−10
−5
0
5
10
15

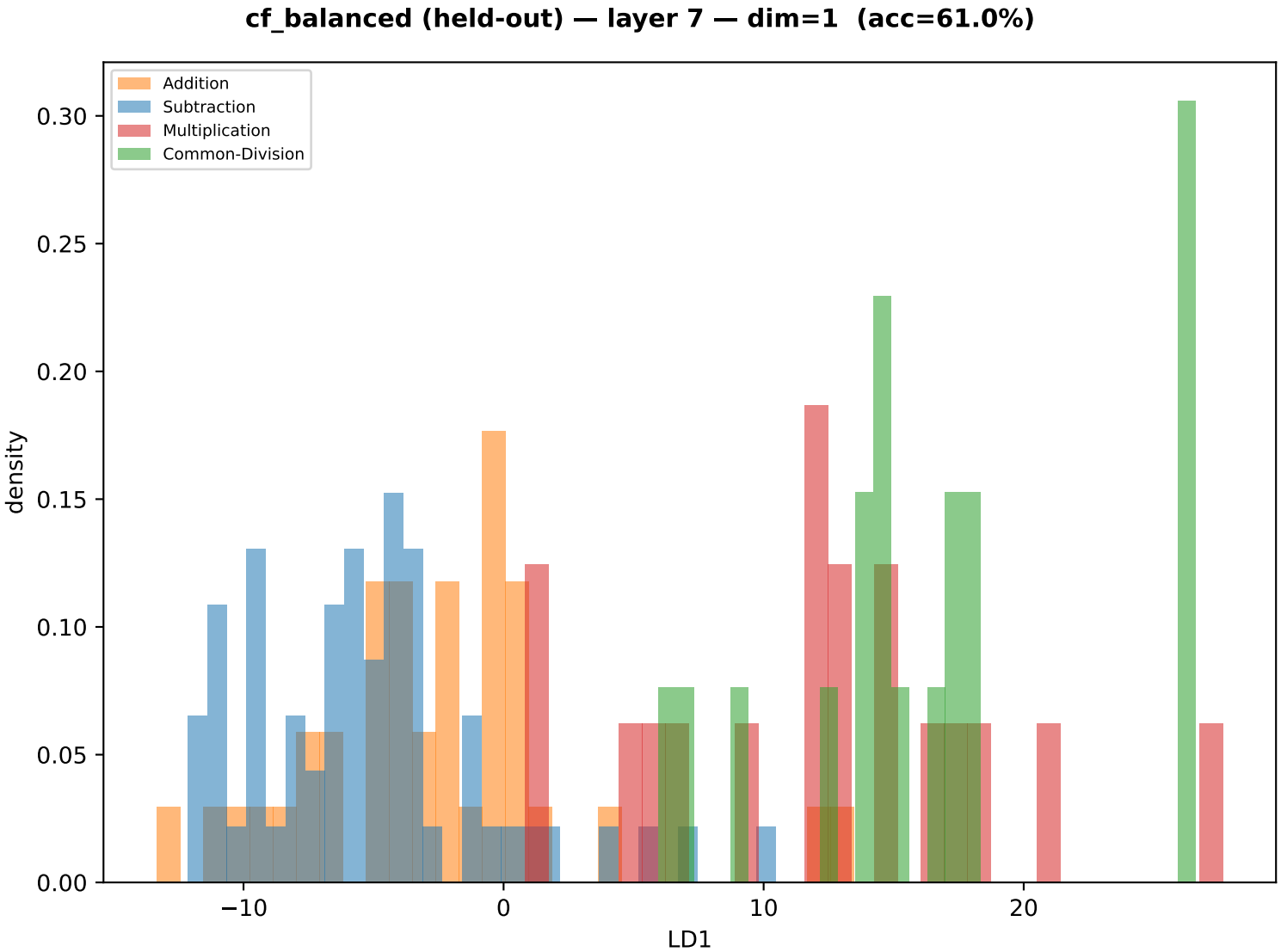

cf_balanced (held-out) — layer 2 — dim=1 (acc=48.5%)
Addition
Subtraction
Multiplication
Common-Division
density
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
−10
−5
0
5
10
15
20
LD1

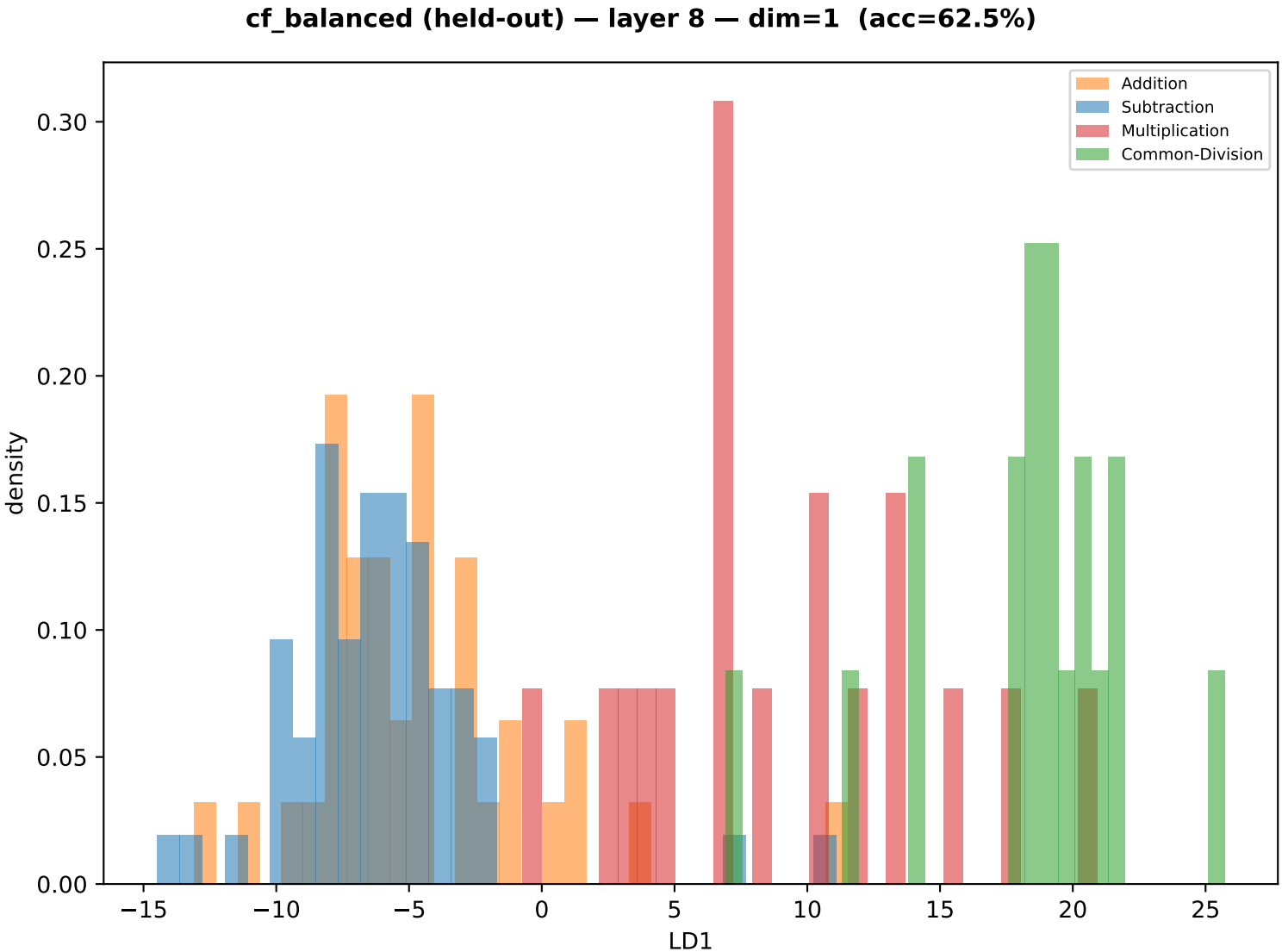

cf_balanced (held-out) — layer 3 — dim=1 (acc=38.2%)
Addition
Subtraction
Multiplication
Common-Division
density
0.00
0.05
0.10
0.15
0.20
0.25
−10
−5
0
5
10
15
LD1

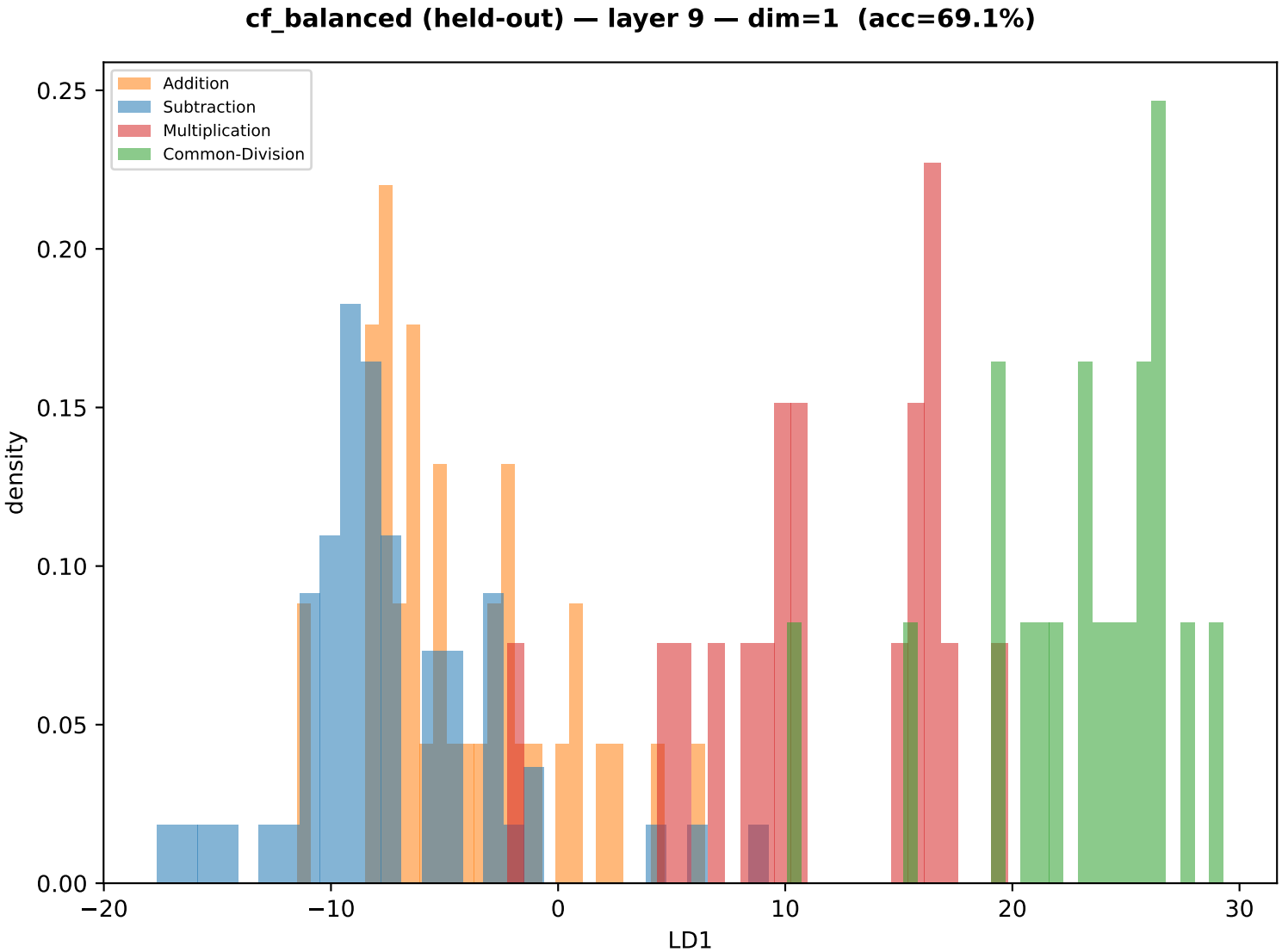

cf_balanced (held-out) — layer 4 — dim=1 (acc=47.8%)
Addition
Subtraction
Multiplication
Common-Division
density
0.00
0.05
0.10
0.15
0.20
−15
−10
−5
0
5
10
15
20
25
LD1

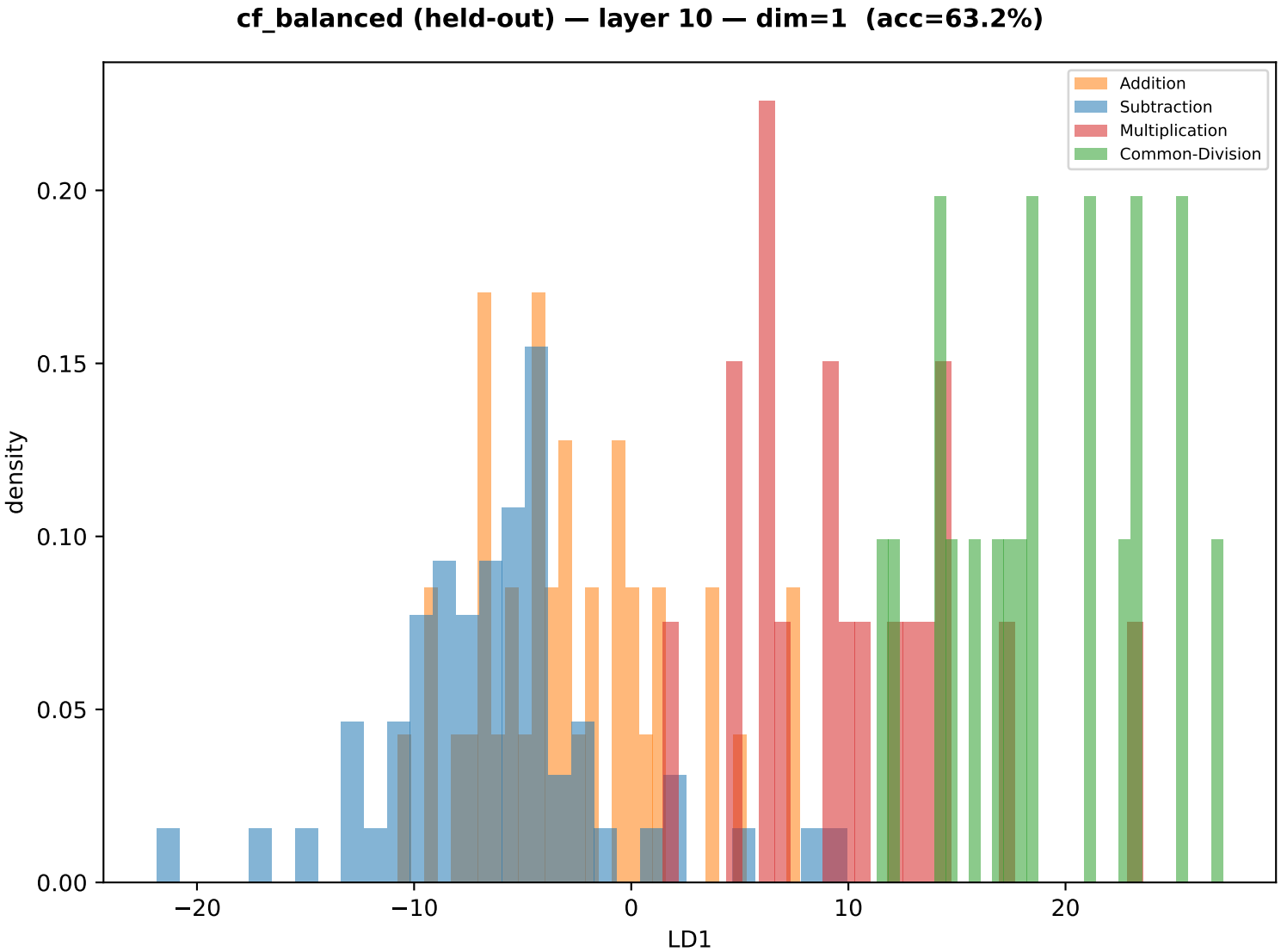

cf_balanced (held-out) — layer 5 — dim=1 (acc=45.6%)
Addition
Subtraction
Multiplication
Common-Division
density
LD1
0.00
0.05
0.10
0.15
0.20
0.25
−15
−10
−5
0
5
10
15
20

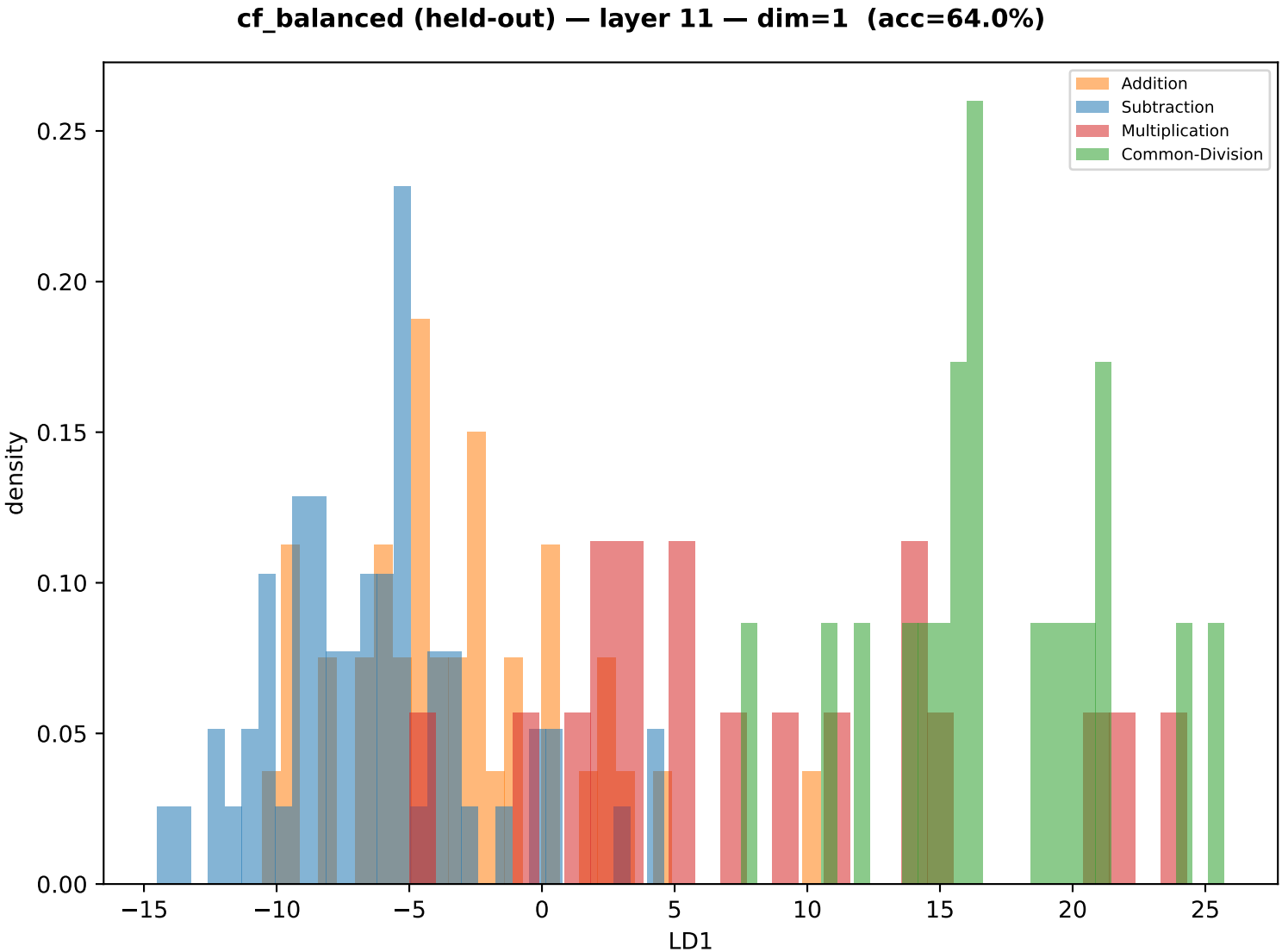

cf_balanced (held-out) — layer 6 — dim=1 (acc=50.0%)
Addition
Subtraction
Multiplication
Common-Division
density
LD1
0.00
0.05
0.10
0.15
0.20
0.25
0.30
−10
−5
0
5
10
15
20
25

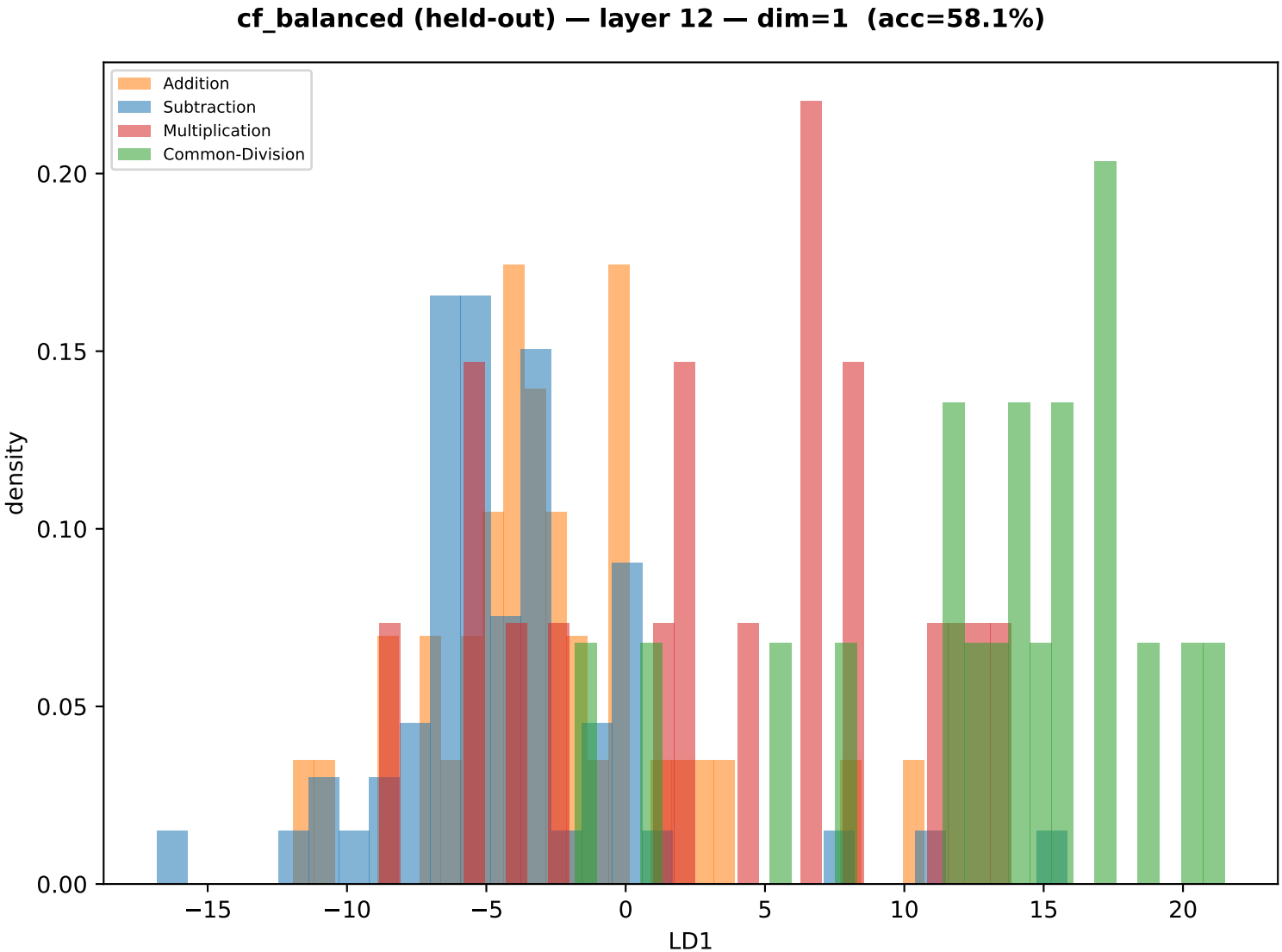

cf_balanced (held-out) — layer 7 — dim=1 (acc=61.0%)
Addition
Subtraction
Multiplication
Common-Division
density
0.00
0.05
0.10
0.15
0.20
0.25
0.30
−10
0
10
20
LD1

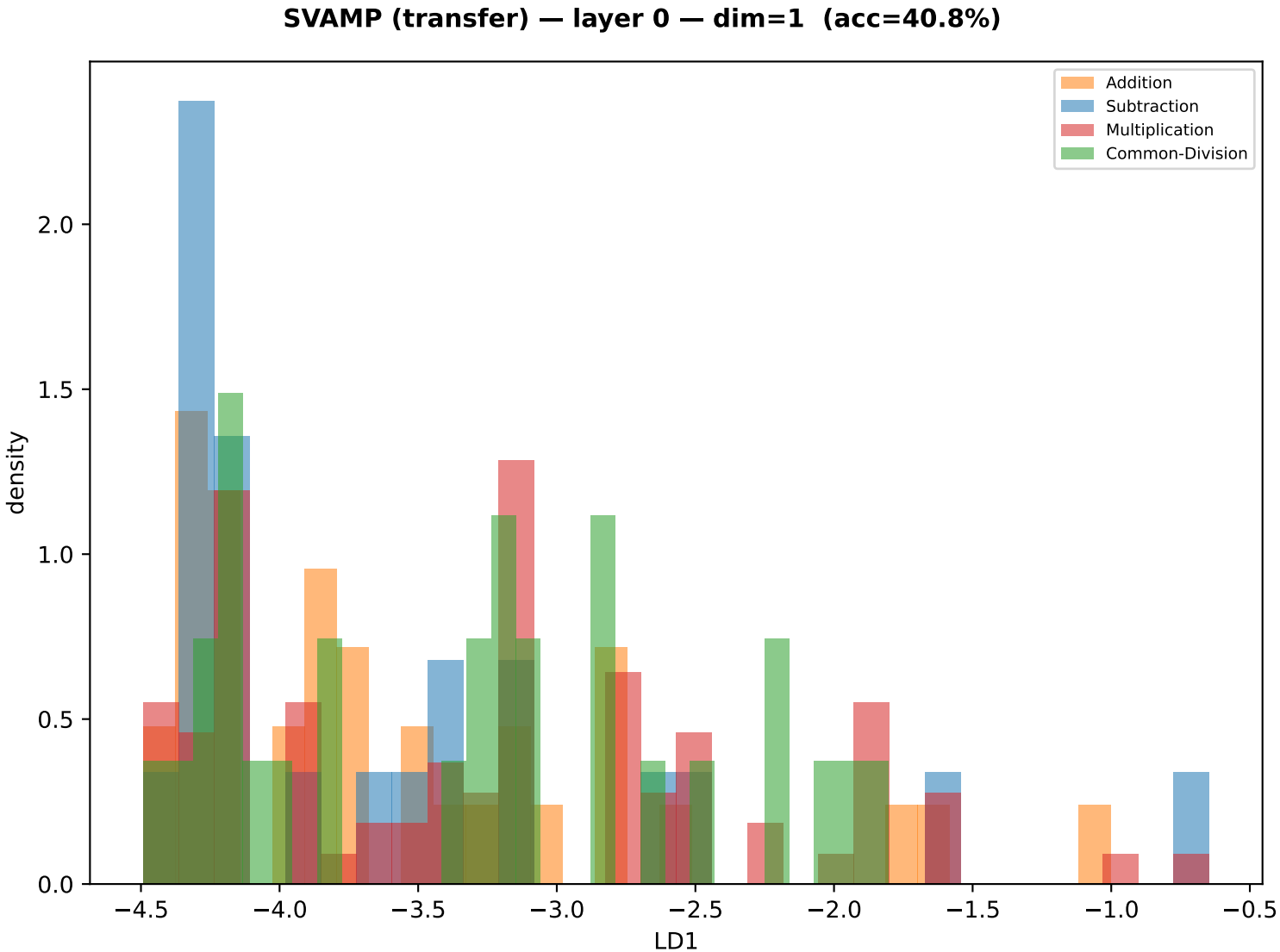

cf_balanced (held-out) — layer 8 — dim=1 (acc=62.5%)
Addition
Subtraction
Multiplication
Common-Division
density
0.00
0.05
0.10
0.15
0.20
0.25
0.30
−15
−10
−5
0
5
10
15
20
25
LD1

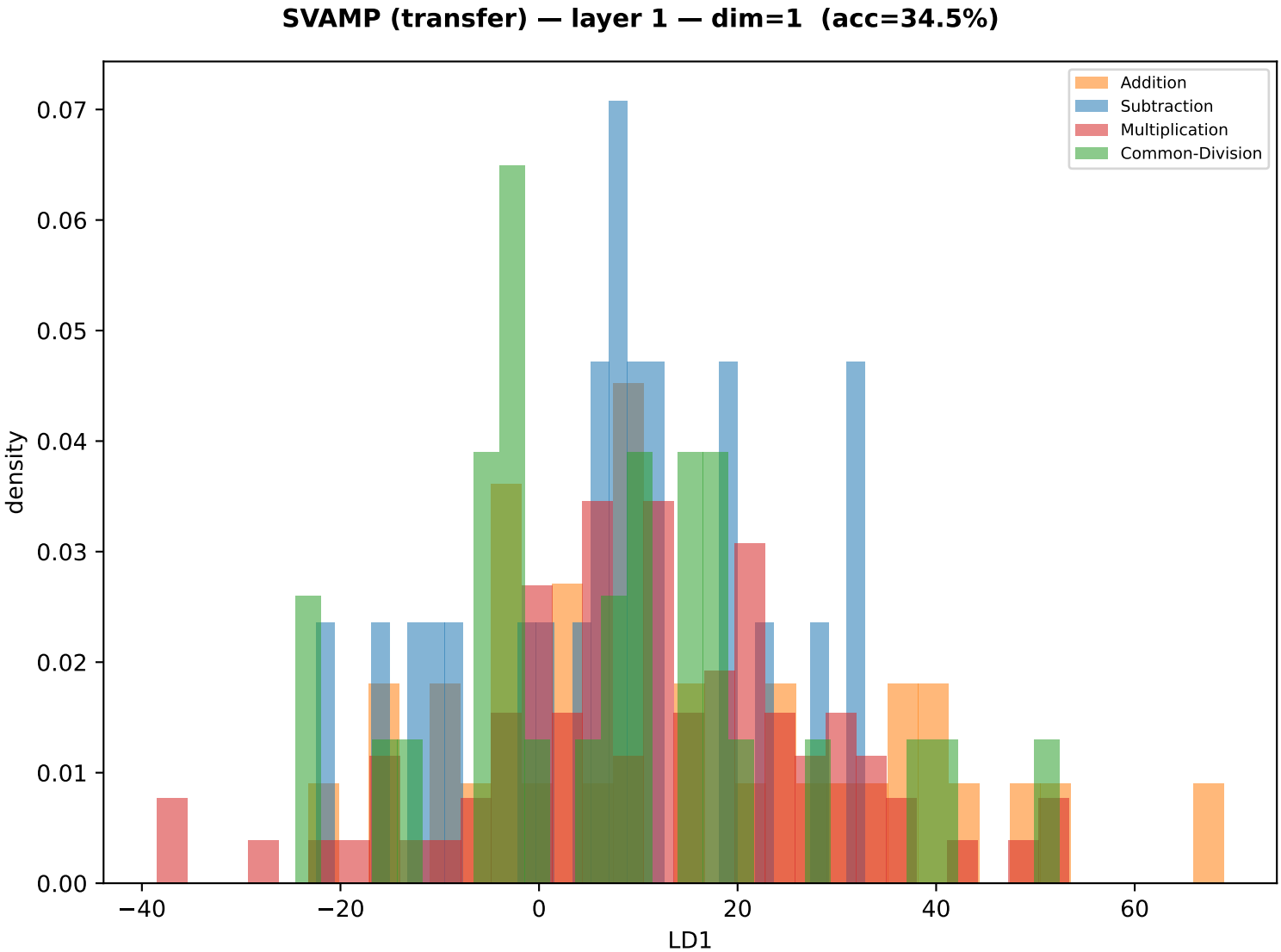

cf_balanced (held-out) — layer 9 — dim=1 (acc=69.1%)
Addition
Subtraction
Multiplication
Common-Division
density
LD1
0.00
0.05
0.10
0.15
0.20
0.25
−20
−10
0
10
20
30

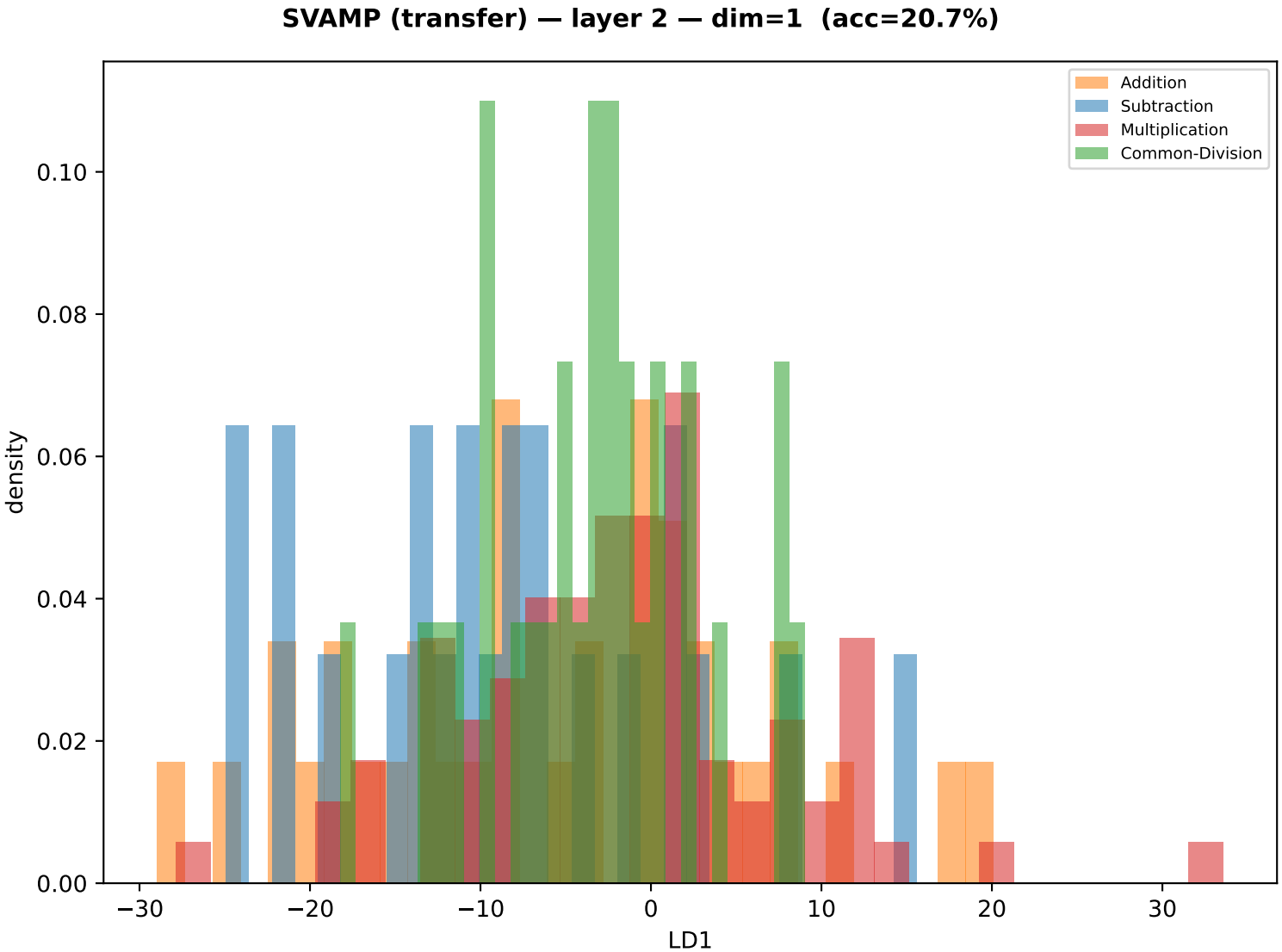

cf_balanced (held-out) — layer 10 — dim=1 (acc=63.2%)
Addition
Subtraction
Multiplication
Common-Division
density
0.00
0.05
0.10
0.15
0.20
−20
−10
0
10
20
LD1

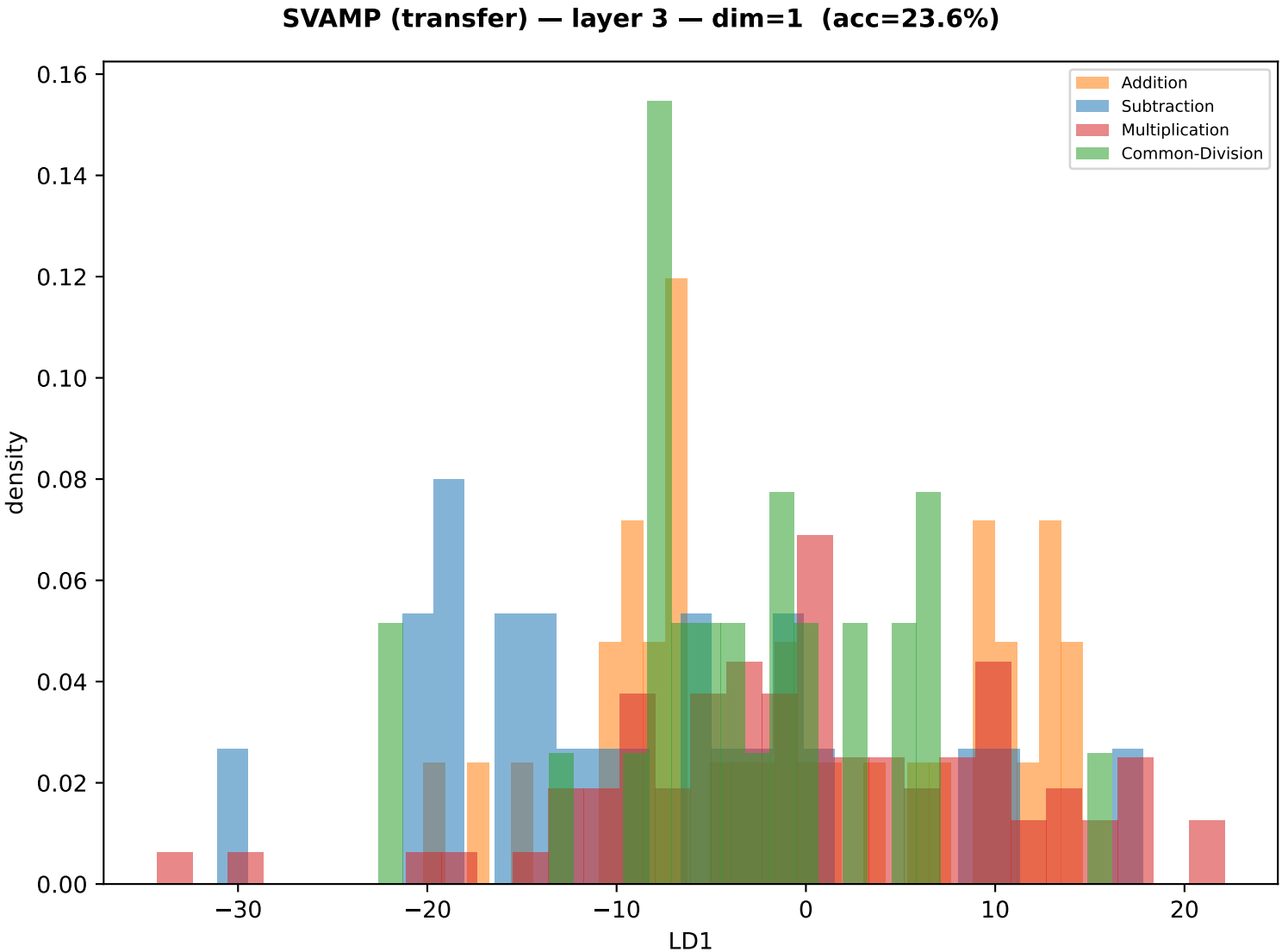

cf_balanced (held-out) — layer 11 — dim=1 (acc=64.0%)
Addition
Subtraction
Multiplication
Common-Division
density
0.00
0.05
0.10
0.15
0.20
0.25
−15
−10
−5
0
5
10
15
20
25
LD1

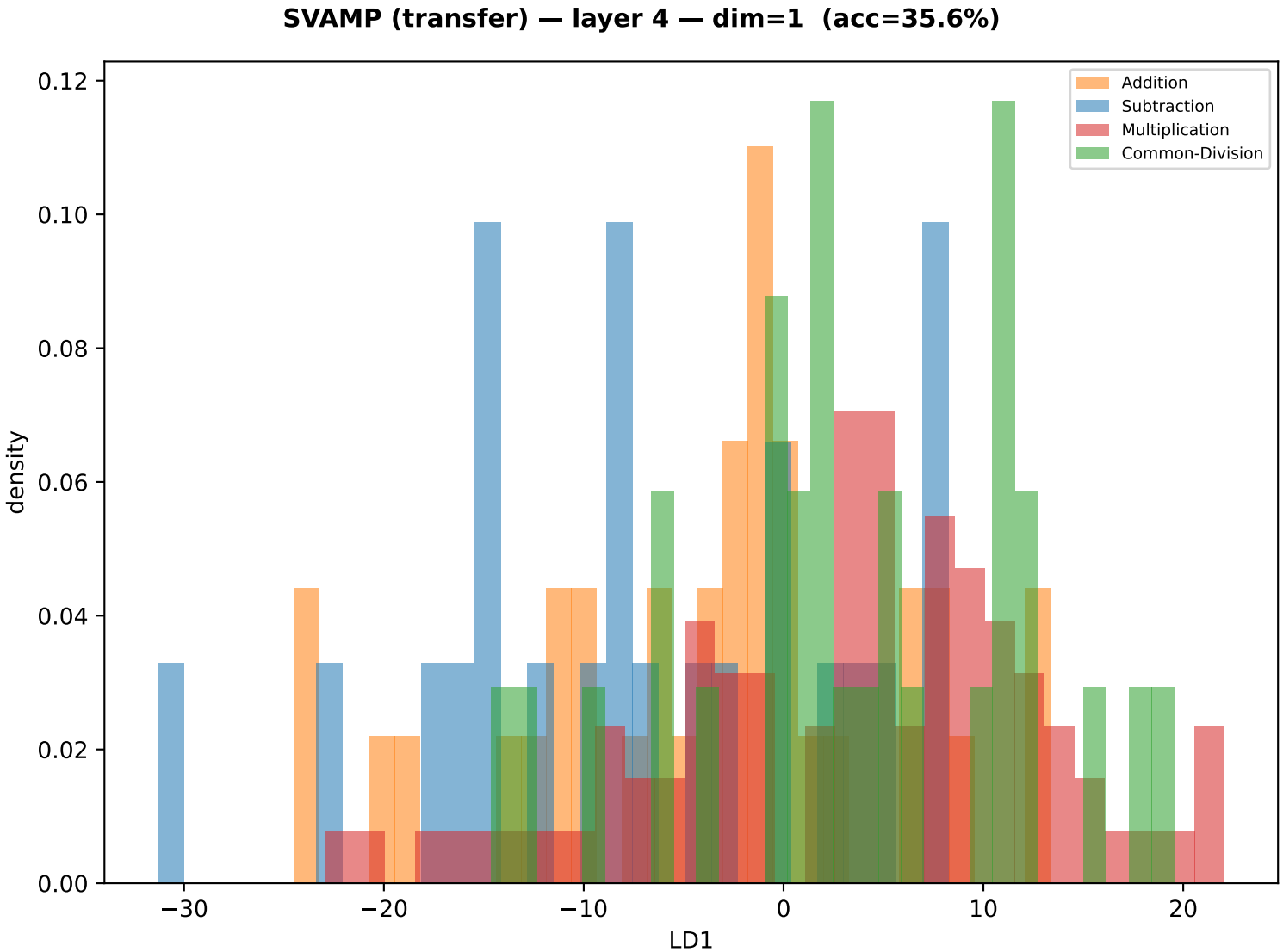

cf_balanced (held-out) — layer 12 — dim=1 (acc=58.1%)
Addition
Subtraction
Multiplication
Common-Division
density
0.00
0.05
0.10
0.15
0.20
−15
−10
−5
0
5
10
15
20
LD1

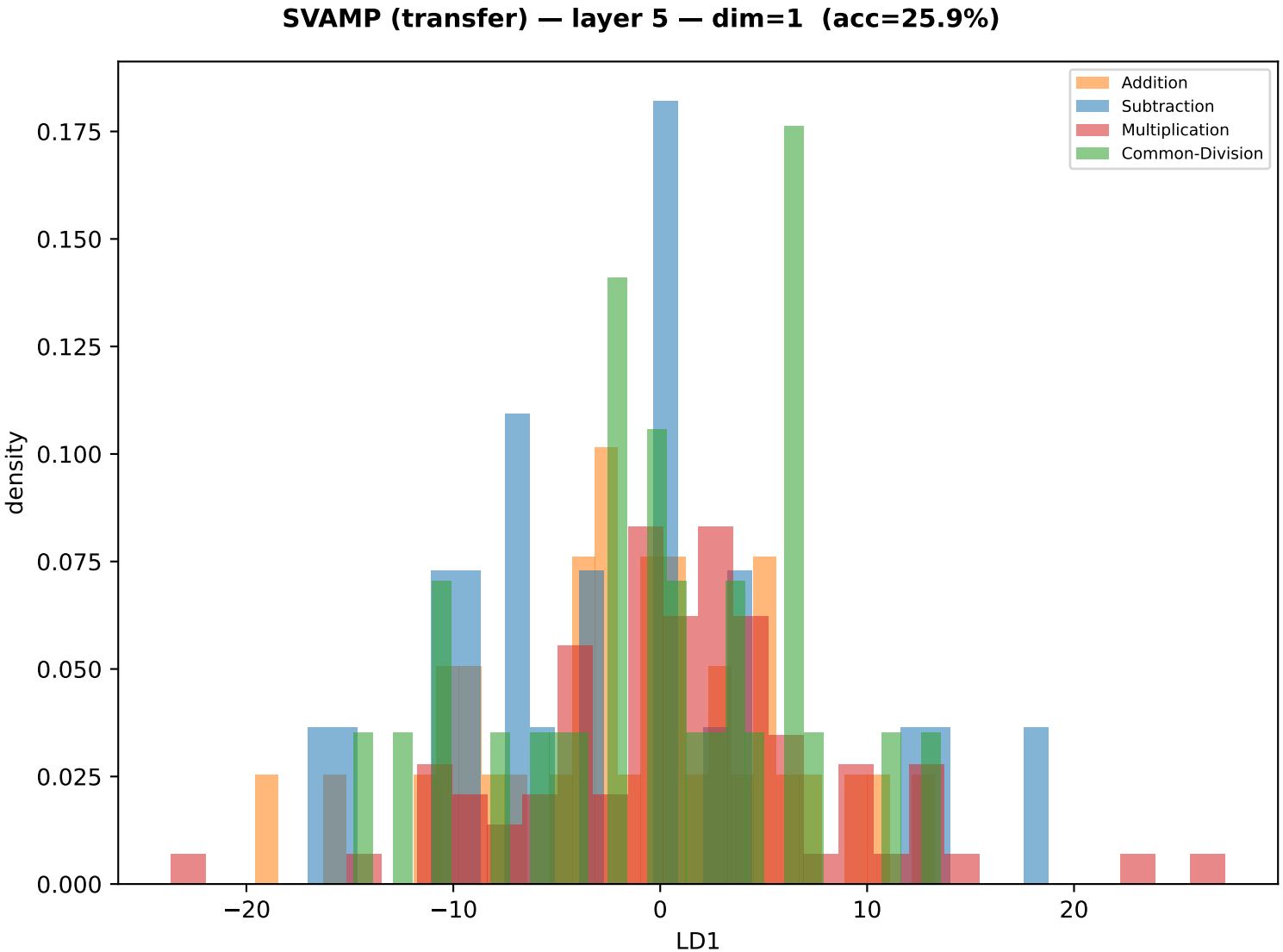

SVAMP (transfer) — layer 0 — dim=1 (acc=40.8%)
Addition
Subtraction
Multiplication
Common-Division
density
LD1
0.0
0.5
1.0
1.5
2.0
−4.5
−4.0
−3.5
−3.0
−2.5
−2.0
−1.5
−1.0
−0.5

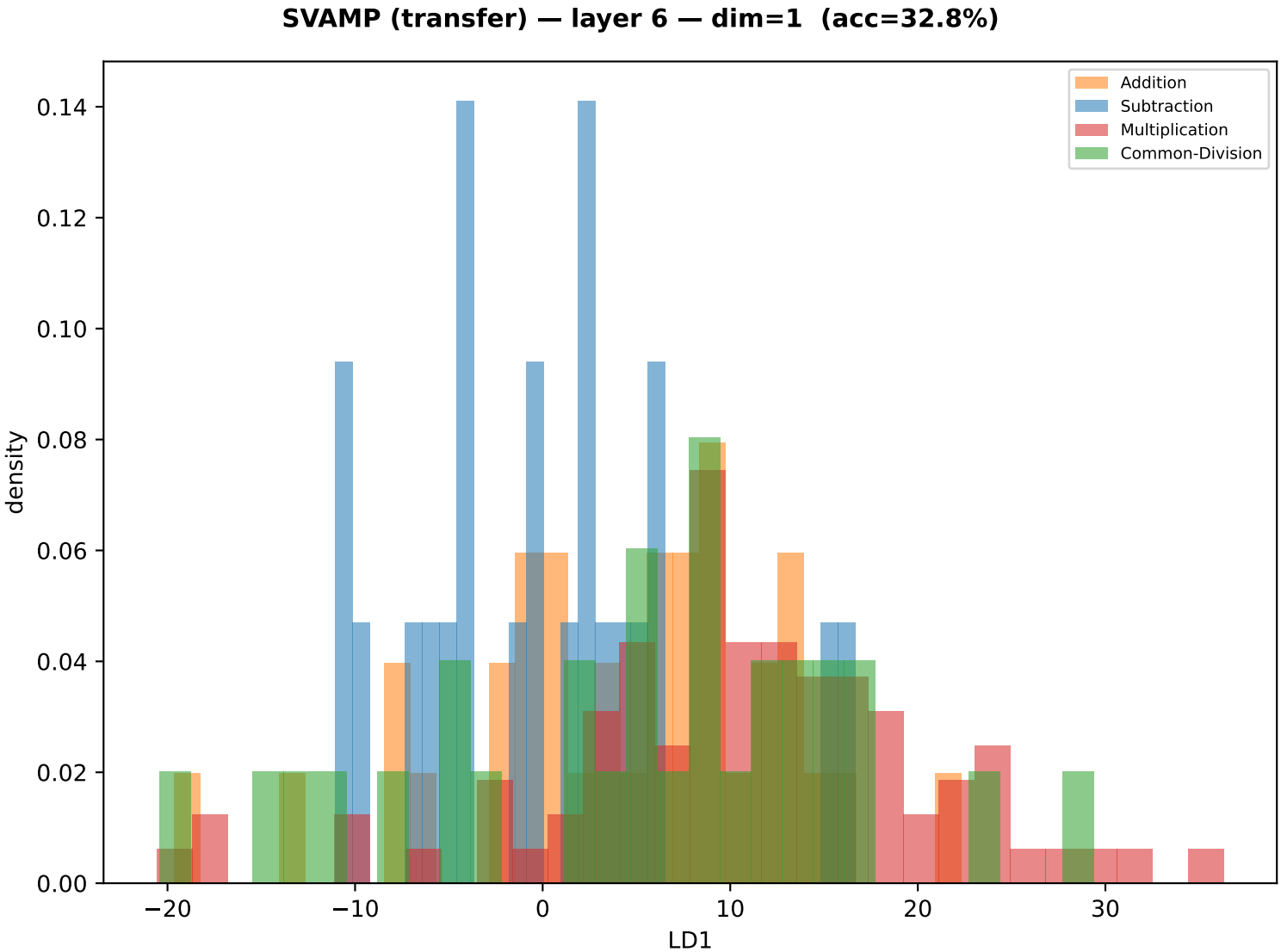

SVAMP (transfer) — layer 1 — dim=1 (acc=34.5%)
Addition
Subtraction
Multiplication
Common-Division
density
LD1
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
−40
−20
0
20
40
60

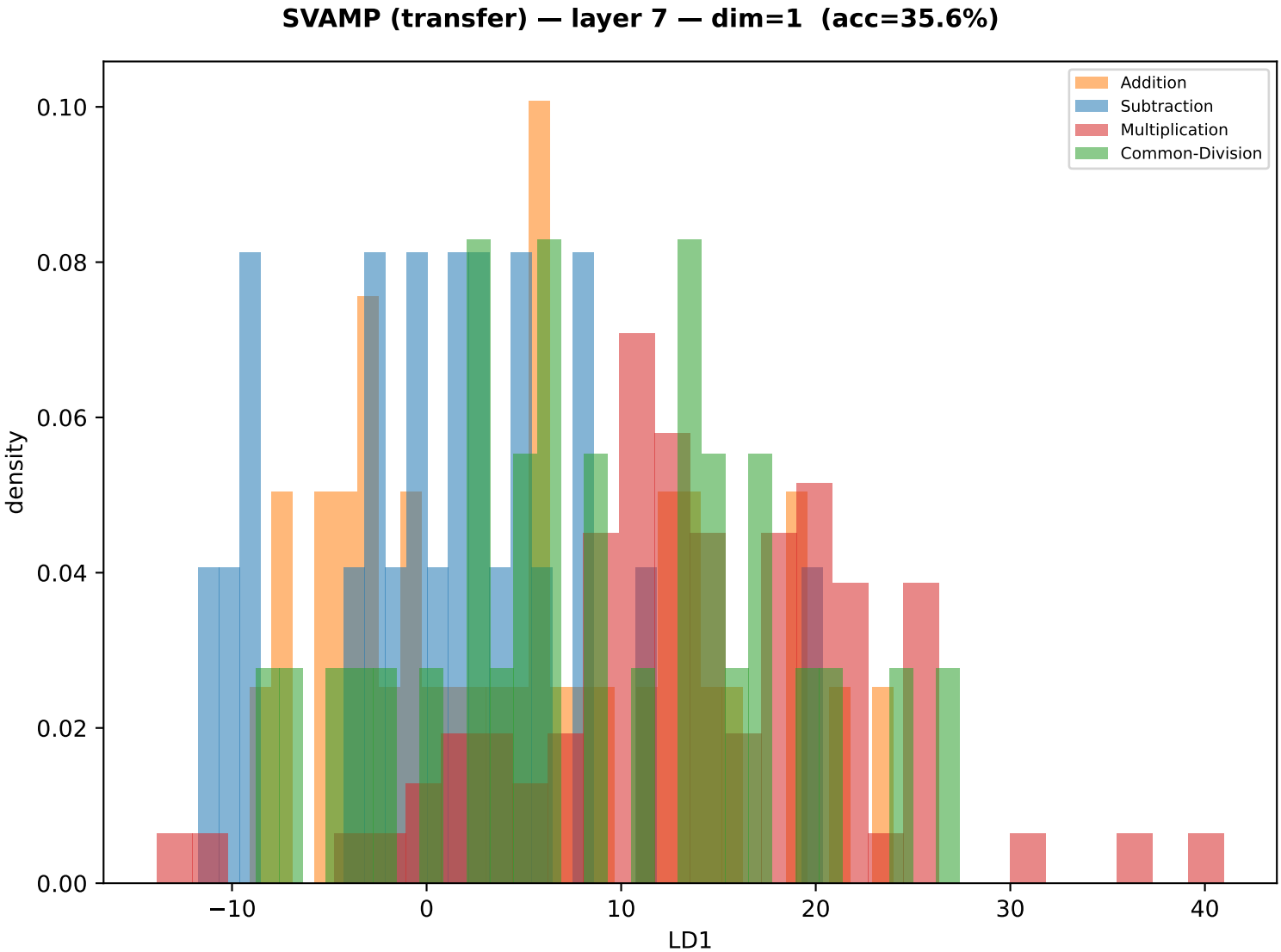

SVAMP (transfer) — layer 2 — dim=1 (acc=20.7%)
Addition
Subtraction
Multiplication
Common-Division
density
0.00
0.02
0.04
0.06
0.08
0.10
−30
−20
−10
0
10
20
30
LD1

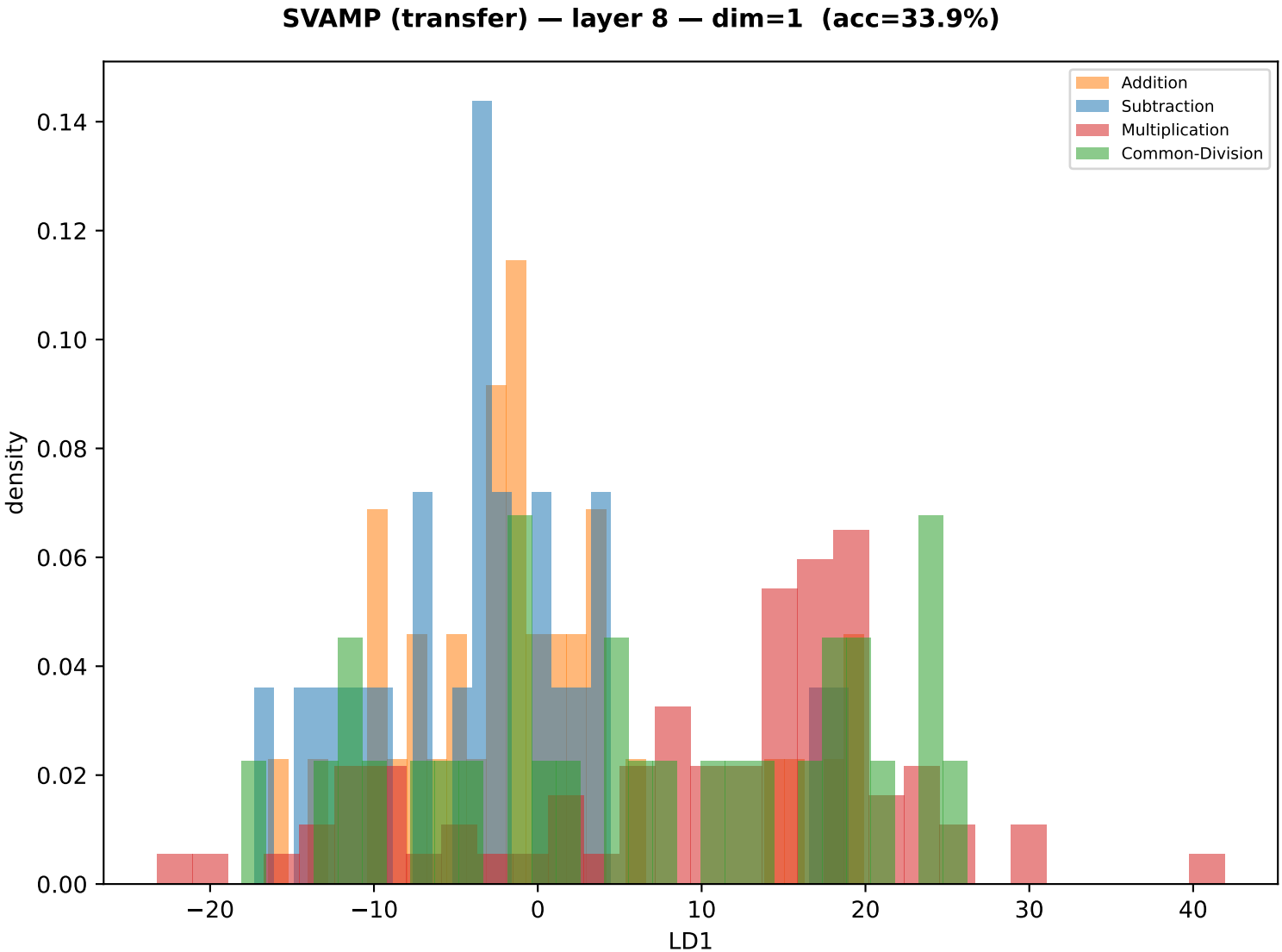

SVAMP (transfer) — layer 3 — dim=1 (acc=23.6%)
Addition
Subtraction
Multiplication
Common-Division
density
LD1
0.16
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
−30
−20
−10
0
10
20

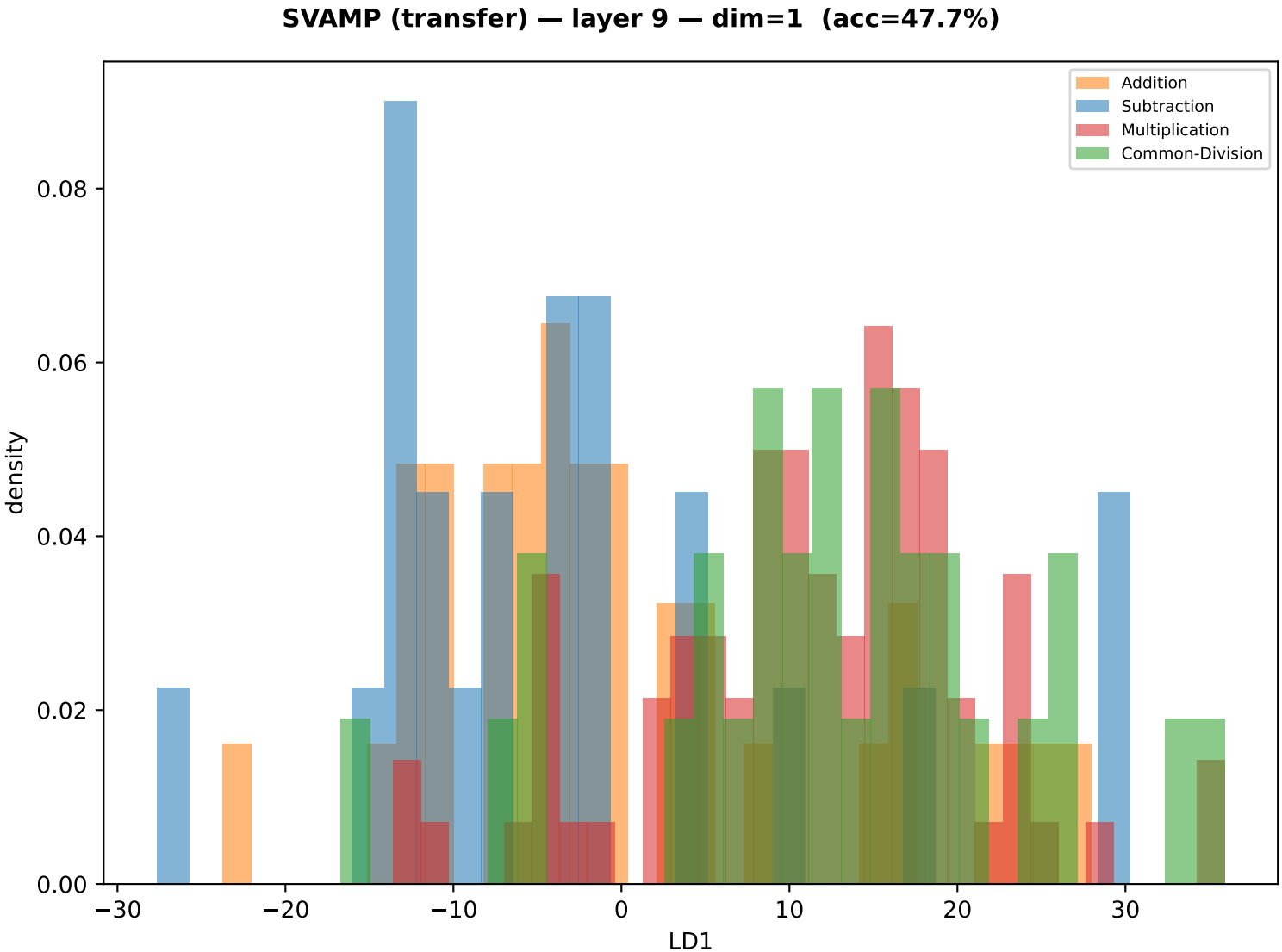

SVAMP (transfer) — layer 4 — dim=1 (acc=35.6%)
Addition
Subtraction
Multiplication
Common-Division
density
LD1
0.00
0.02
0.04
0.06
0.08
0.10
0.12
−30
−20
−10
0
10
20

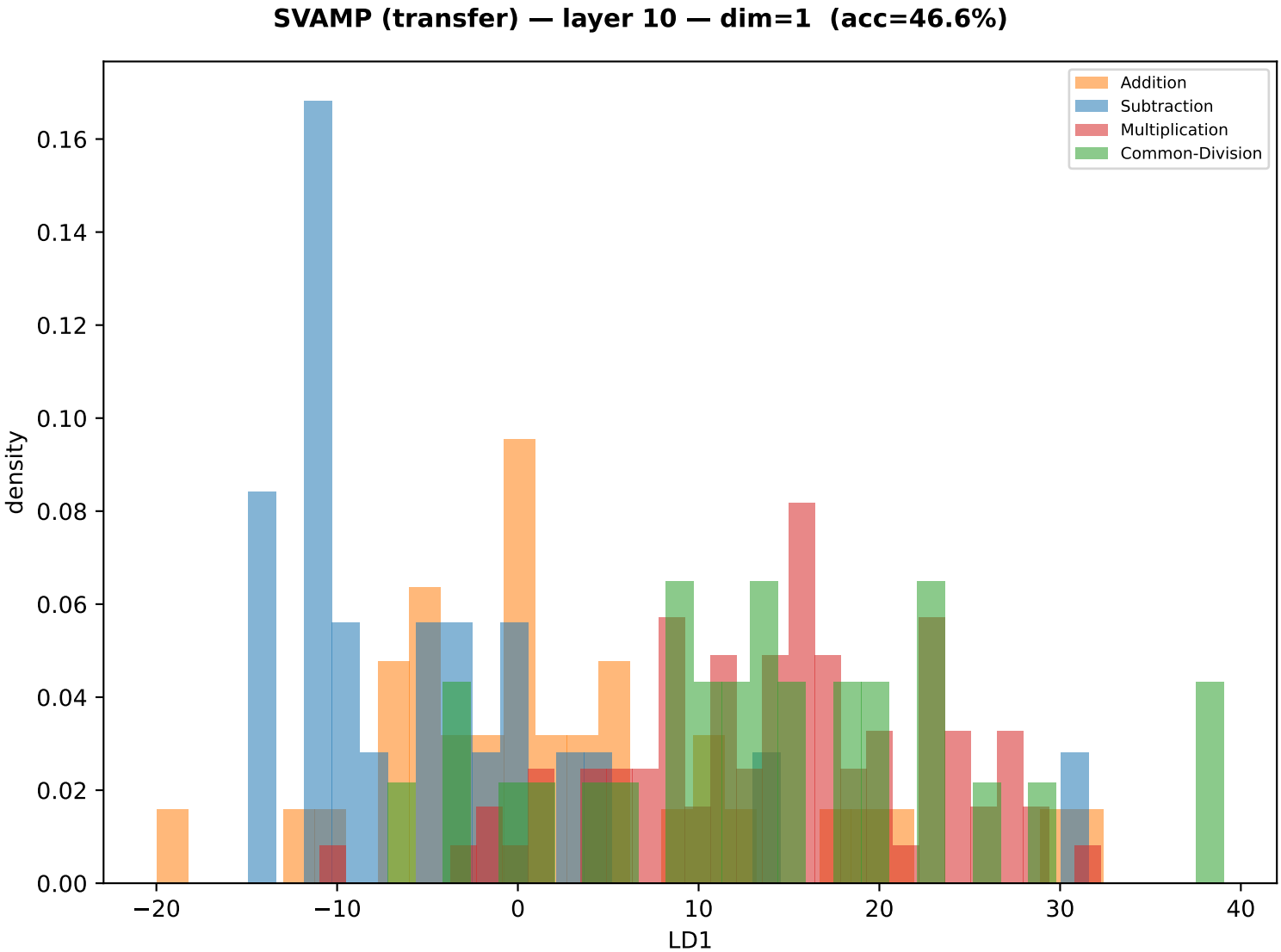

SVAMP (transfer) — layer 5 — dim=1 (acc=25.9%)
Addition
Subtraction
Multiplication
Common-Division
density
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
−20
−10
0
10
20
LD1

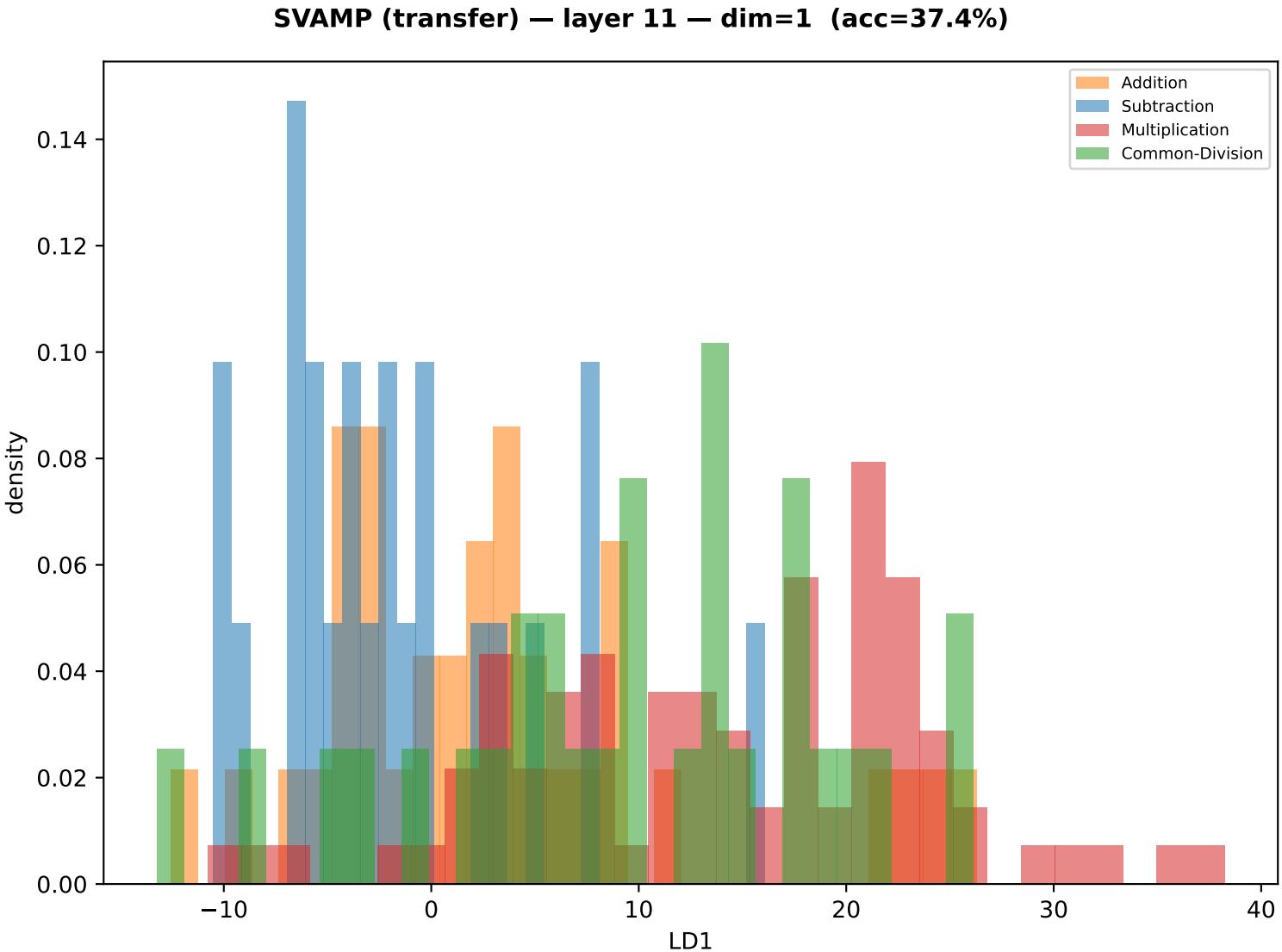

SVAMP (transfer) — layer 6 — dim=1 (acc=32.8%)
Addition
Subtraction
Multiplication
Common-Division
density
LD1
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
−20
−10
0
10
20
30

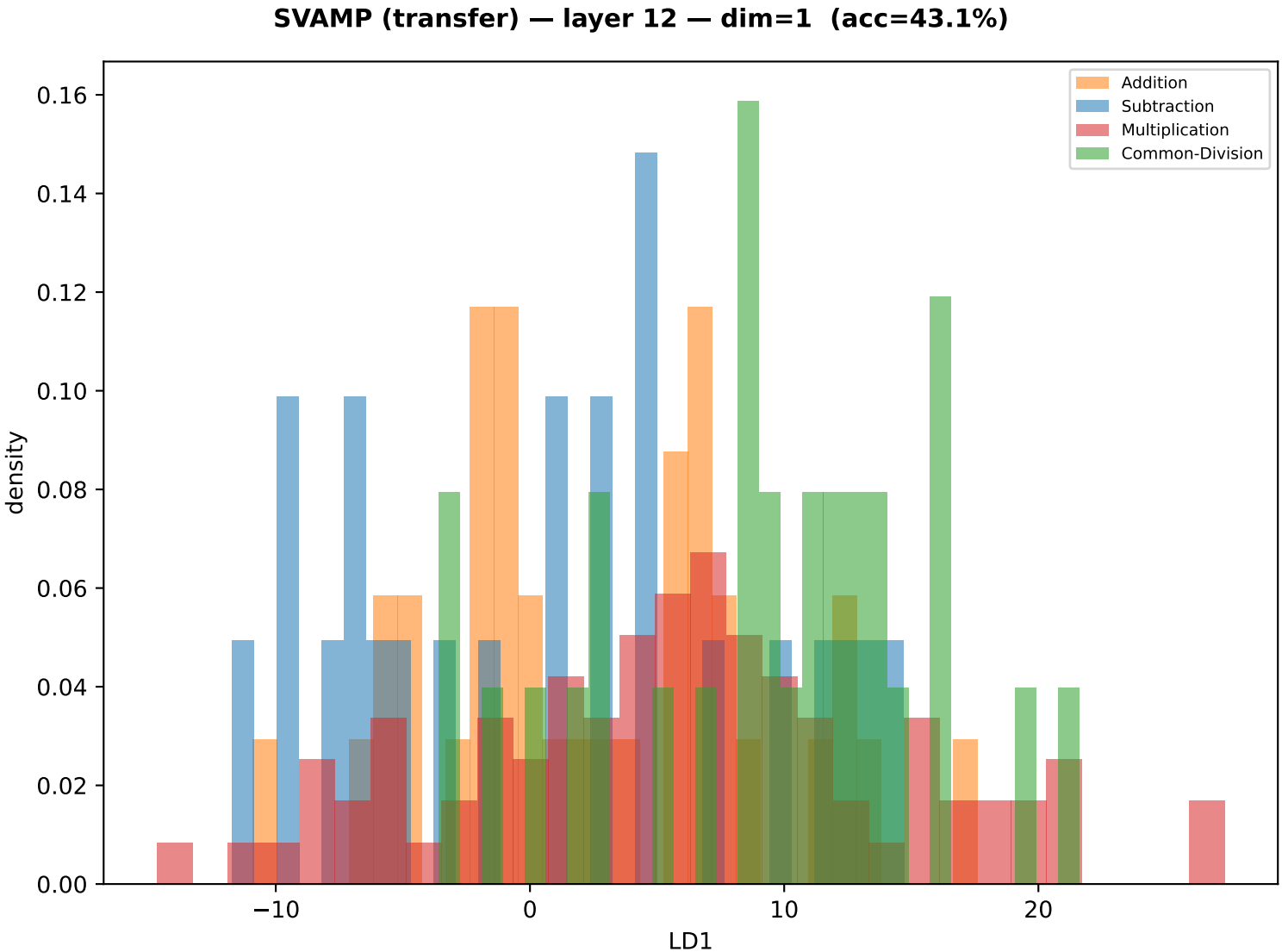

SVAMP (transfer) — layer 7 — dim=1 (acc=35.6%)
Addition
Subtraction
Multiplication
Common-Division
density
0.00
0.02
0.04
0.06
0.08
0.10
−10
0
10
20
30
40
LD1

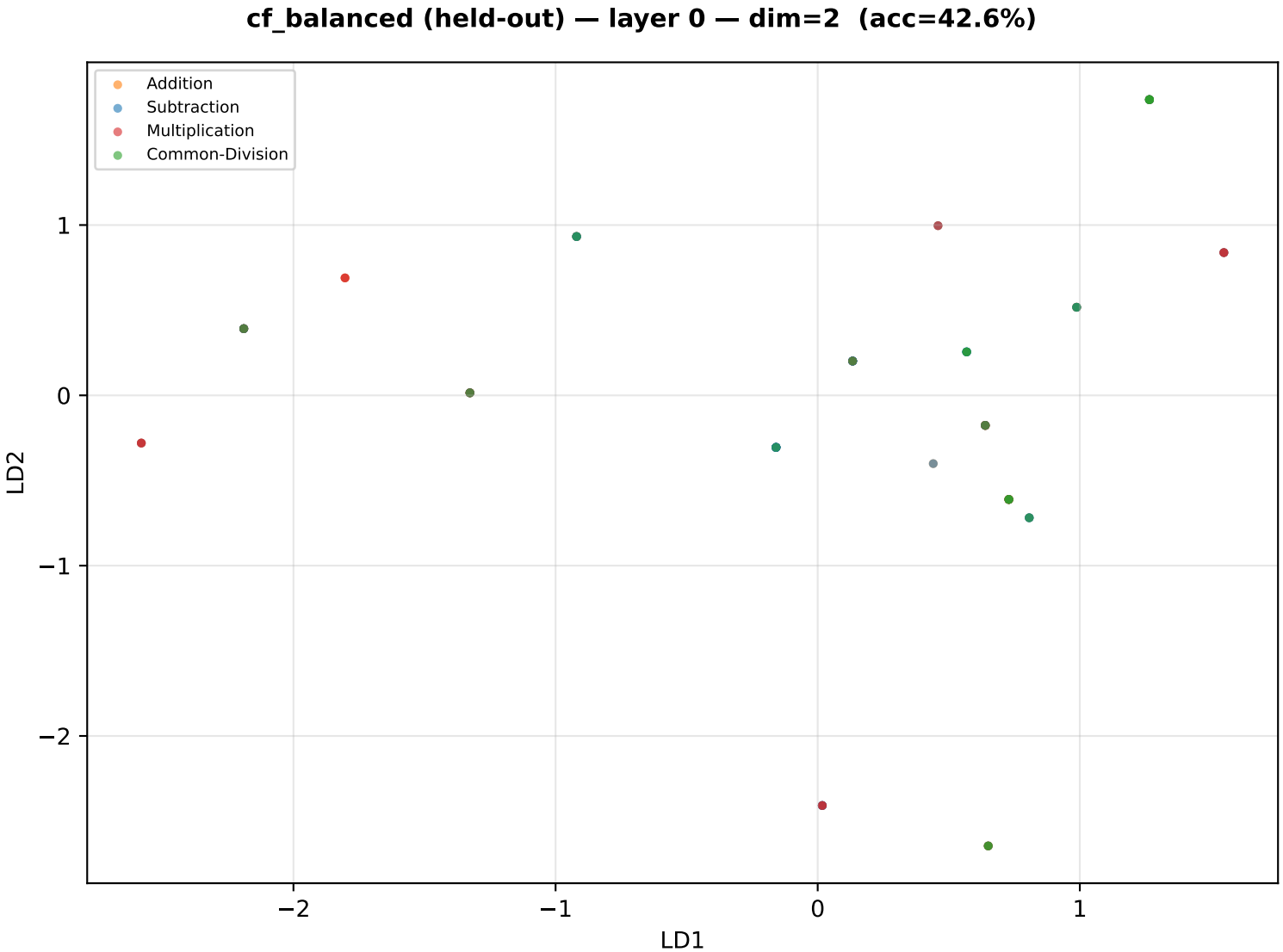

SVAMP (transfer) — layer 8 — dim=1 (acc=33.9%)
Addition
Subtraction
Multiplication
Common-Division
density
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
−20
−10
0
10
20
30
40
LD1

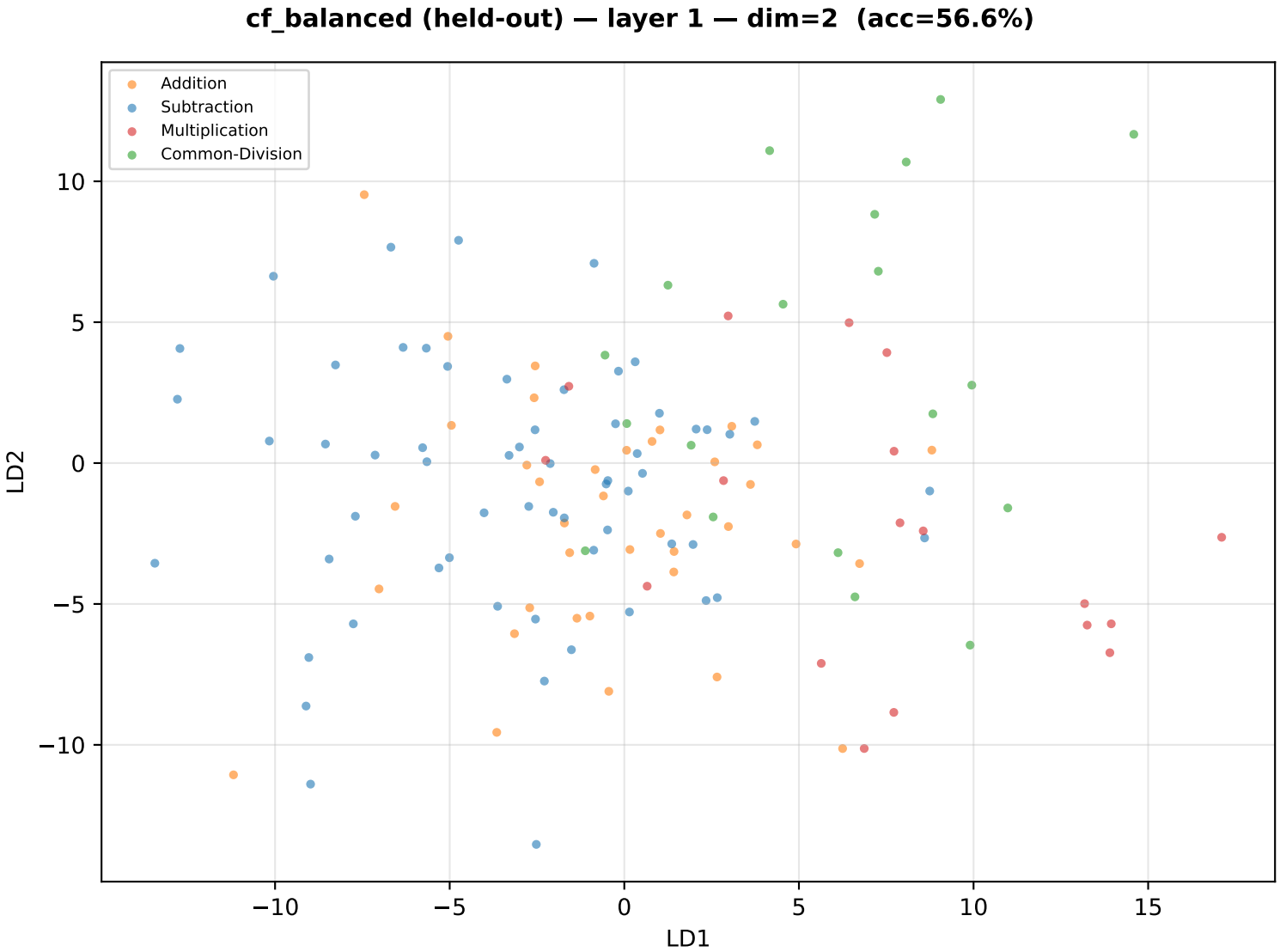

SVAMP (transfer) — layer 9 — dim=1 (acc=47.7%)
Addition
Subtraction
Multiplication
Common-Division
density
0.00
0.02
0.04
0.06
0.08
−30
−20
−10
0
10
20
30
LD1

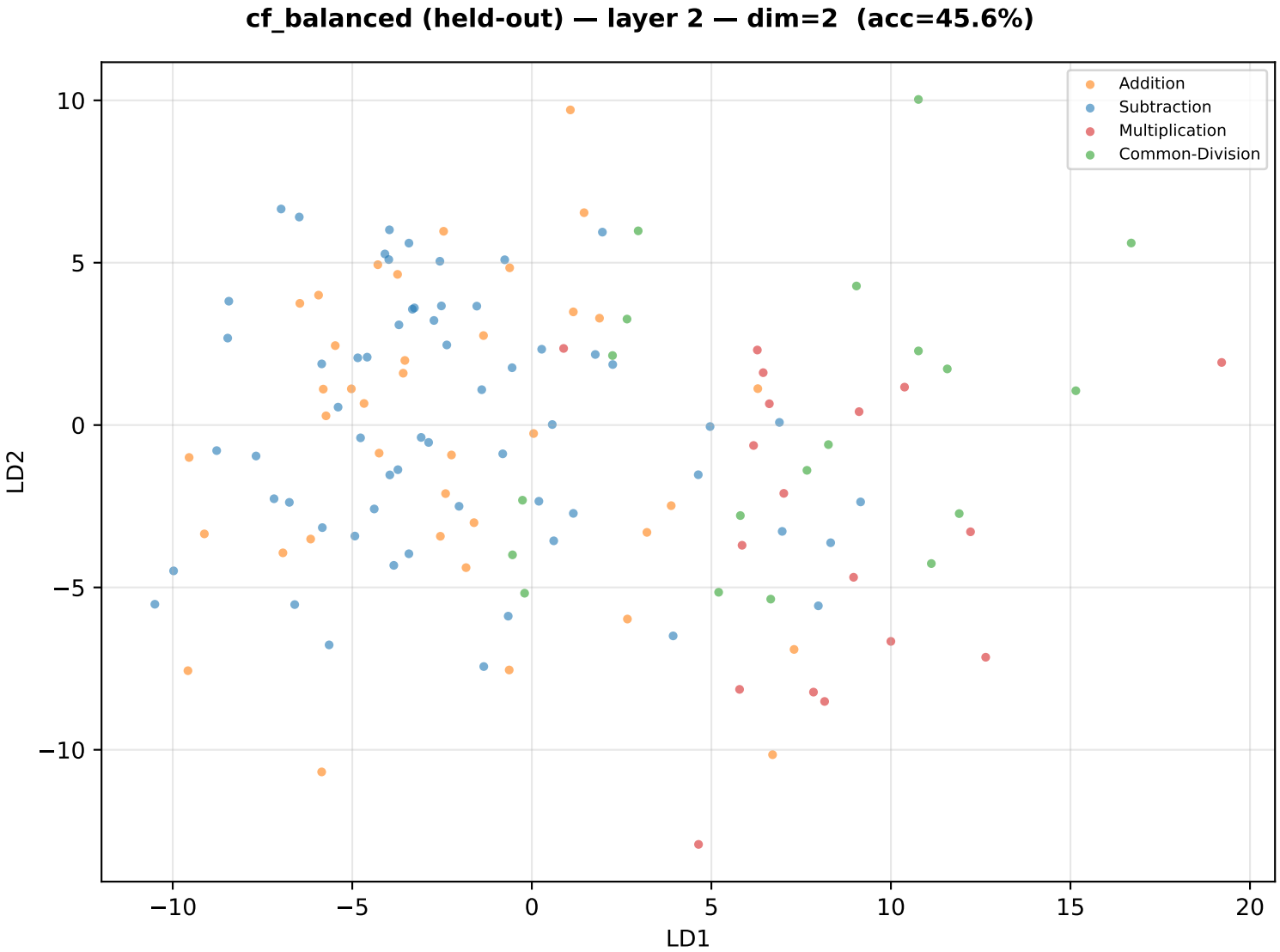

SVAMP (transfer) — layer 10 — dim=1 (acc=46.6%)
Addition
Subtraction
Multiplication
Common-Division
density
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
0.16
−20
−10
0
10
20
30
40
LD1

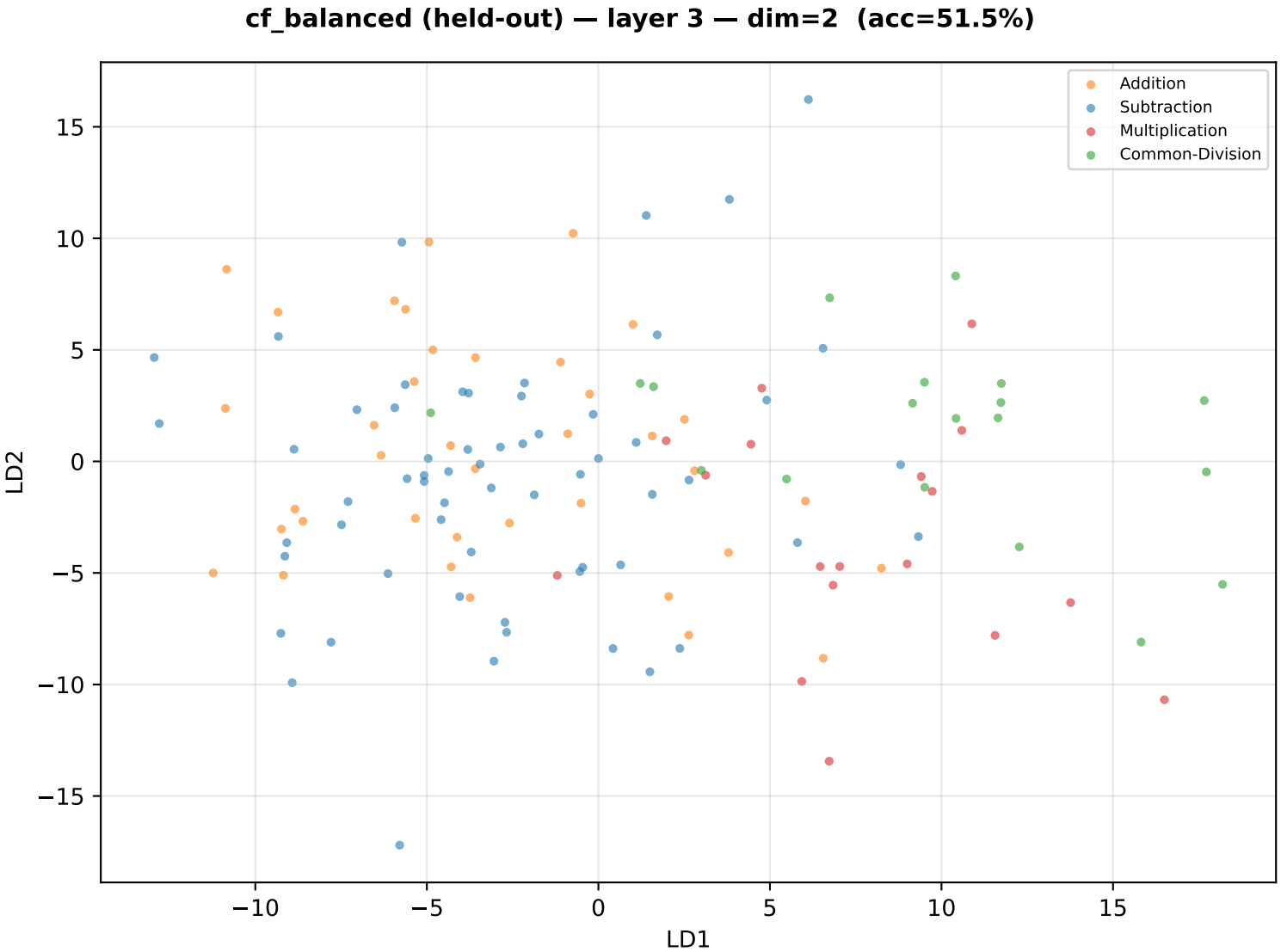

SVAMP (transfer) — layer 11 — dim=1 (acc=37.4%)
Addition
Subtraction
Multiplication
Common-Division
density
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
−10
0
10
20
30
40
LD1

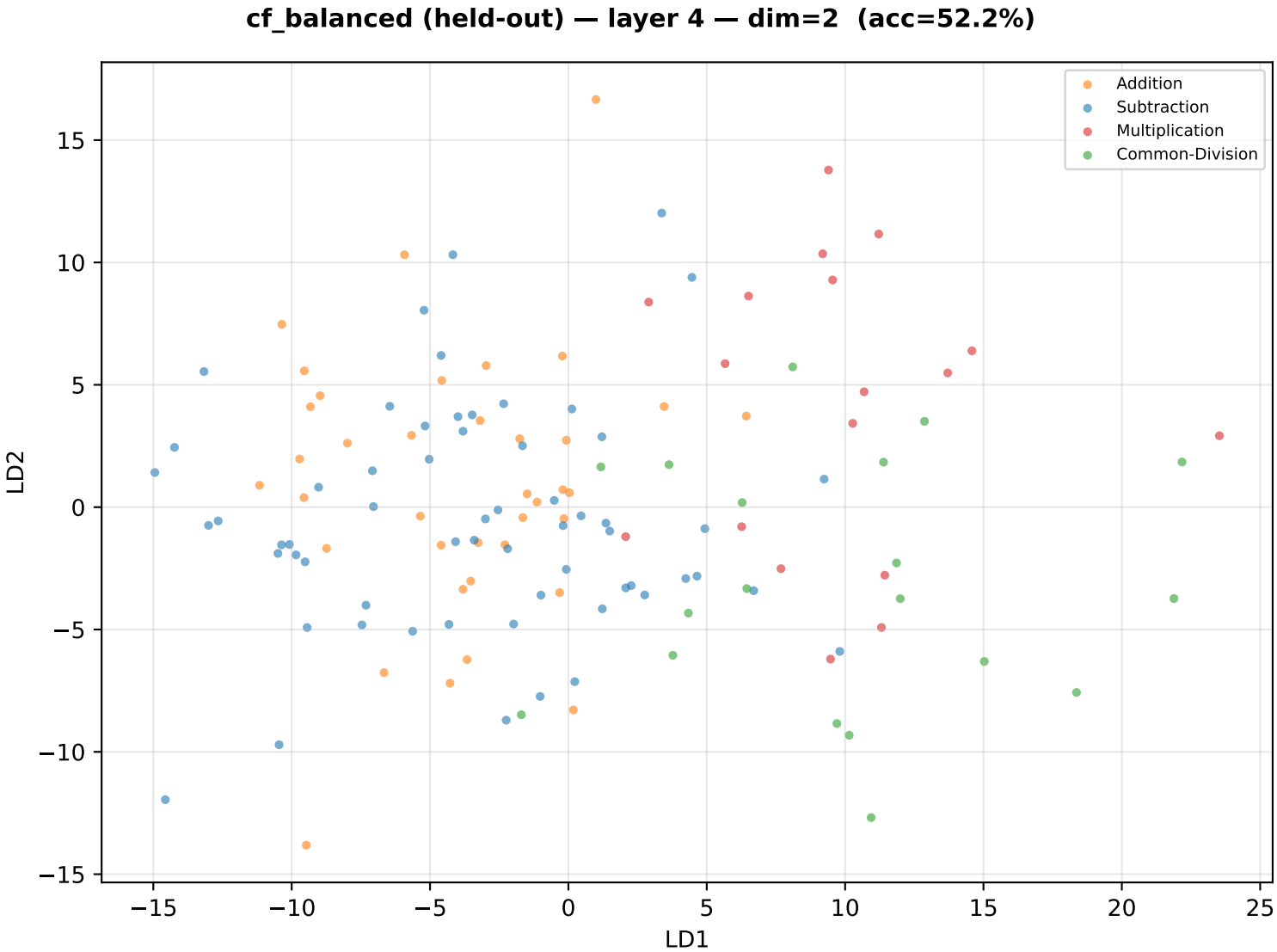

SVAMP (transfer) — layer 12 — dim=1 (acc=43.1%)
Addition
Subtraction
Multiplication
Common-Division
density
LD1
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
0.16
−10
0
10
20

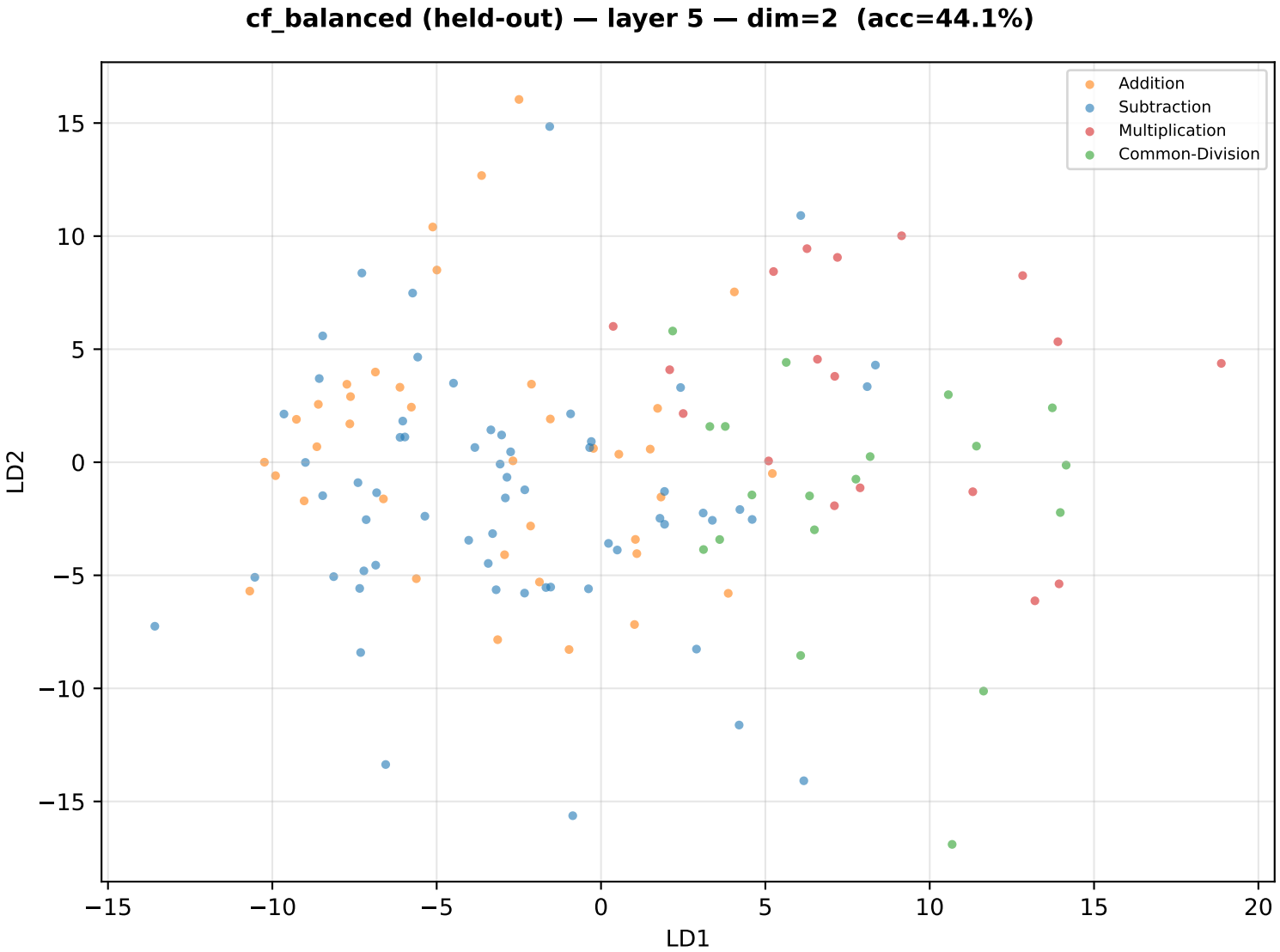

cf_balanced (held-out) — layer 0 — dim=2 (acc=42.6%)
Addition
Subtraction
Multiplication
Common-Division
LD2
LD1
1
0
−1
−2
−2
−1
0
1

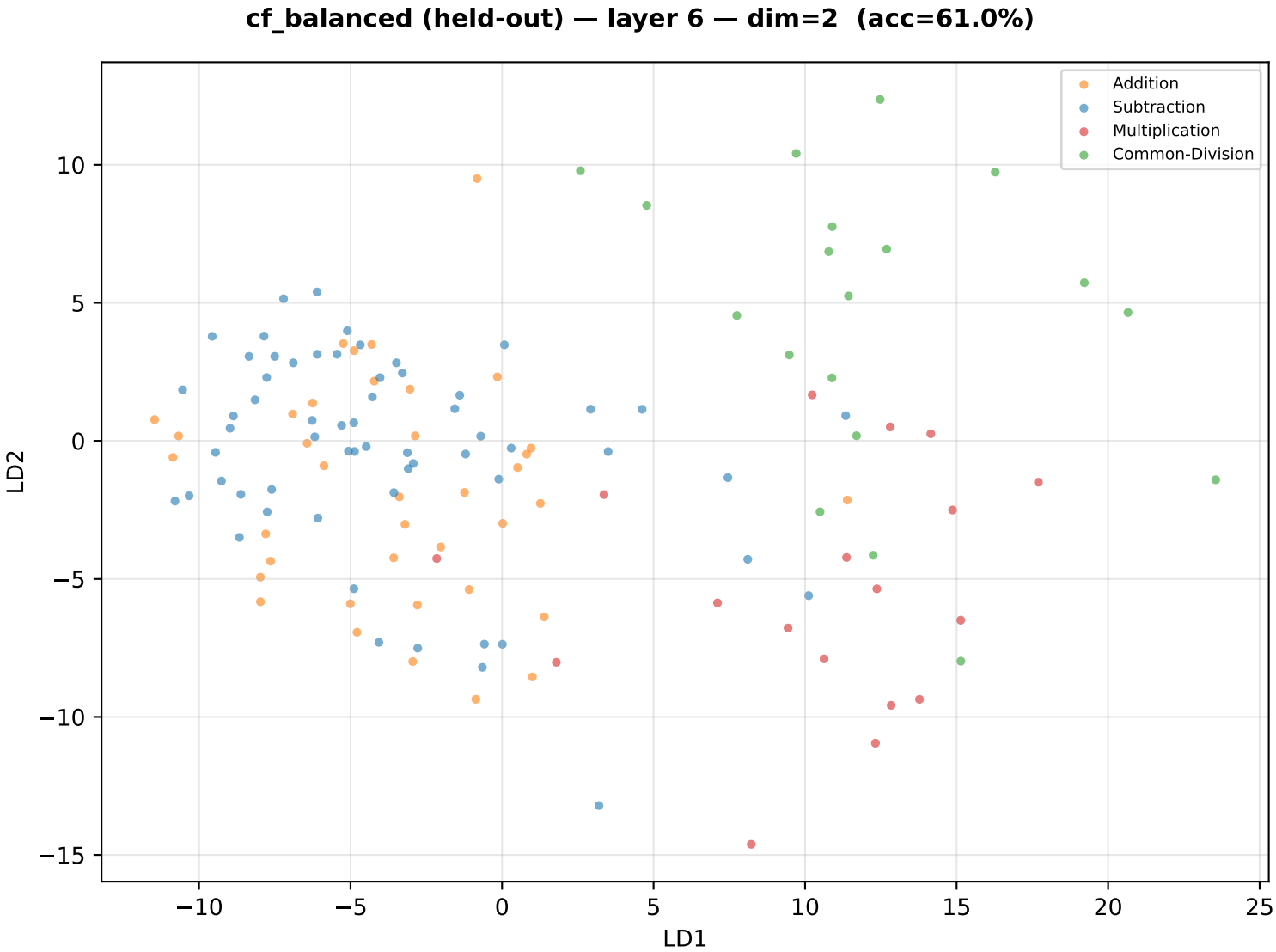

cf_balanced (held-out) — layer 1 — dim=2 (acc=56.6%)
Addition
Subtraction
Multiplication
Common-Division
LD1
LD2
−10
−5
0
5
10
15
−10
−5
0
5
10

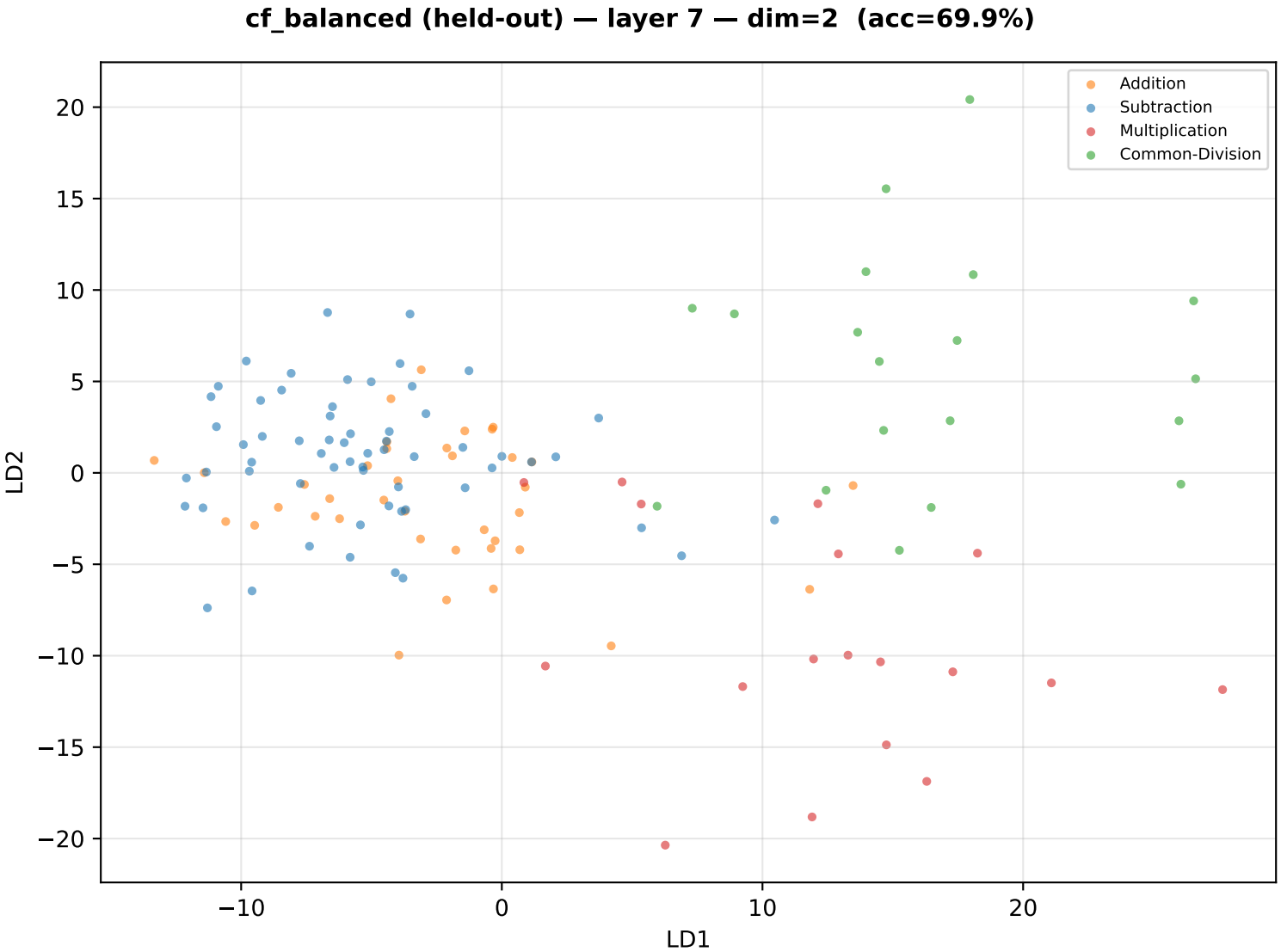

cf_balanced (held-out) — layer 2 — dim=2 (acc=45.6%)
Addition
Subtraction
Multiplication
Common-Division
LD1
LD2
−10
−5
0
5
10
15
20

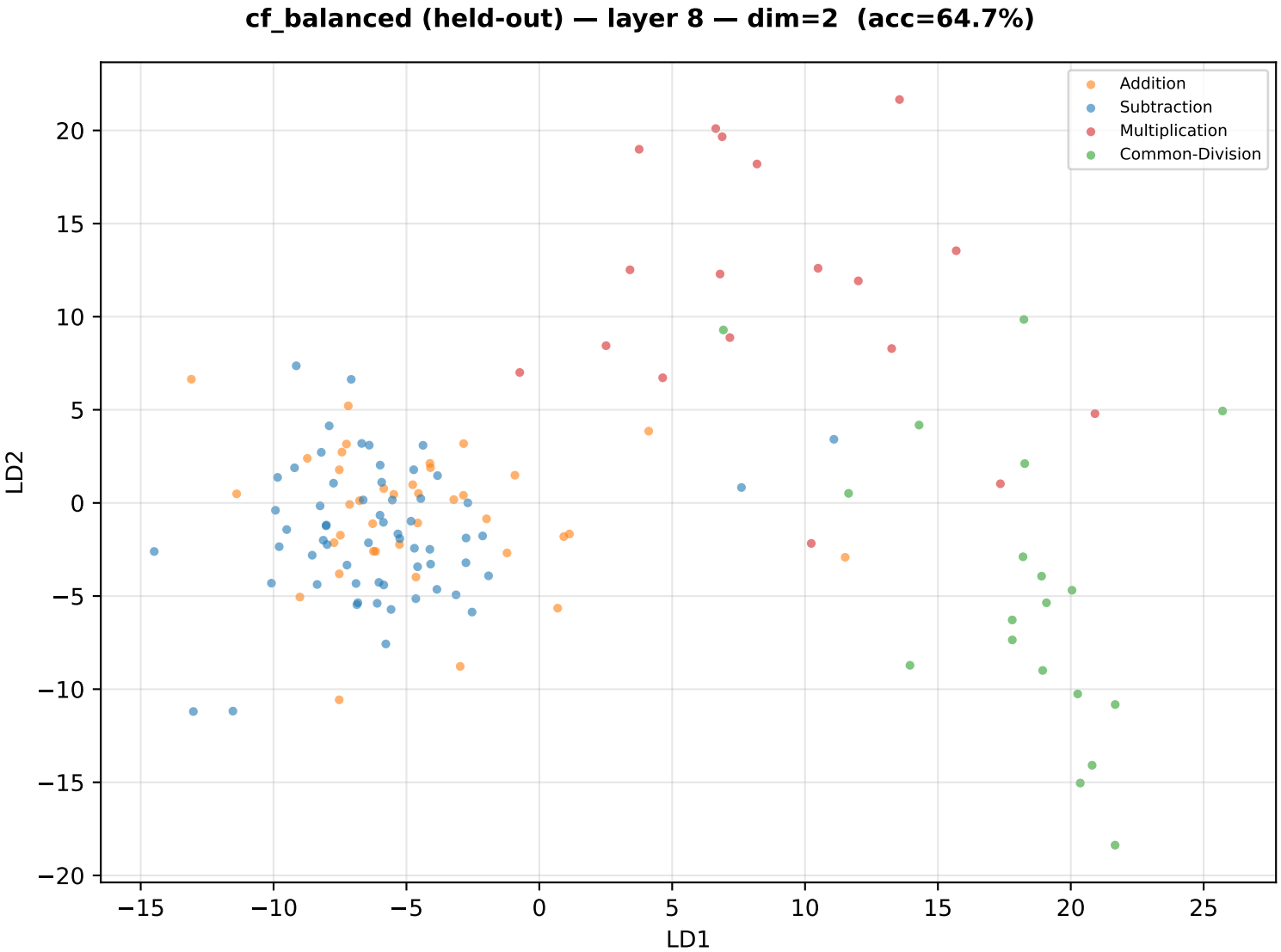

cf_balanced (held-out) — layer 3 — dim=2 (acc=51.5%)
Addition
Subtraction
Multiplication
Common-Division
LD1
LD2
−10
−5
0
5
10
15
−15

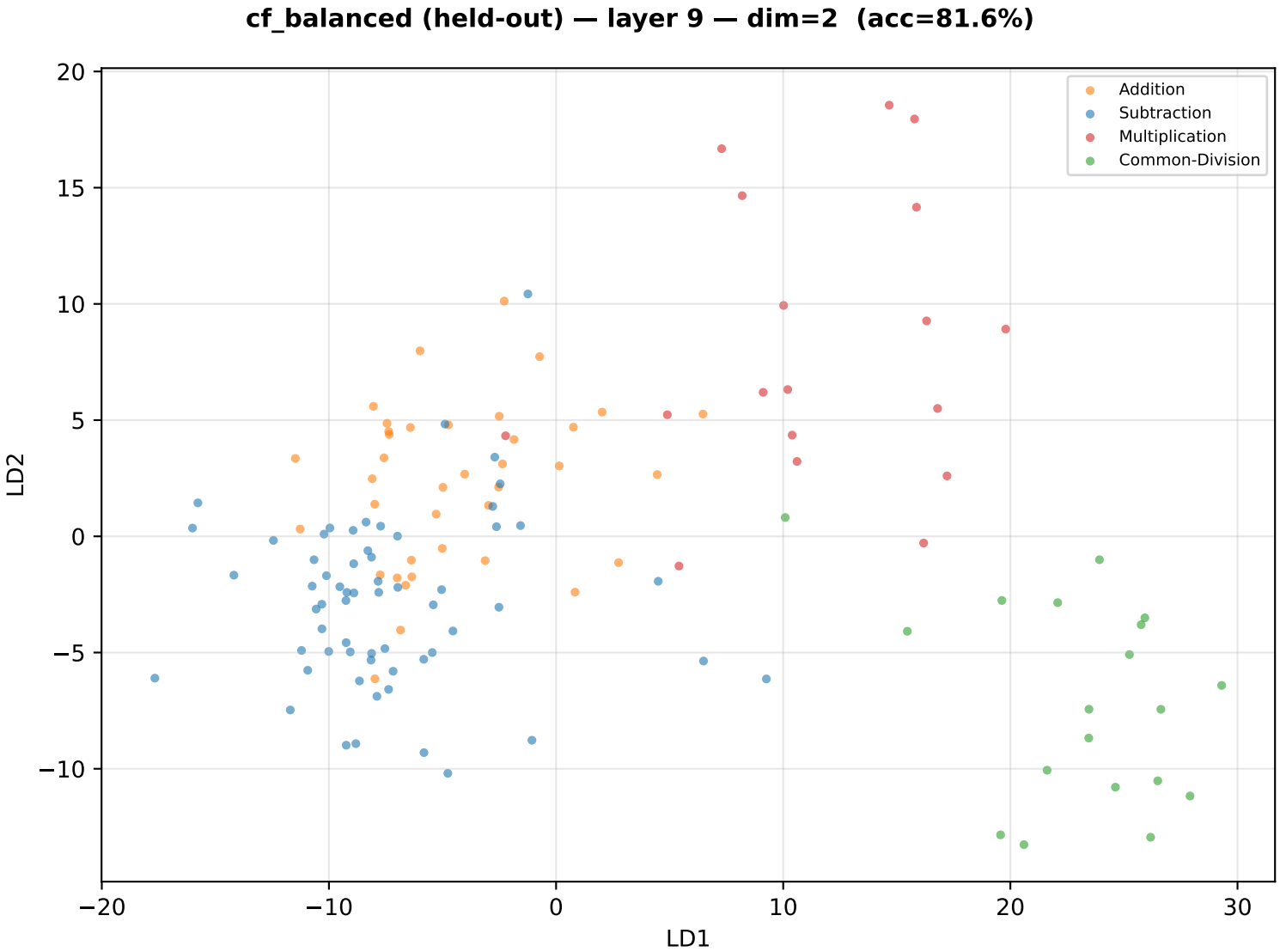

cf_balanced (held-out) — layer 4 — dim=2 (acc=52.2%)
Addition
Subtraction
Multiplication
Common-Division
LD1
LD2
−15
−10
−5
0
5
10
15
20
25

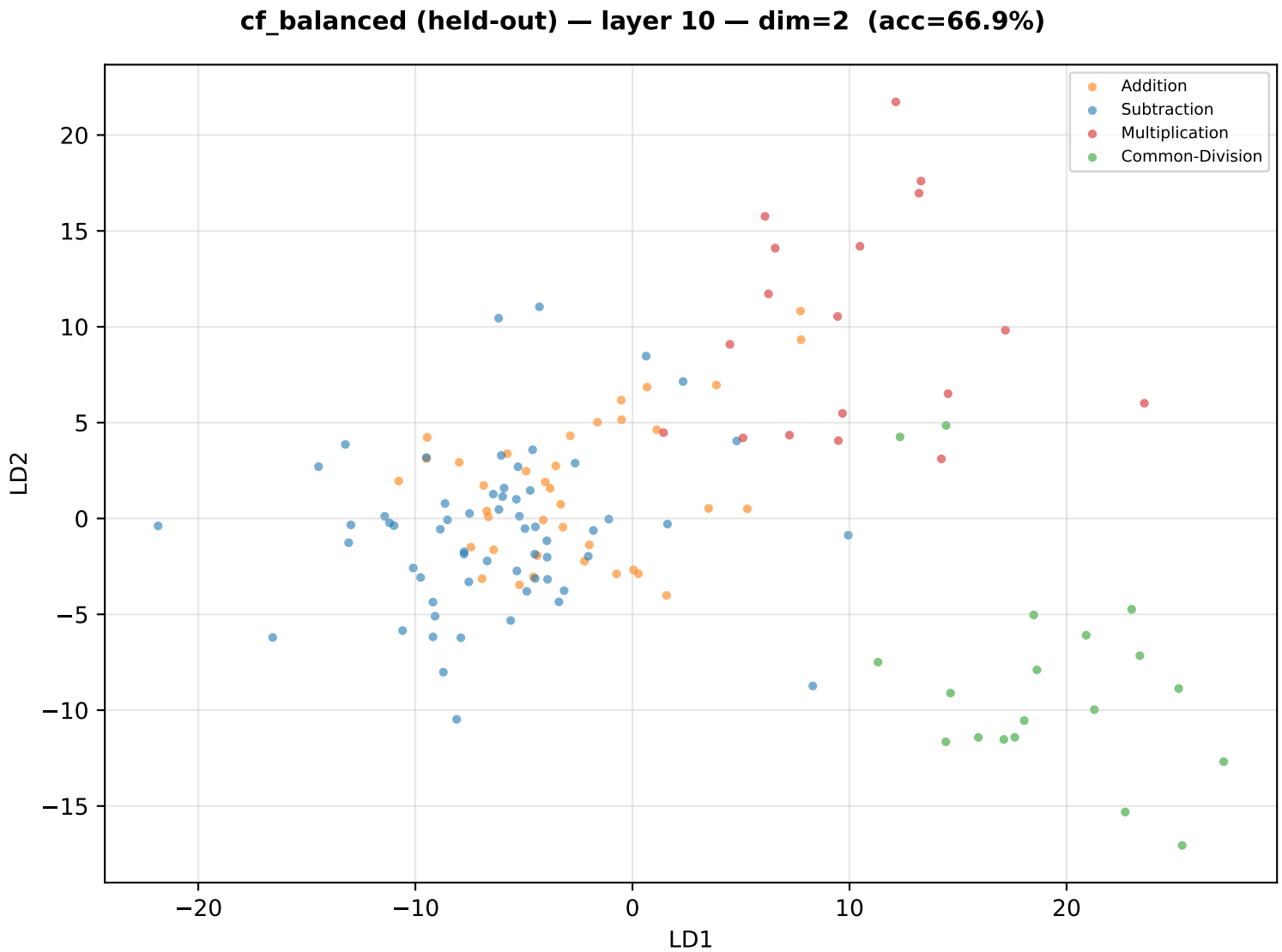

cf_balanced (held-out) — layer 5 — dim=2 (acc=44.1%)
Addition
Subtraction
Multiplication
Common-Division
LD1
LD2
−15
−10
−5
0
5
10
15
20

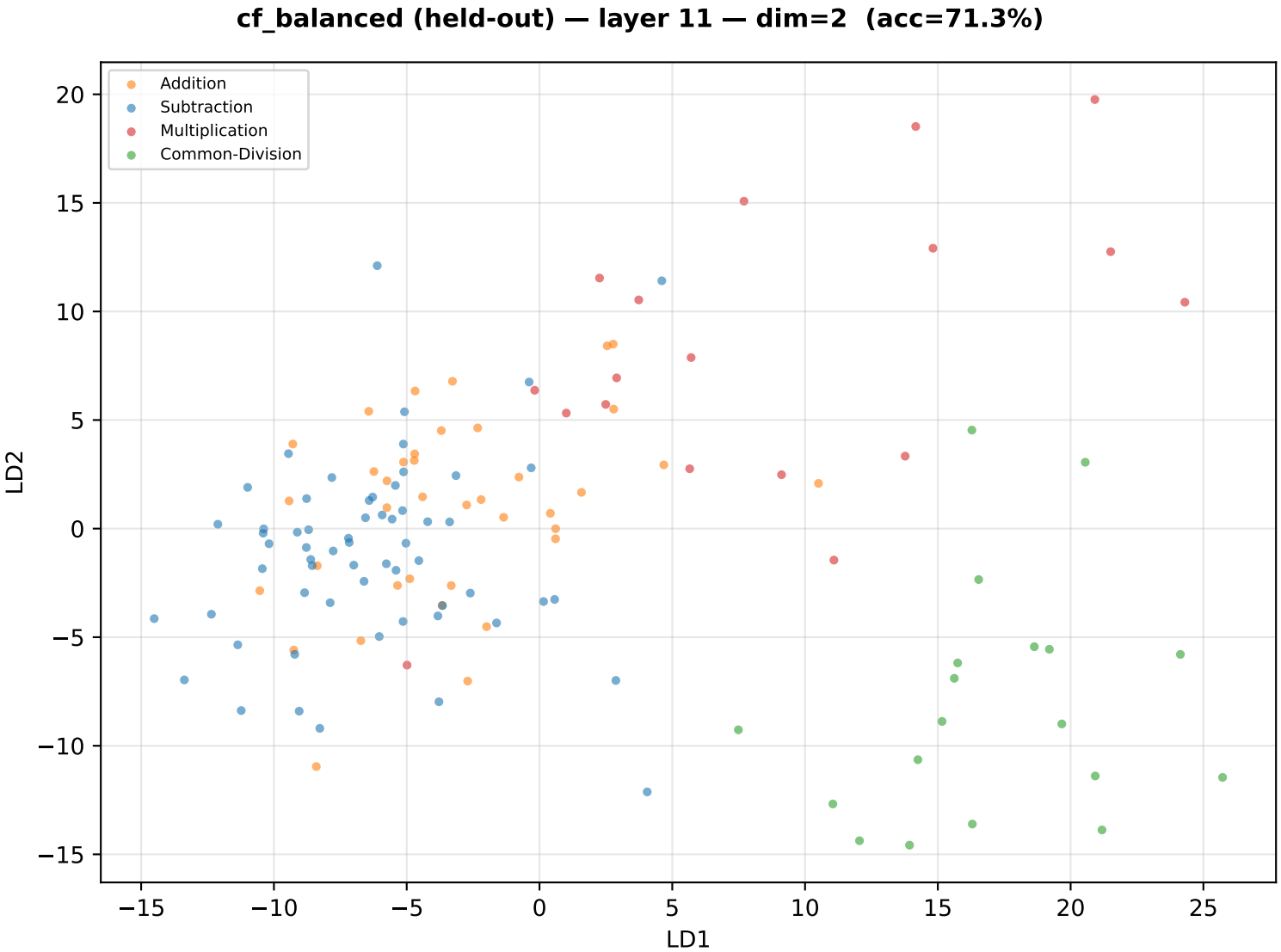

cf_balanced (held-out) — layer 6 — dim=2 (acc=61.0%)
Addition
Subtraction
Multiplication
Common-Division
LD1
LD2
−10
−5
0
5
10
15
20
25
−15

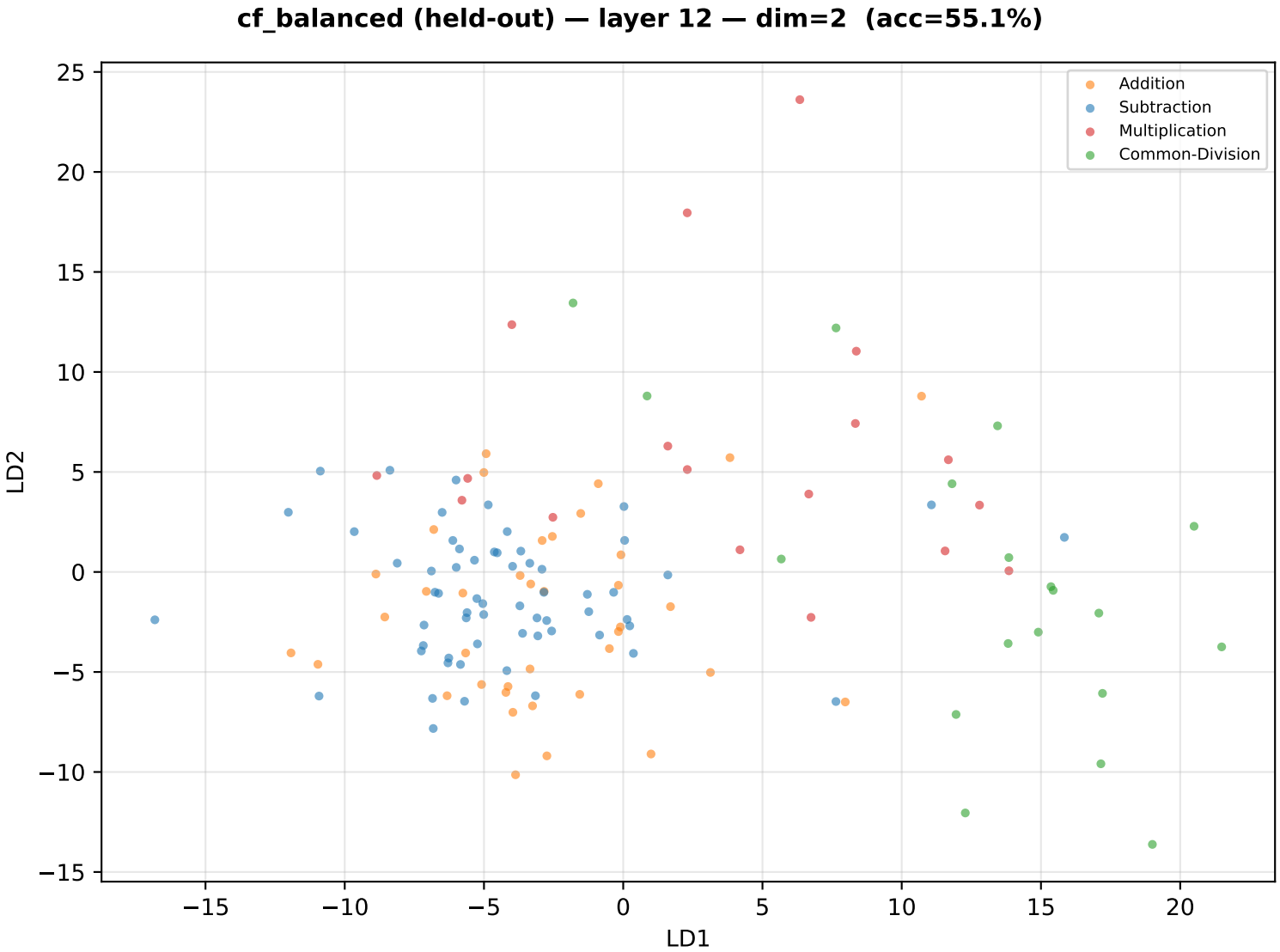

cf_balanced (held-out) — layer 7 — dim=2 (acc=69.9%)
Addition
Subtraction
Multiplication
Common-Division
LD1
LD2
−10
0
10
20
−20
−15
−10
−5
0
5
10
15
20

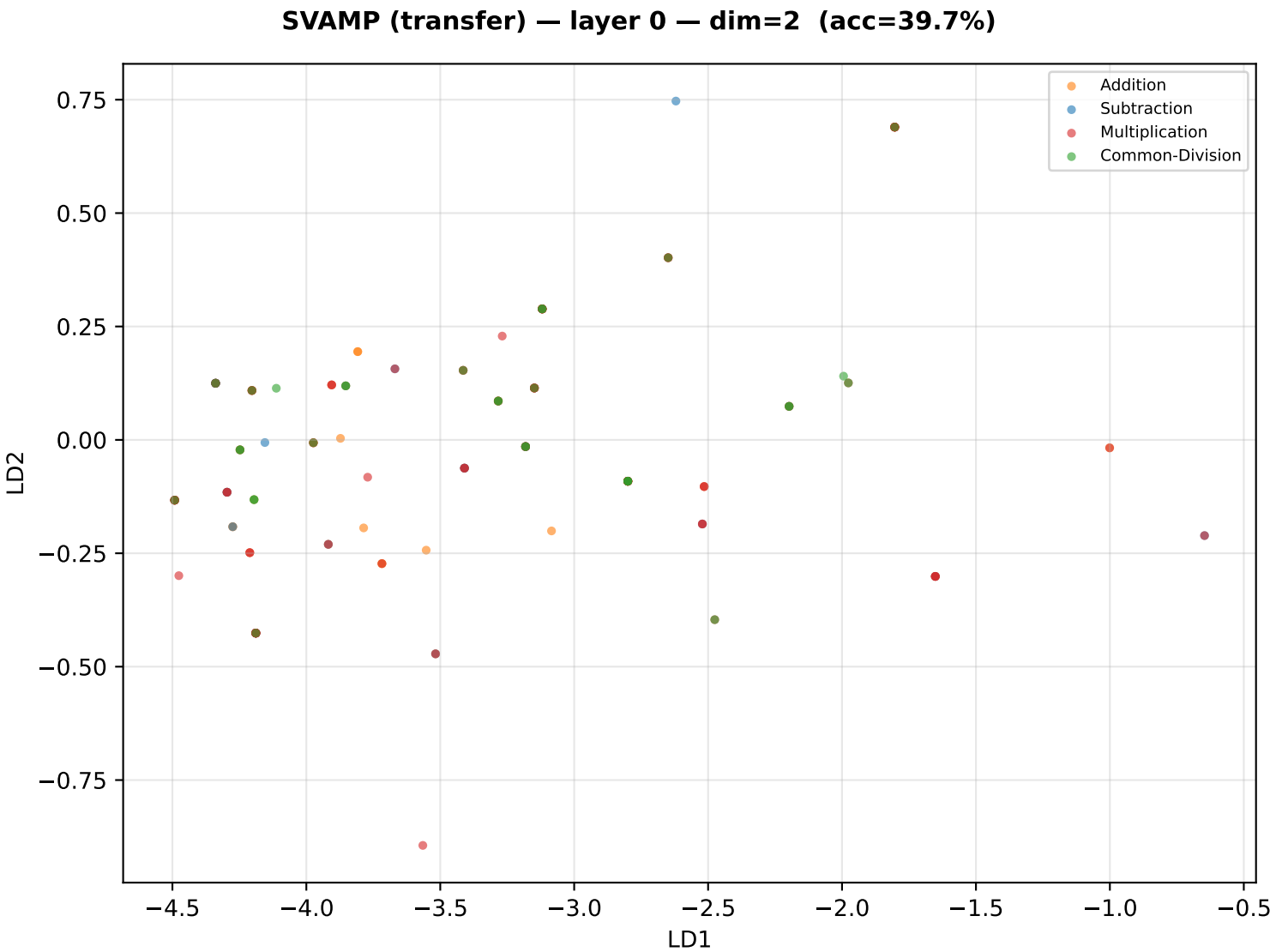

cf_balanced (held-out) — layer 8 — dim=2 (acc=64.7%)
Addition
Subtraction
Multiplication
Common-Division
LD1
LD2
−15
−10
−5
0
5
10
15
20
25
−20
−15
−10
−5
0
5
10
15
20

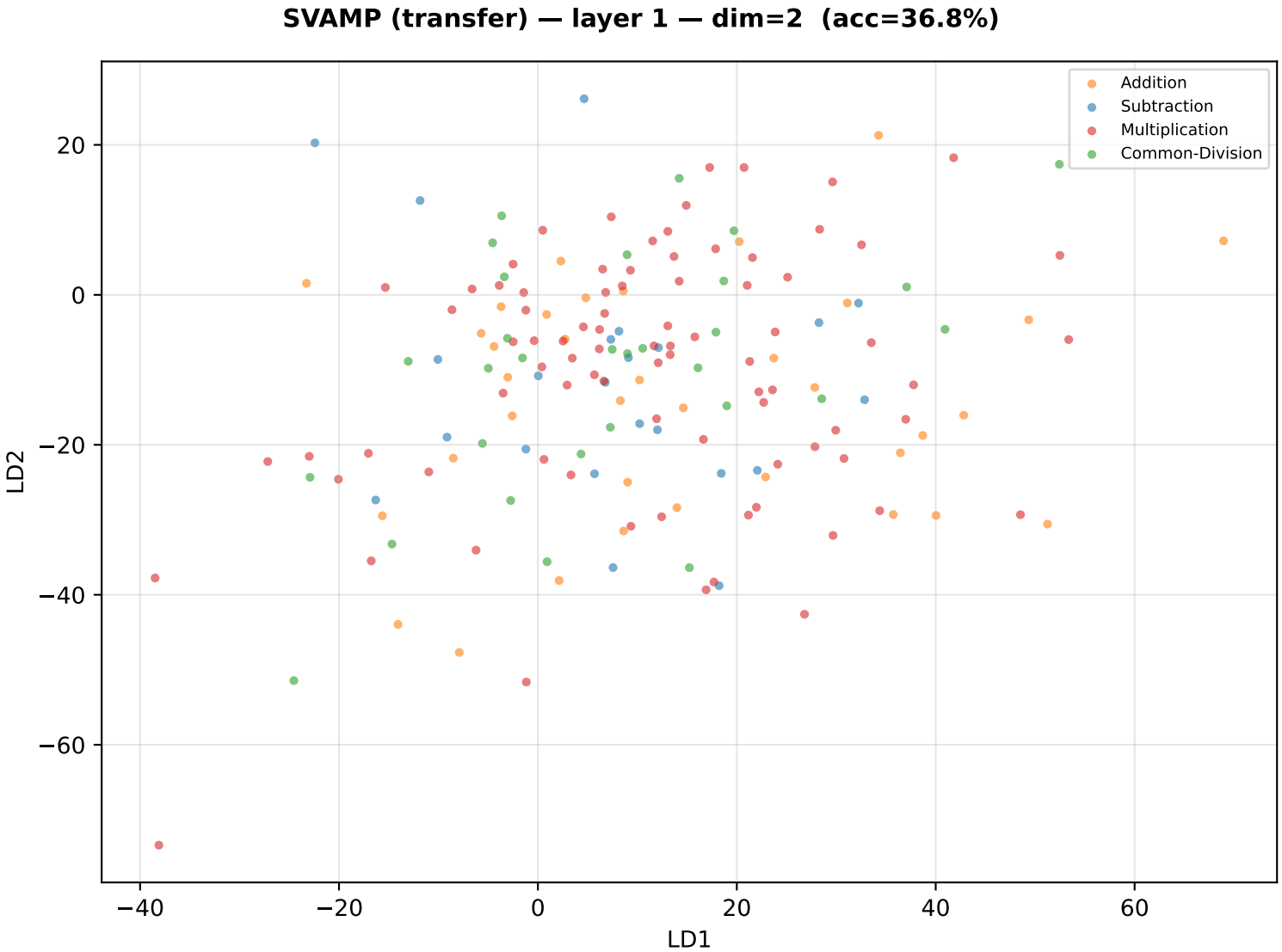

cf_balanced (held-out) — layer 9 — dim=2 (acc=81.6%)
Addition
Subtraction
Multiplication
Common-Division
LD1
LD2
20
15
10
5
0
−5
−10
−20
−10
0
10
20
30

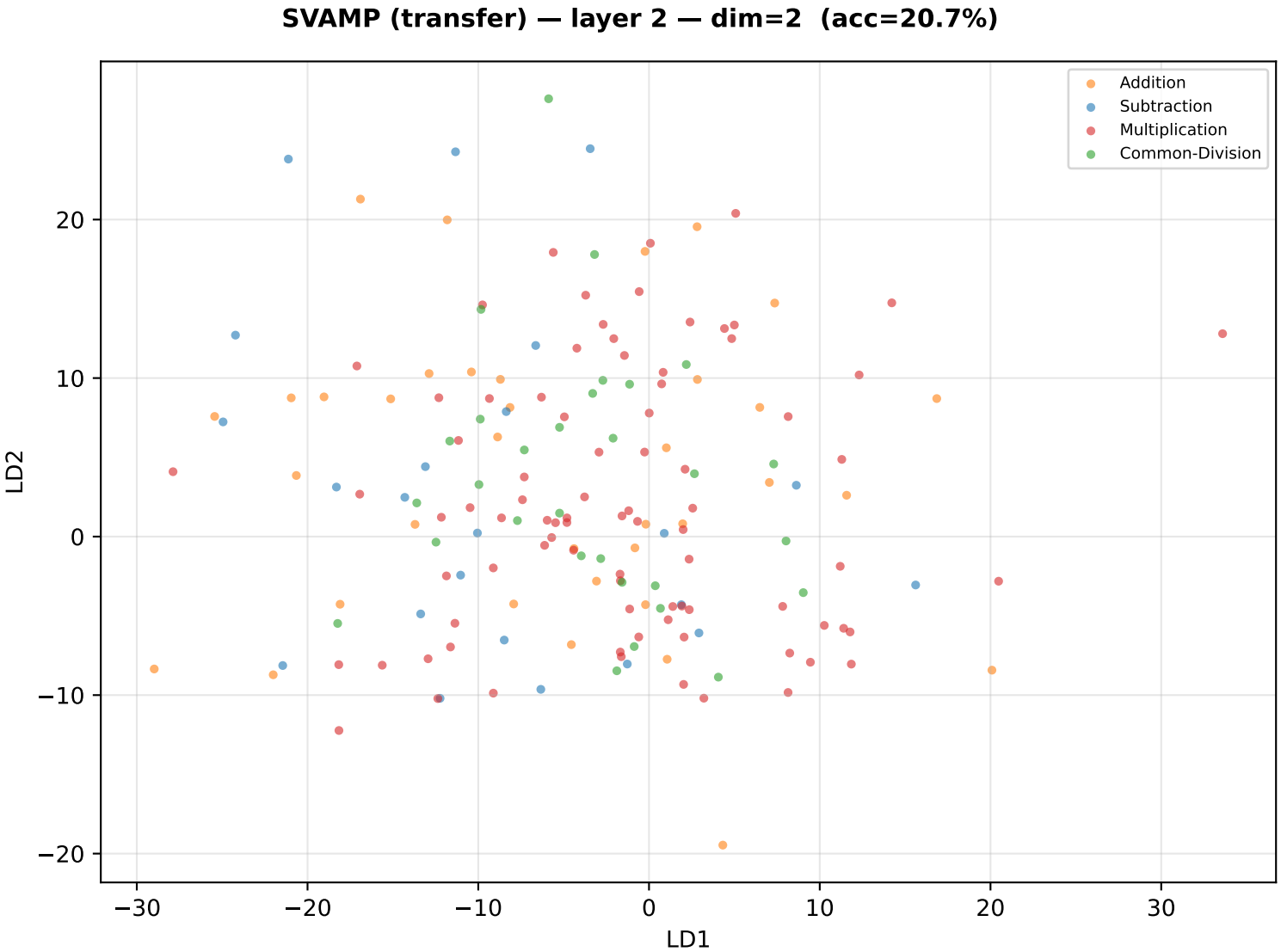

cf_balanced (held-out) — layer 10 — dim=2 (acc=66.9%)
Addition
Subtraction
Multiplication
Common-Division
LD1
LD2
−20
−10
0
10
20
−15
−10
−5
0
5
10
15
20

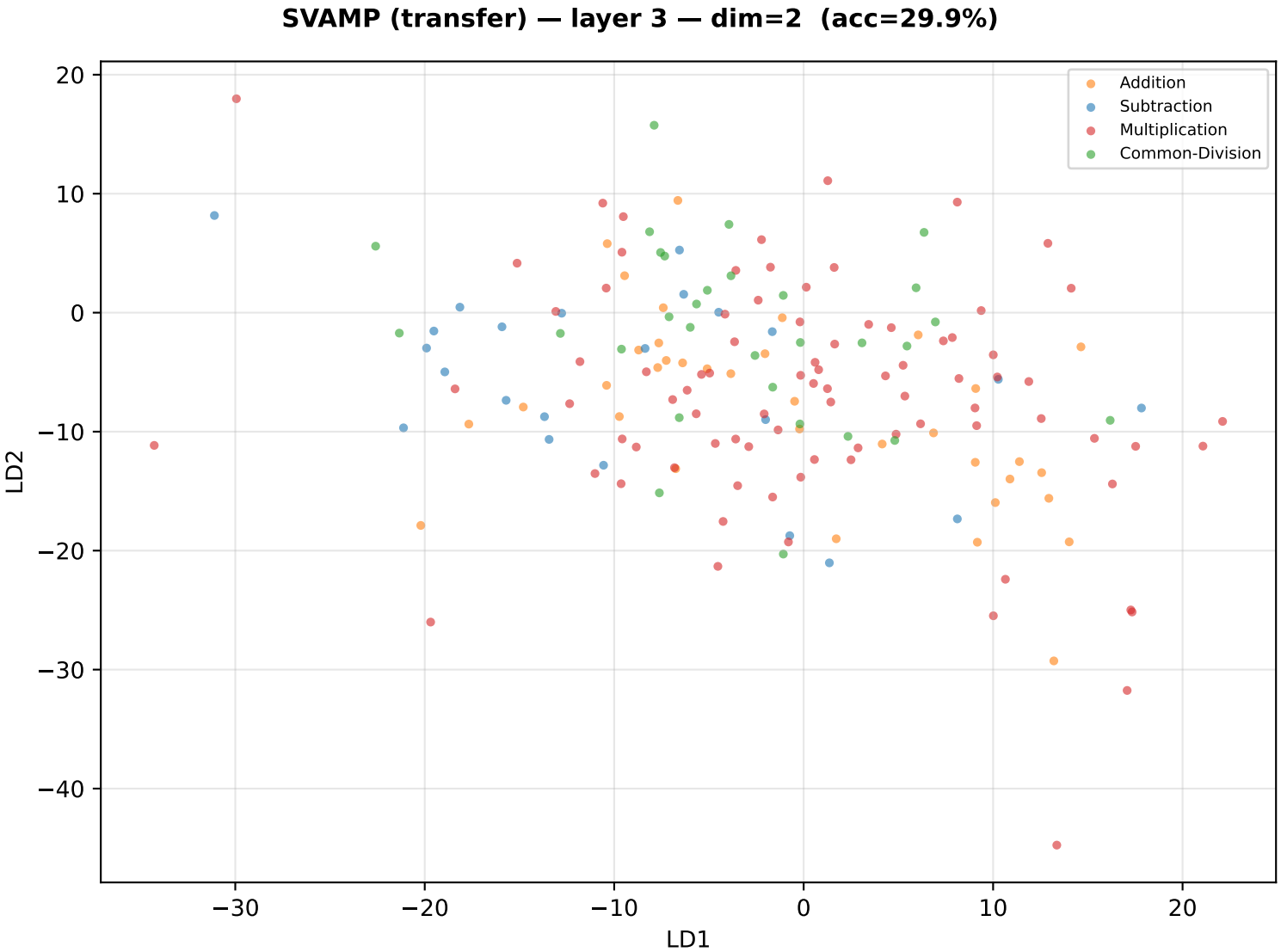

cf_balanced (held-out) — layer 11 — dim=2 (acc=71.3%)
Addition
Subtraction
Multiplication
Common-Division
LD1
LD2
−15
−10
−5
0
5
10
15
20
25

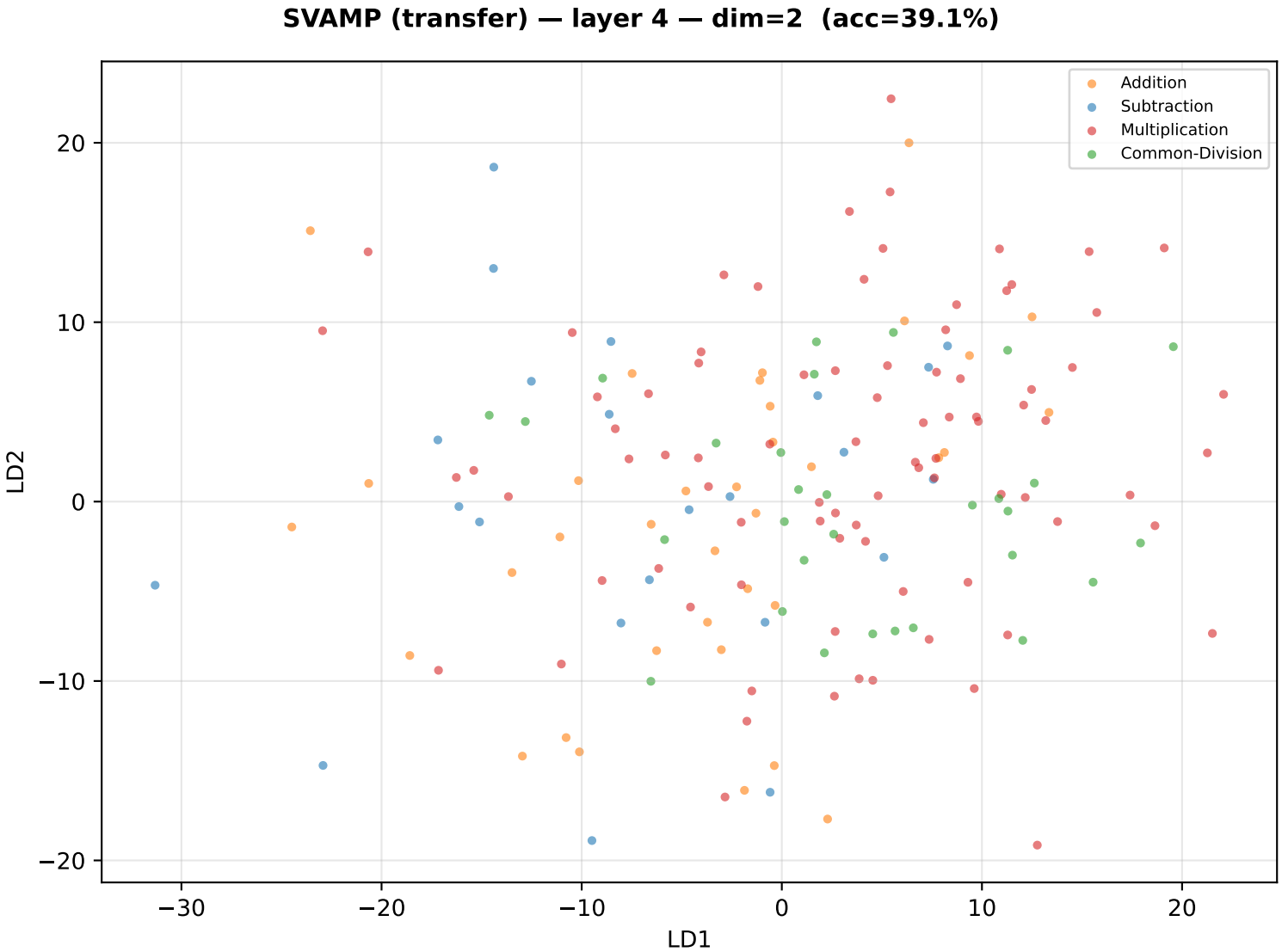

cf_balanced (held-out) — layer 12 — dim=2 (acc=55.1%)
Addition
Subtraction
Multiplication
Common-Division
LD1
LD2
−15
−10
−5
0
5
10
15
20
25

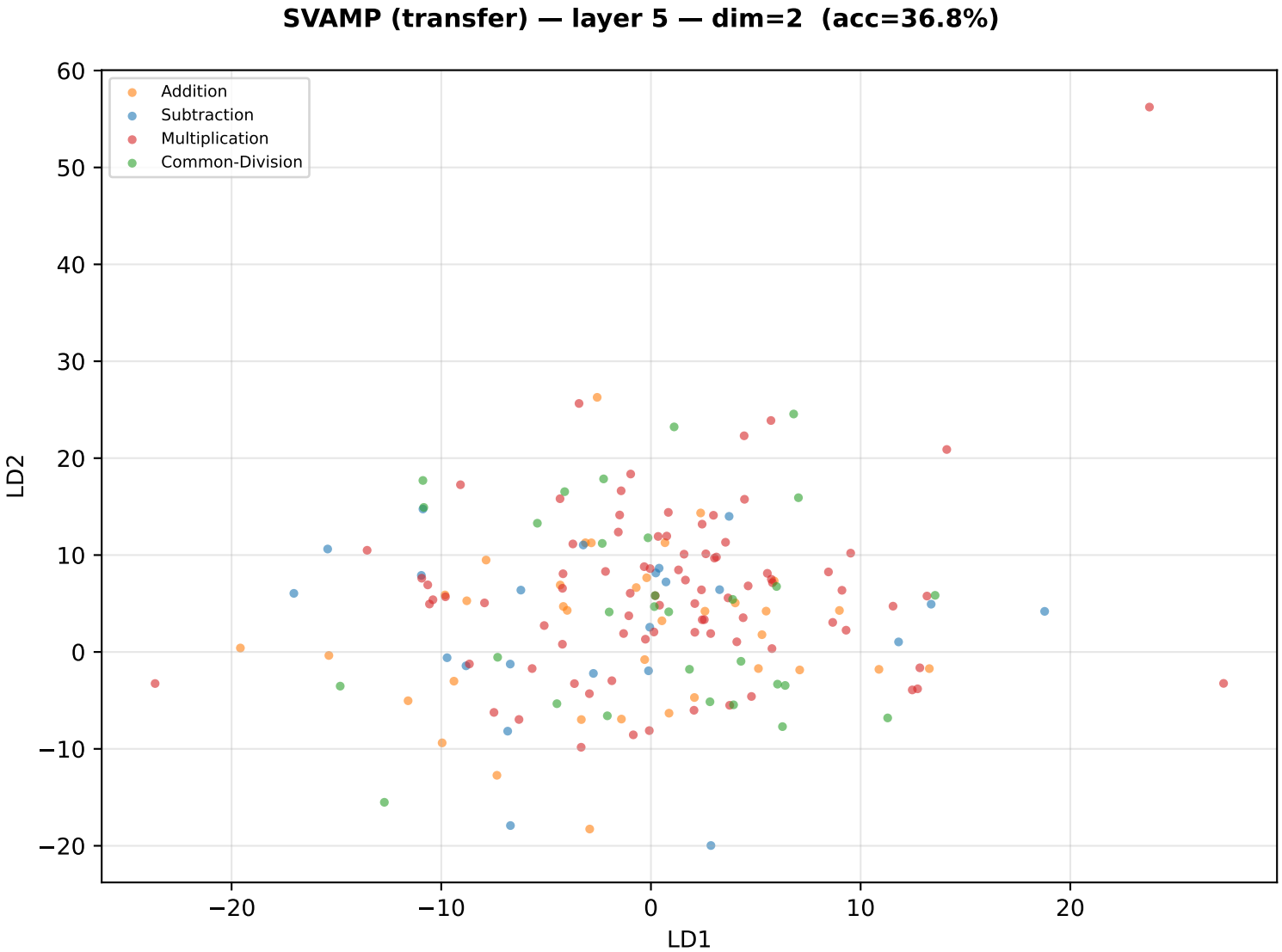

SVAMP (transfer) — layer 0 — dim=2 (acc=39.7%)
Addition
Subtraction
Multiplication
Common-Division
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
LD2
−4.5
−4.0
−3.5
−3.0
−2.5
−2.0
−1.5
−1.0
−0.5
LD1

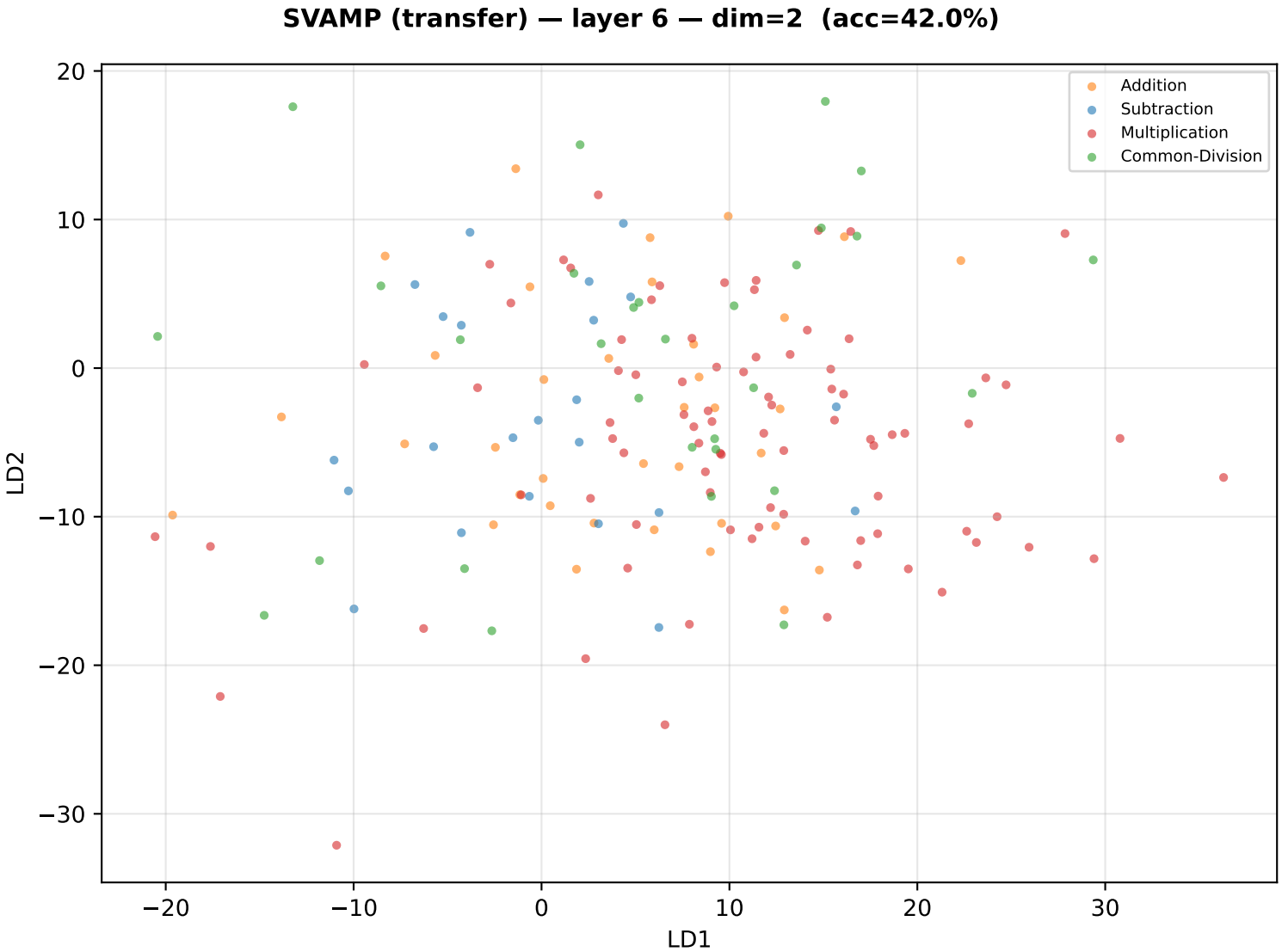

SVAMP (transfer) — layer 1 — dim=2 (acc=36.8%)
Addition
Subtraction
Multiplication
Common-Division
LD1
LD2
−40
−20
0
20
40
60
20
0
−20
−40
−60

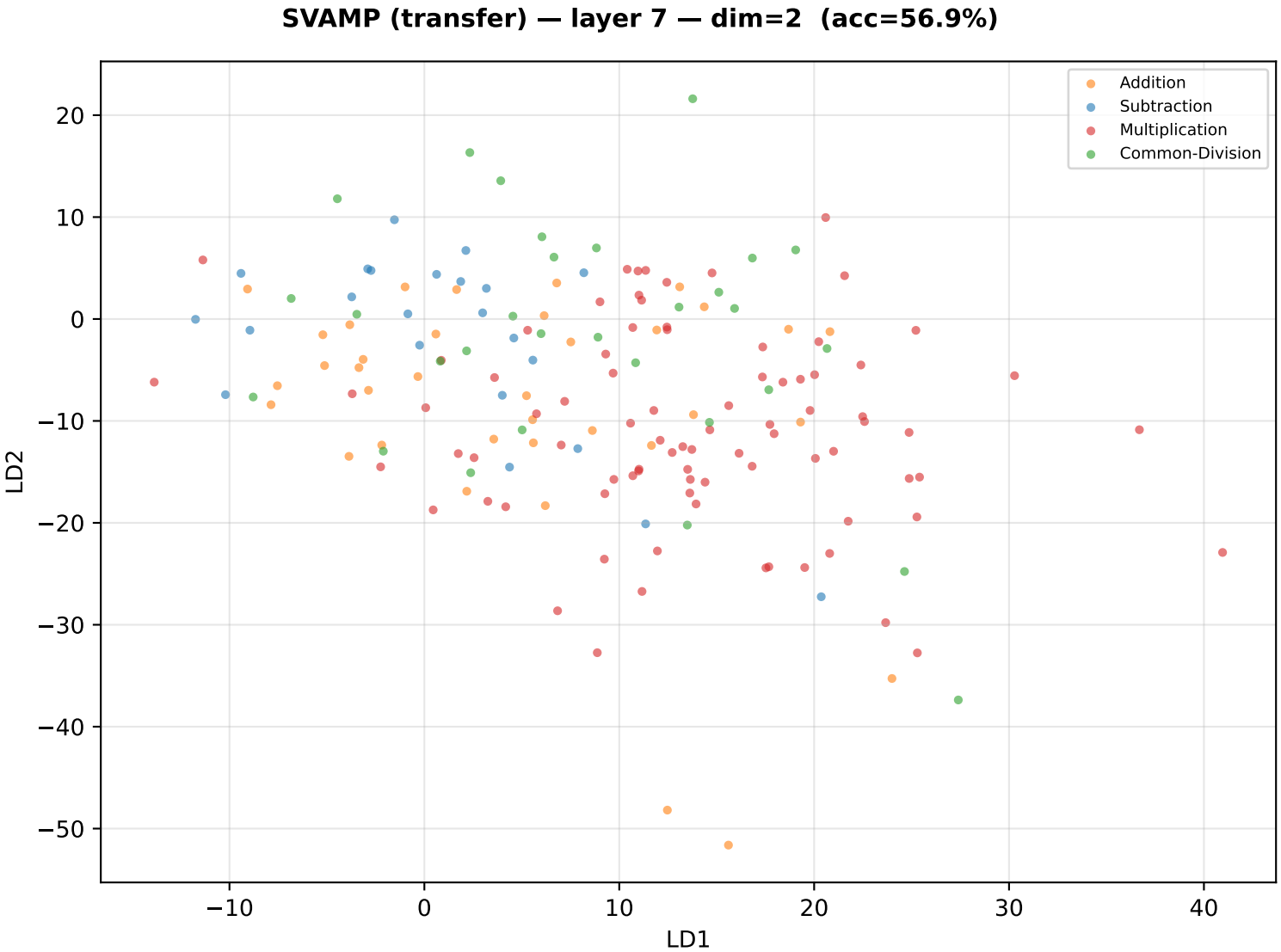

SVAMP (transfer) — layer 2 — dim=2 (acc=20.7%)
Addition
Subtraction
Multiplication
Common-Division
LD2
LD1
20
10
0
−10
−20
−30
−20
−10
0
10
20
30

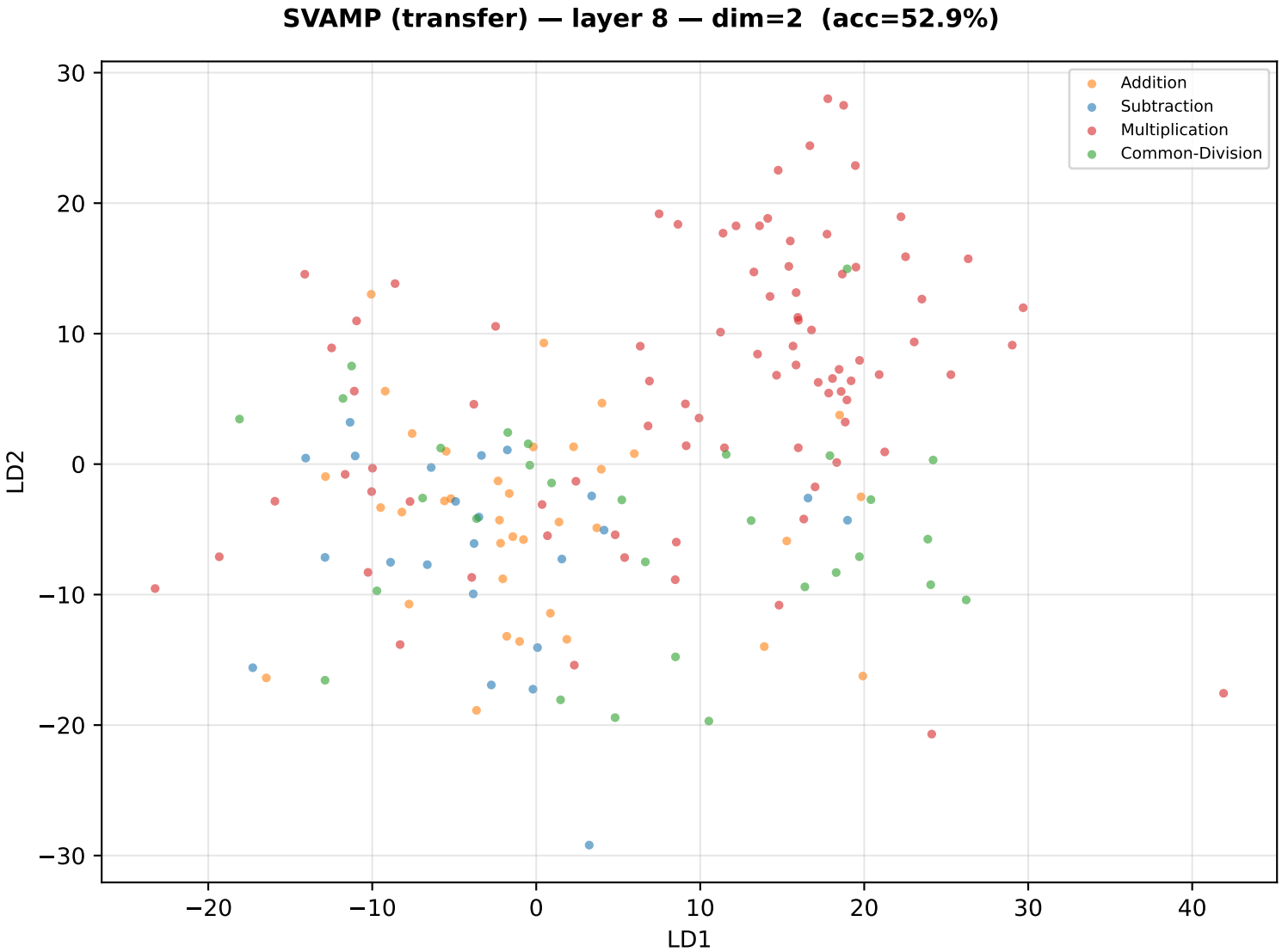

SVAMP (transfer) — layer 3 — dim=2 (acc=29.9%)
Addition
Subtraction
Multiplication
Common-Division
LD1
LD2
20
10
0
−10
−20
−30
−40
−30
−20
−10
0
10
20

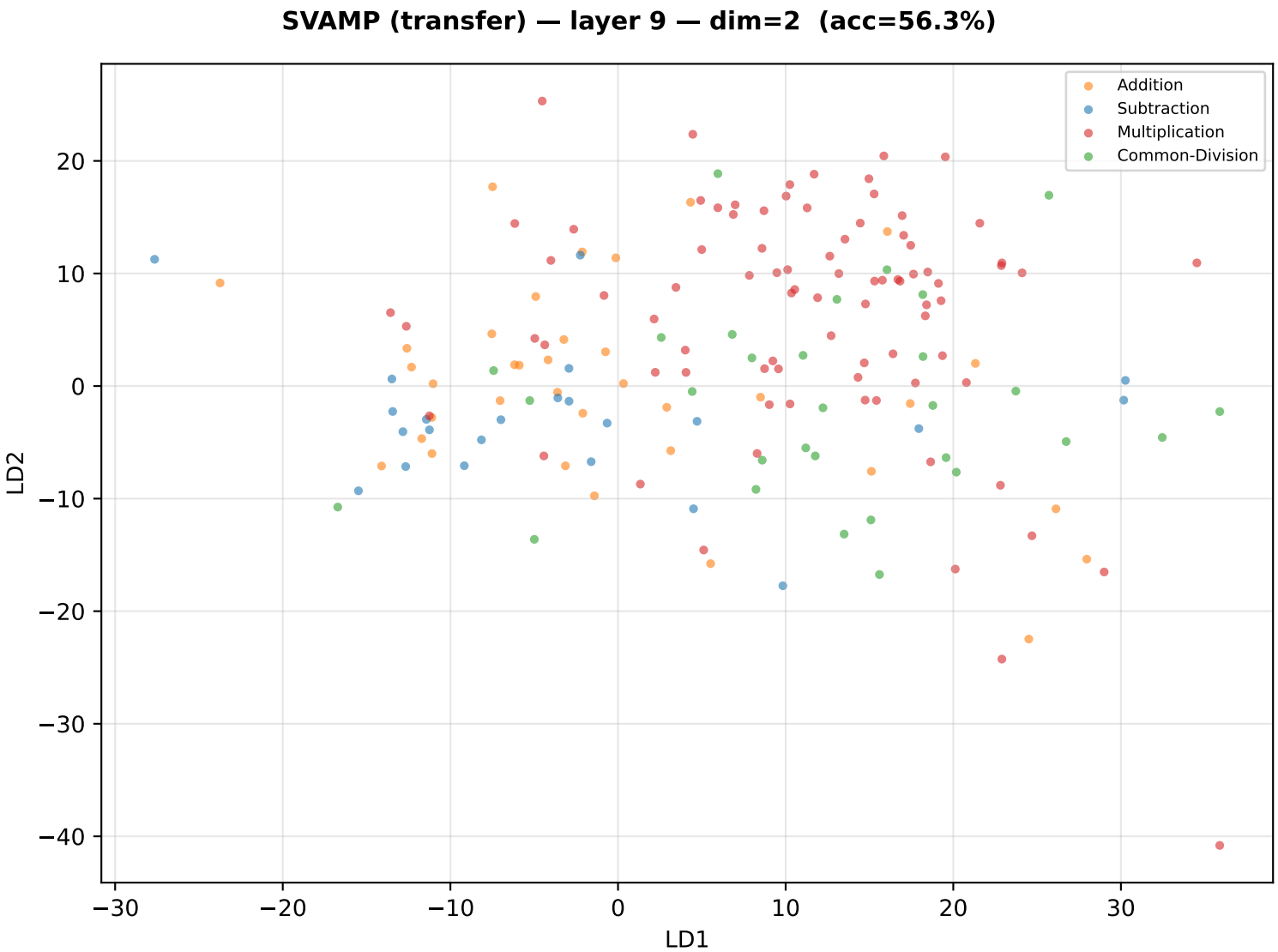

SVAMP (transfer) — layer 4 — dim=2 (acc=39.1%)
Addition
Subtraction
Multiplication
Common-Division
LD2
LD1
20
10
0
−10
−20
−30
−20
−10
0
10
20

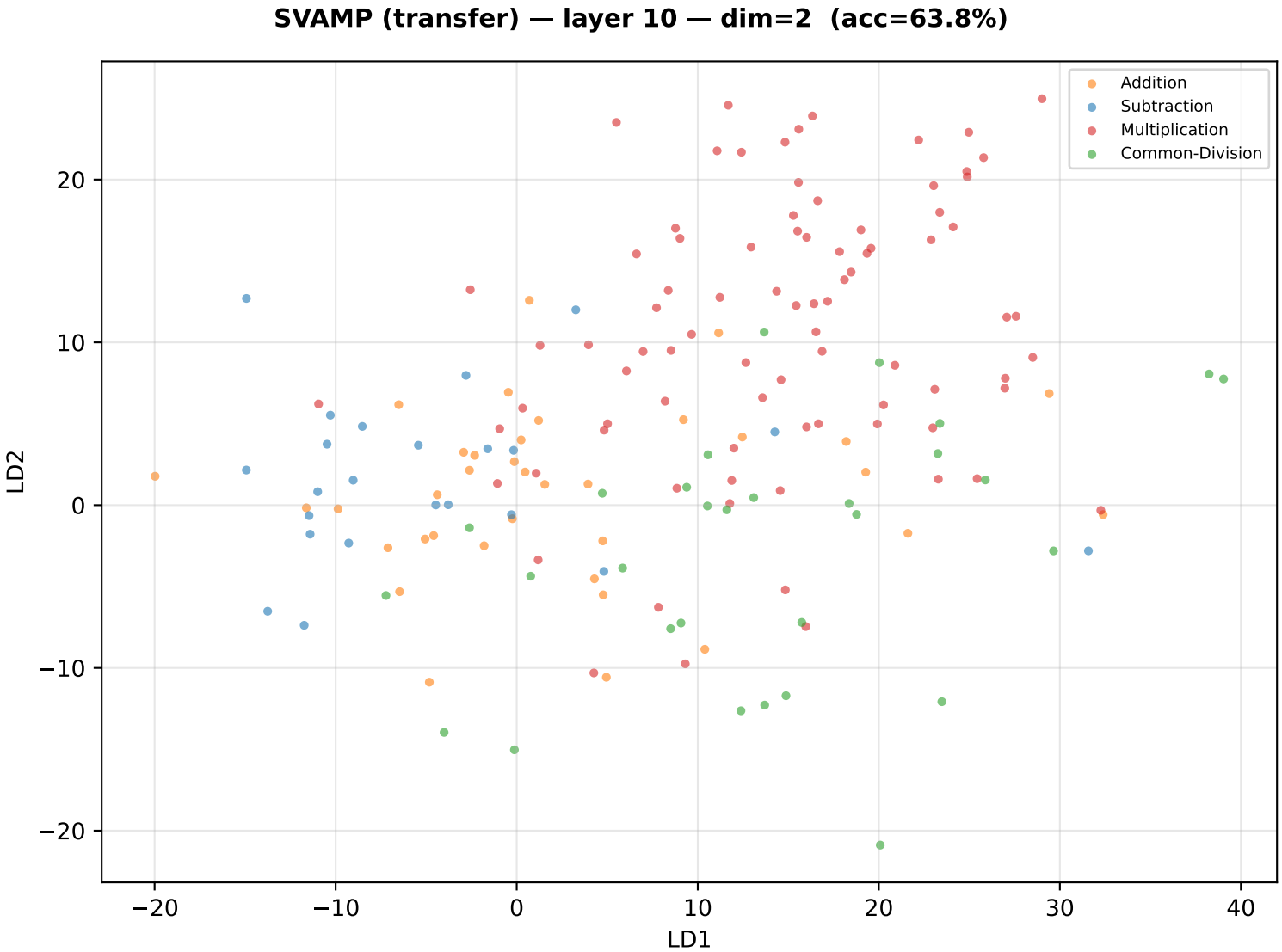

SVAMP (transfer) — layer 5 — dim=2 (acc=36.8%)
Addition
Subtraction
Multiplication
Common-Division
LD2
LD1
60
50
40
30
20
10
0
−10
−20
−20
−10
0
10
20

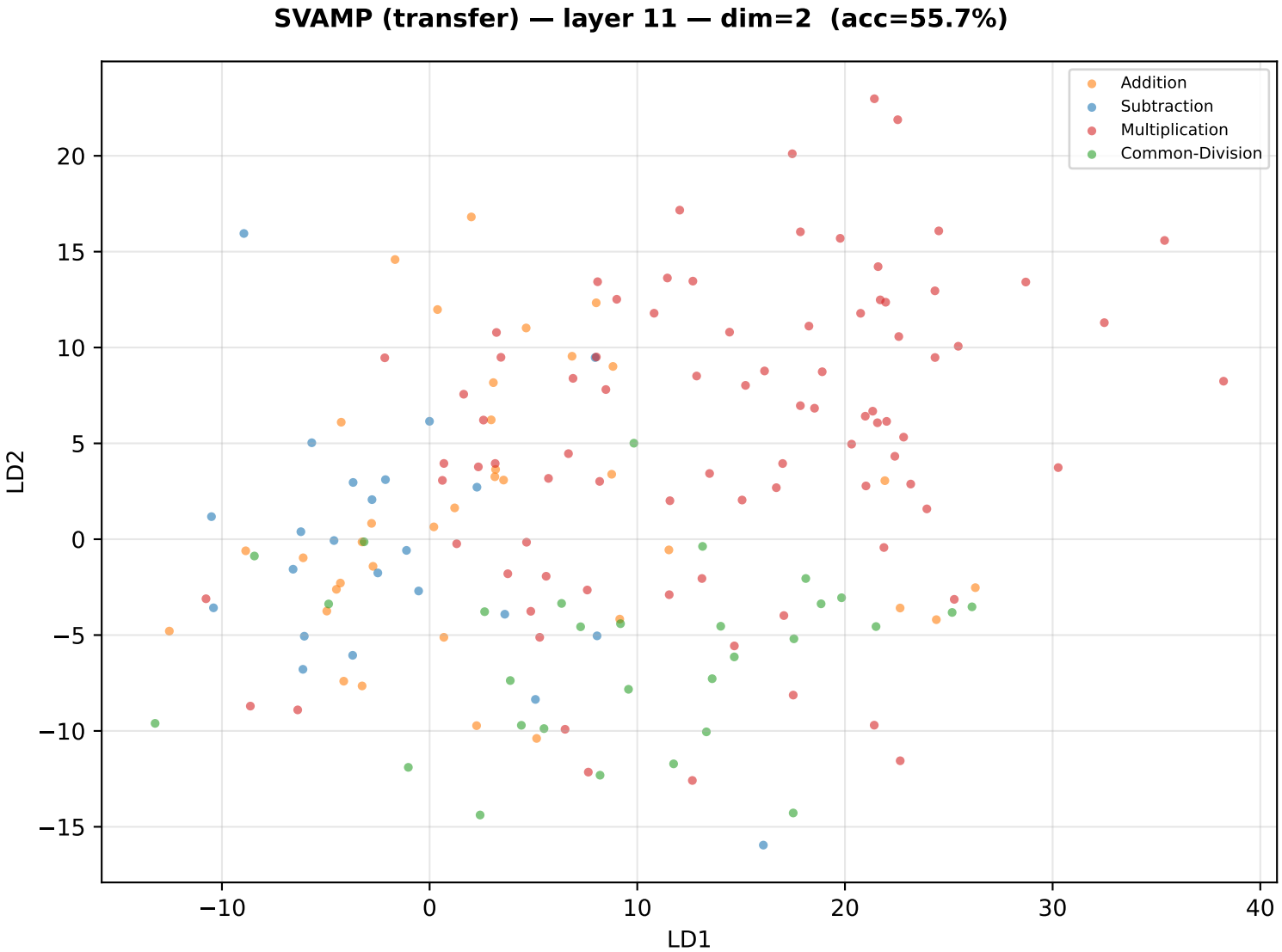

SVAMP (transfer) — layer 6 — dim=2 (acc=42.0%)
Addition
Subtraction
Multiplication
Common-Division
LD1
LD2
20
10
0
−10
−20
−30
−20
−10
0
10
20
30

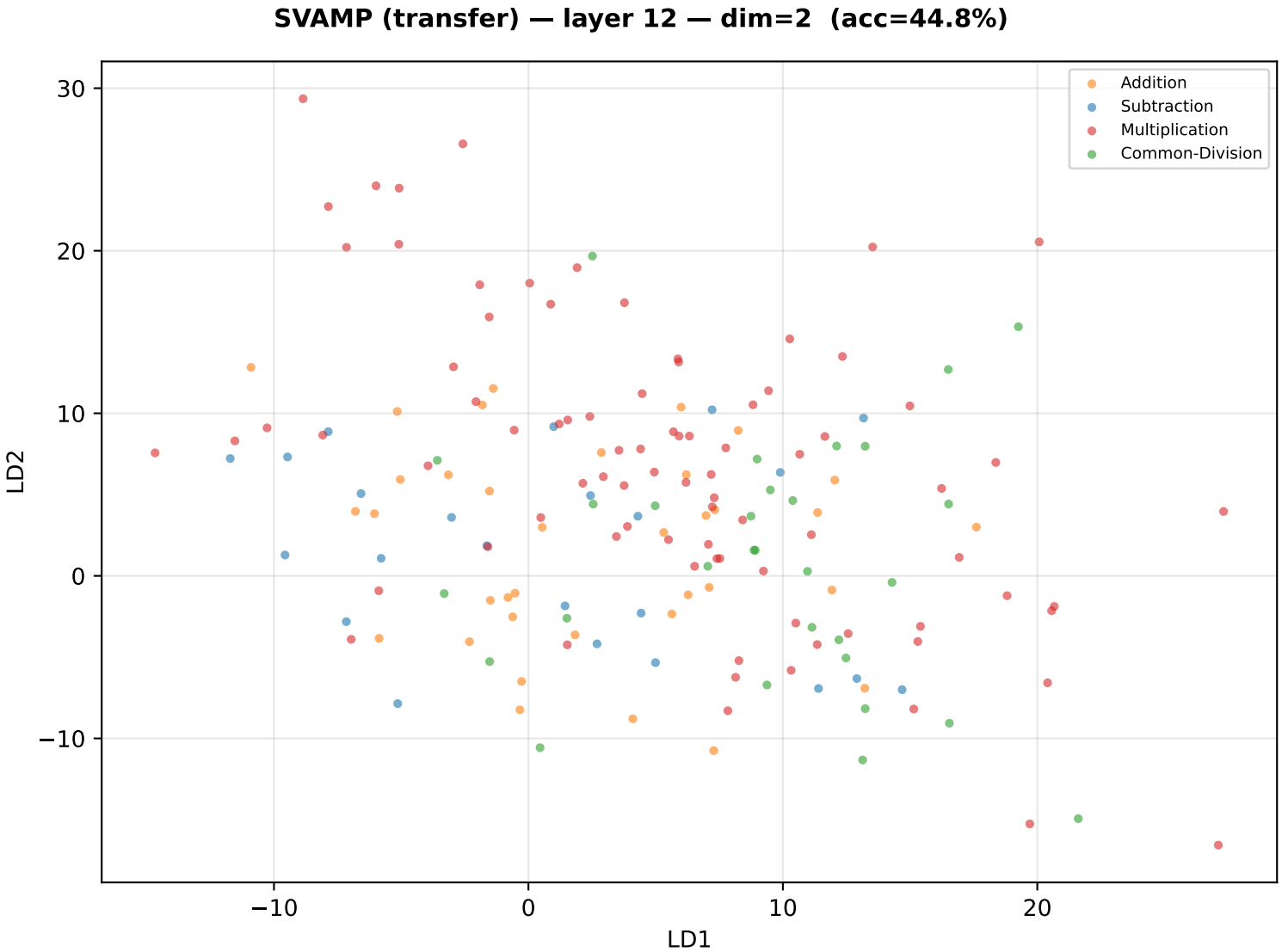

SVAMP (transfer) — layer 7 — dim=2 (acc=56.9%)
Addition
Subtraction
Multiplication
Common-Division
LD1
LD2
−10
0
10
20
30
40
20
10
0
−10
−20
−30
−40
−50

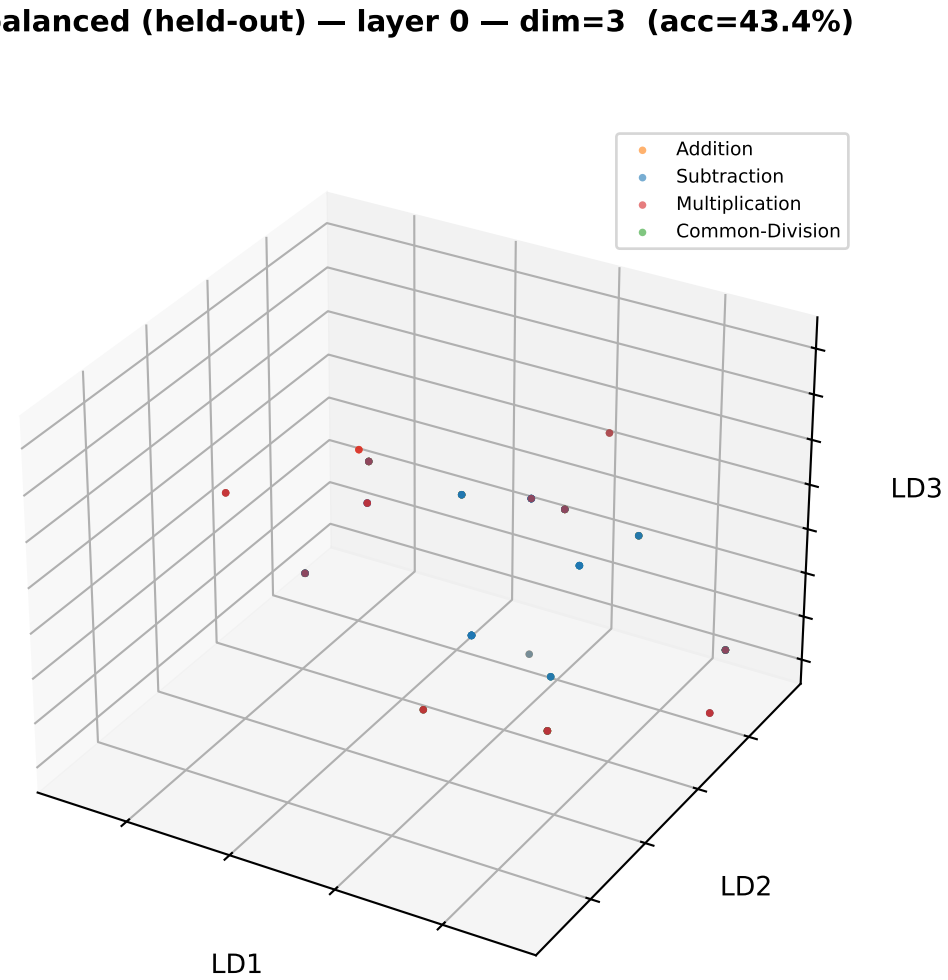

SVAMP (transfer) — layer 8 — dim=2 (acc=52.9%)
Addition
Subtraction
Multiplication
Common-Division
LD1
LD2
30
20
10
0
−10
−20
−30
−20
−10
0
10
20
30
40

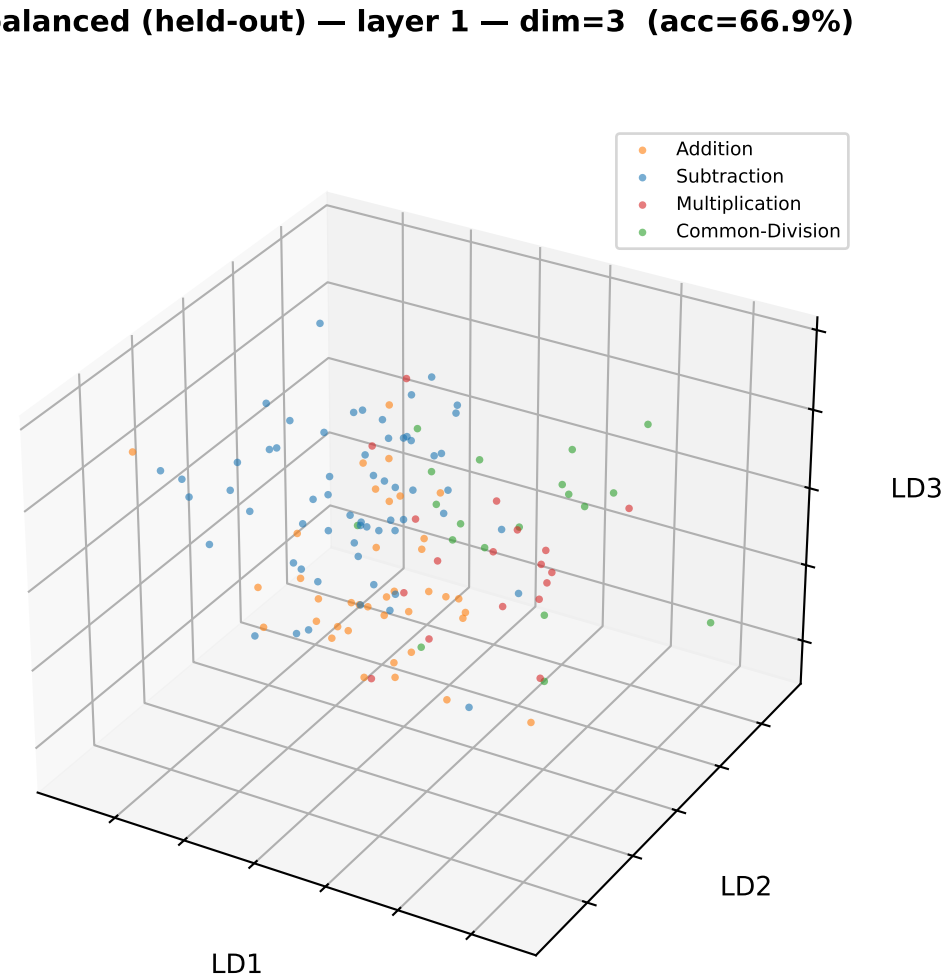

SVAMP (transfer) — layer 9 — dim=2 (acc=56.3%)
Addition
Subtraction
Multiplication
Common-Division
LD1
LD2
−30
−20
−10
0
10
20
30
−40
−30
−20
−10
0
10
20

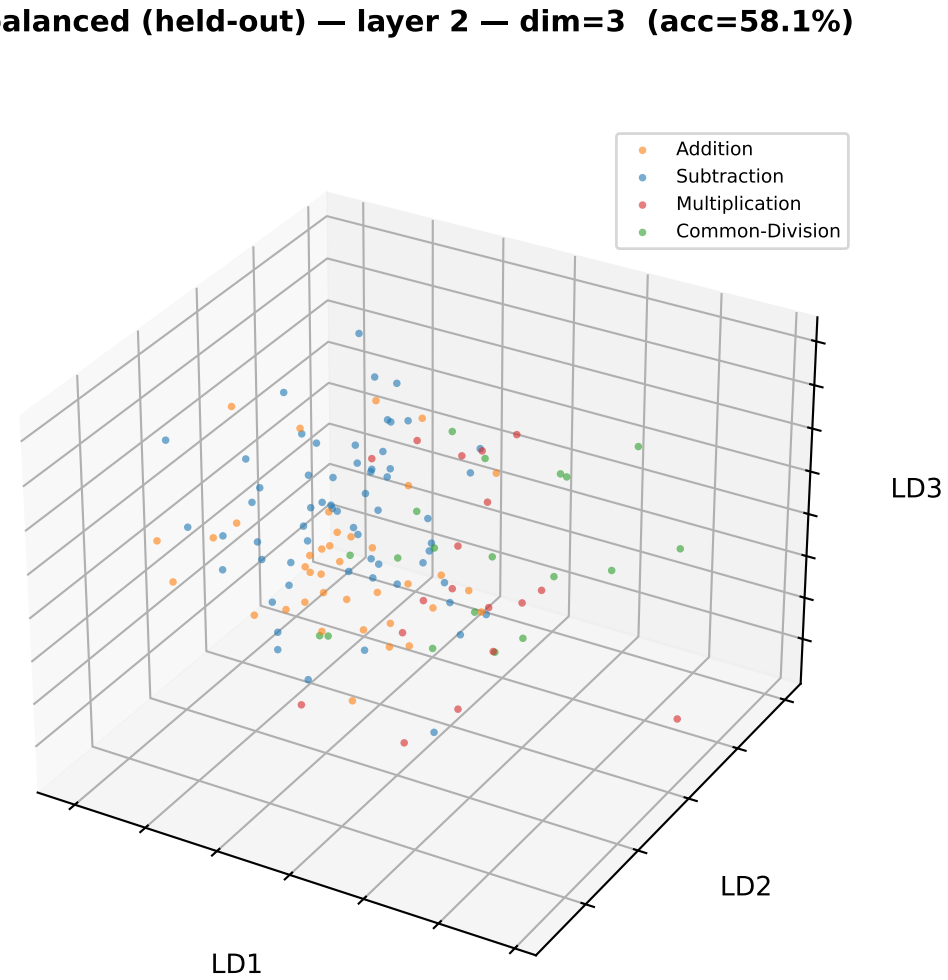

SVAMP (transfer) — layer 10 — dim=2 (acc=63.8%)
Addition
Subtraction
Multiplication
Common-Division
LD2
LD1
20
10
0
−10
−20
−20
−10
0
10
20
30
40

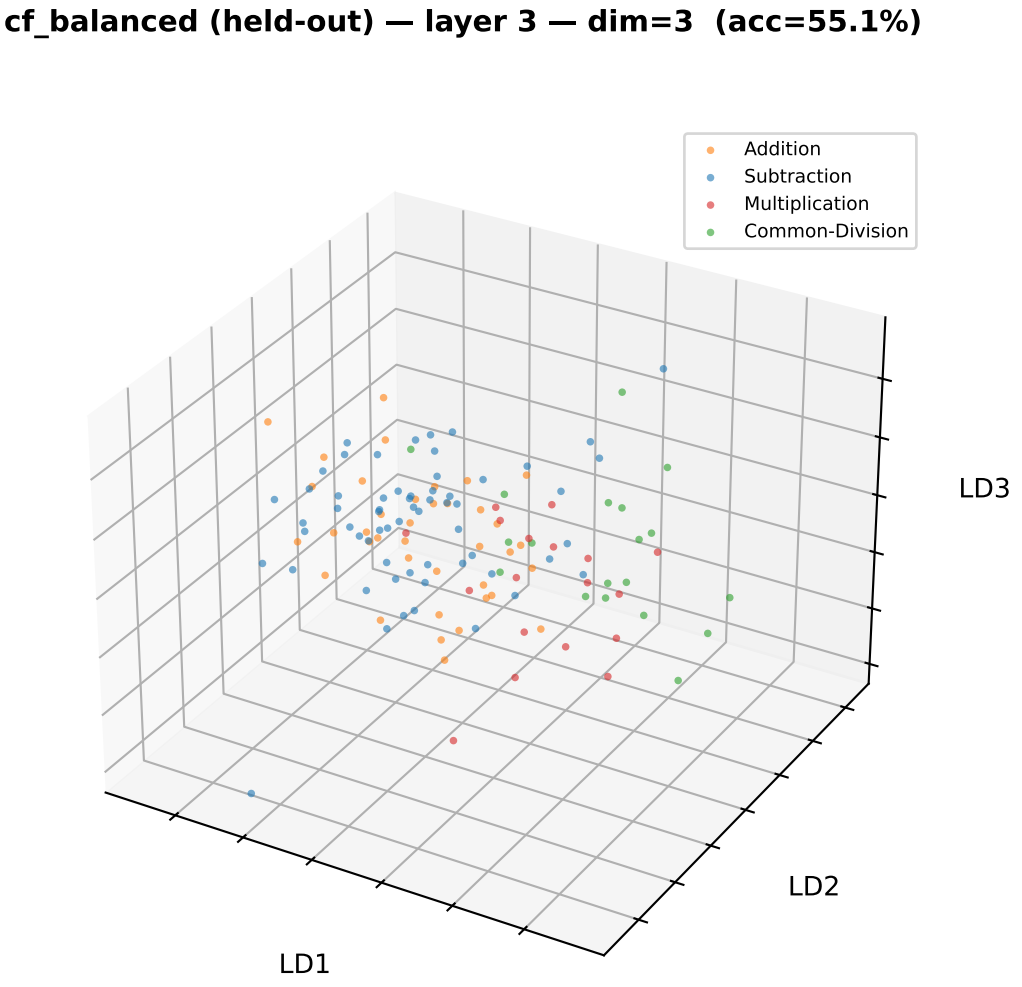

SVAMP (transfer) — layer 11 — dim=2 (acc=55.7%)
Addition
Subtraction
Multiplication
Common-Division
LD2
LD1
20
15
10
5
0
−5
−10
−15
−10
0
10
20
30
40

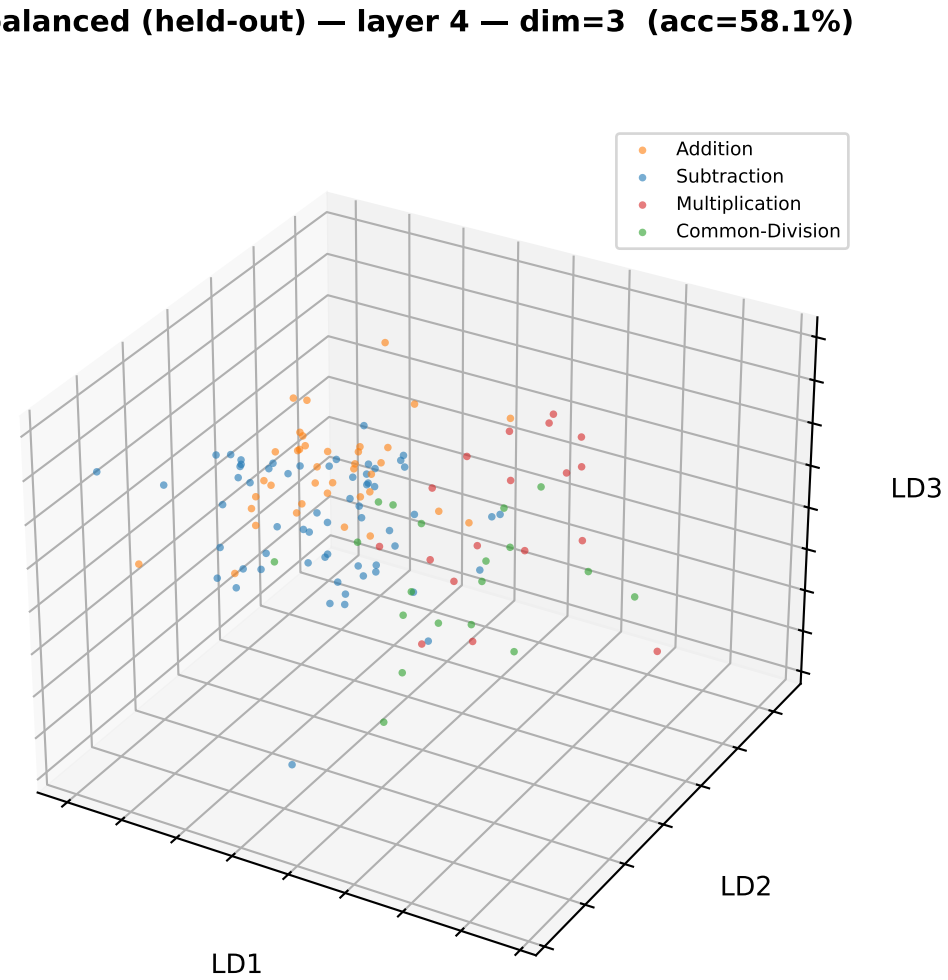

SVAMP (transfer) — layer 12 — dim=2 (acc=44.8%)
Addition
Subtraction
Multiplication
Common-Division
LD2
LD1
30
20
10
0
−10
−10
0
10
20

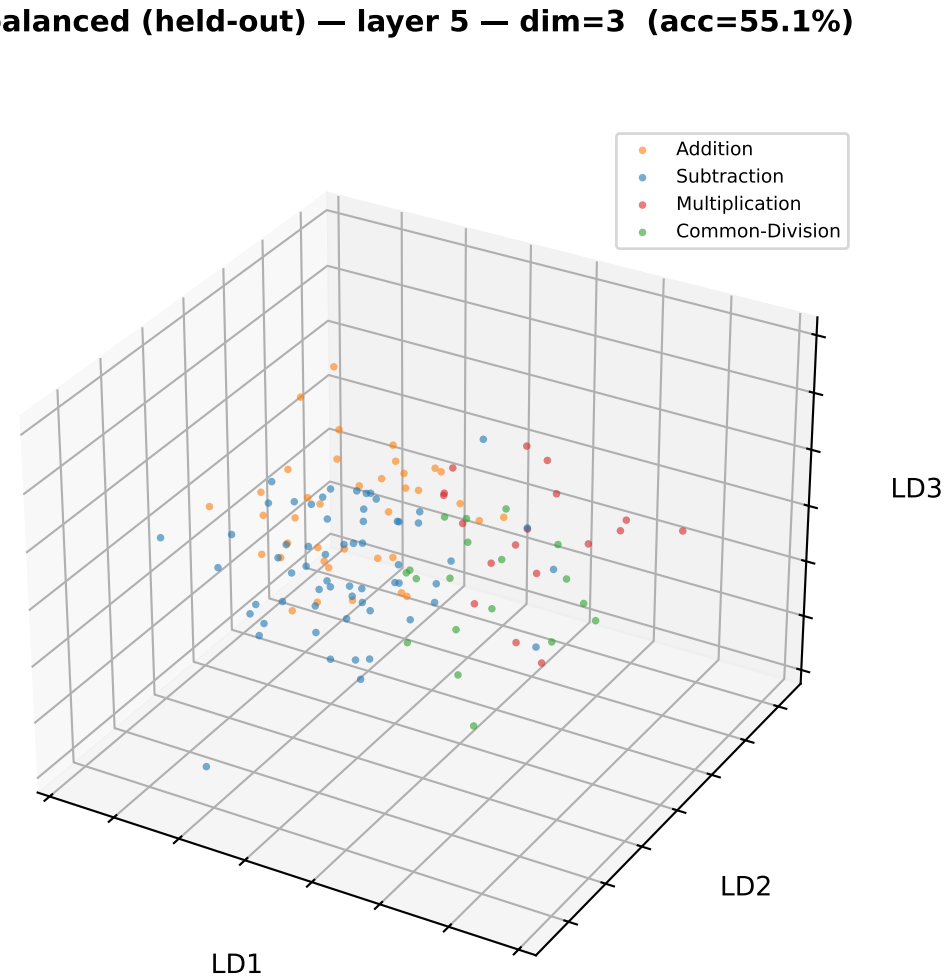

cf_balanced (held-out) — layer 0 — dim=3 (acc=43.4%)
Addition
Subtraction
Multiplication
Common-Division
LD1
LD2
LD3

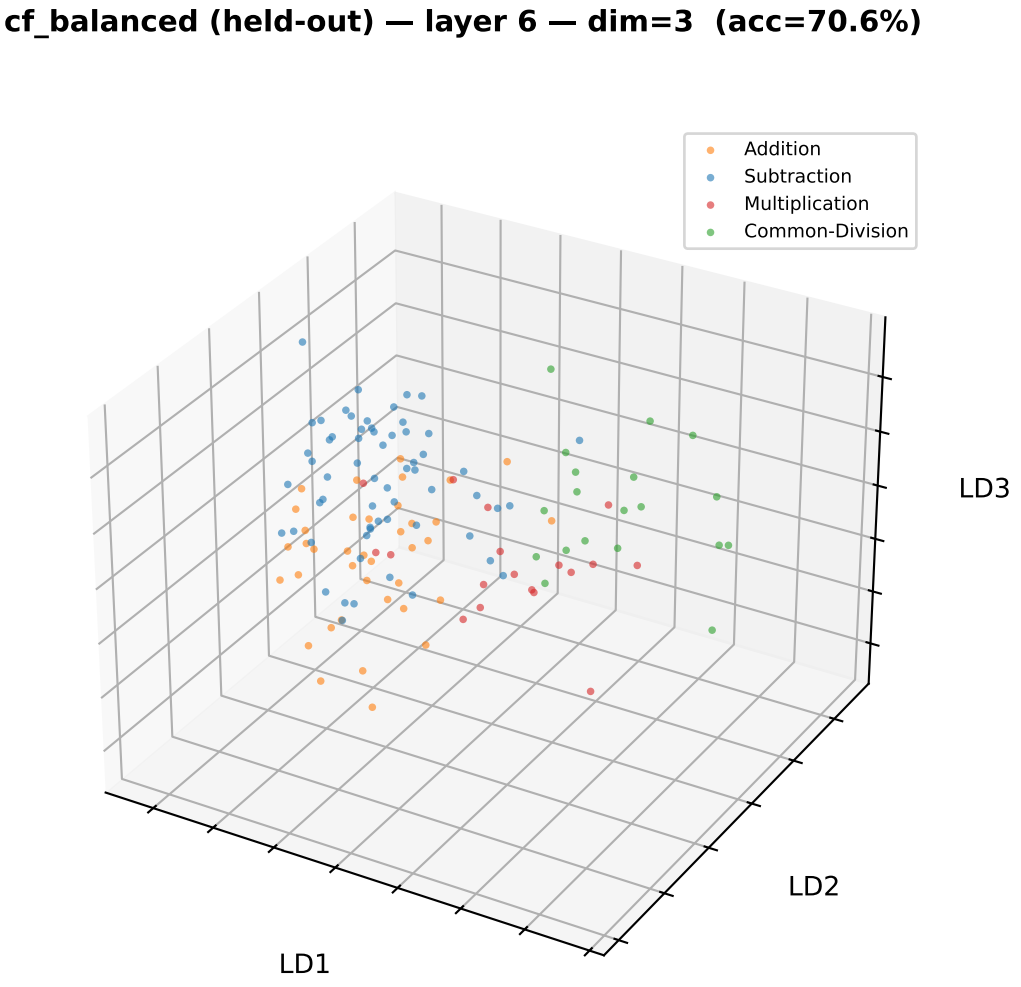

cf_balanced (held-out) — layer 1 — dim=3 (acc=66.9%)
Addition
Subtraction
Multiplication
Common-Division
LD1
LD2
LD3

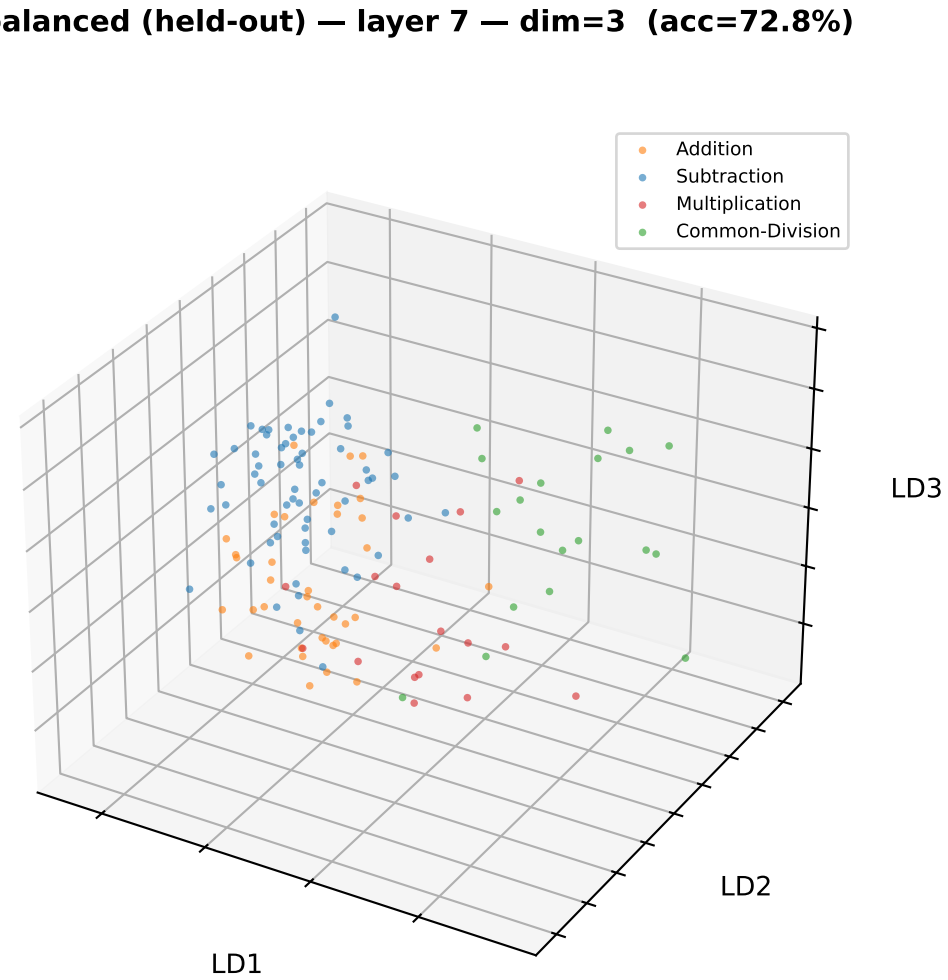

cf_balanced (held-out) — layer 2 — dim=3 (acc=58.1%)
Addition
Subtraction
Multiplication
Common-Division
LD1
LD2
LD3

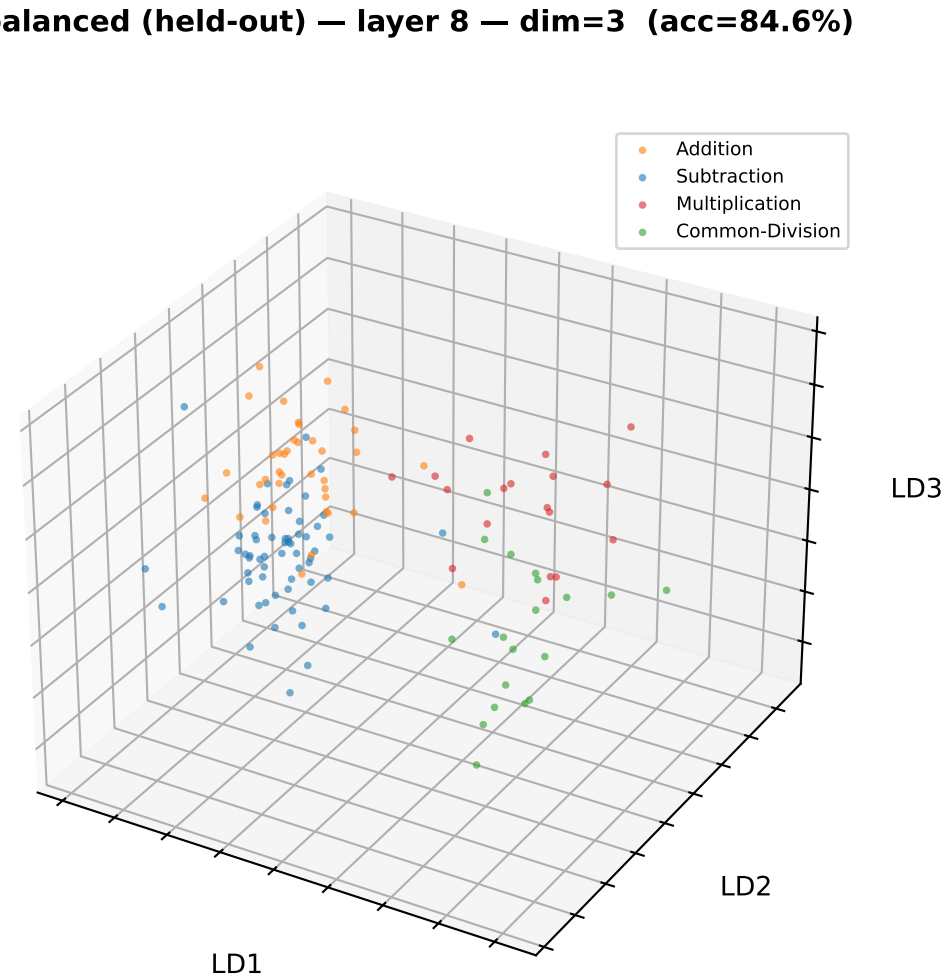

cf_balanced (held-out) — layer 3 — dim=3 (acc=55.1%)
Addition
Subtraction
Multiplication
Common-Division
LD1
LD2
LD3

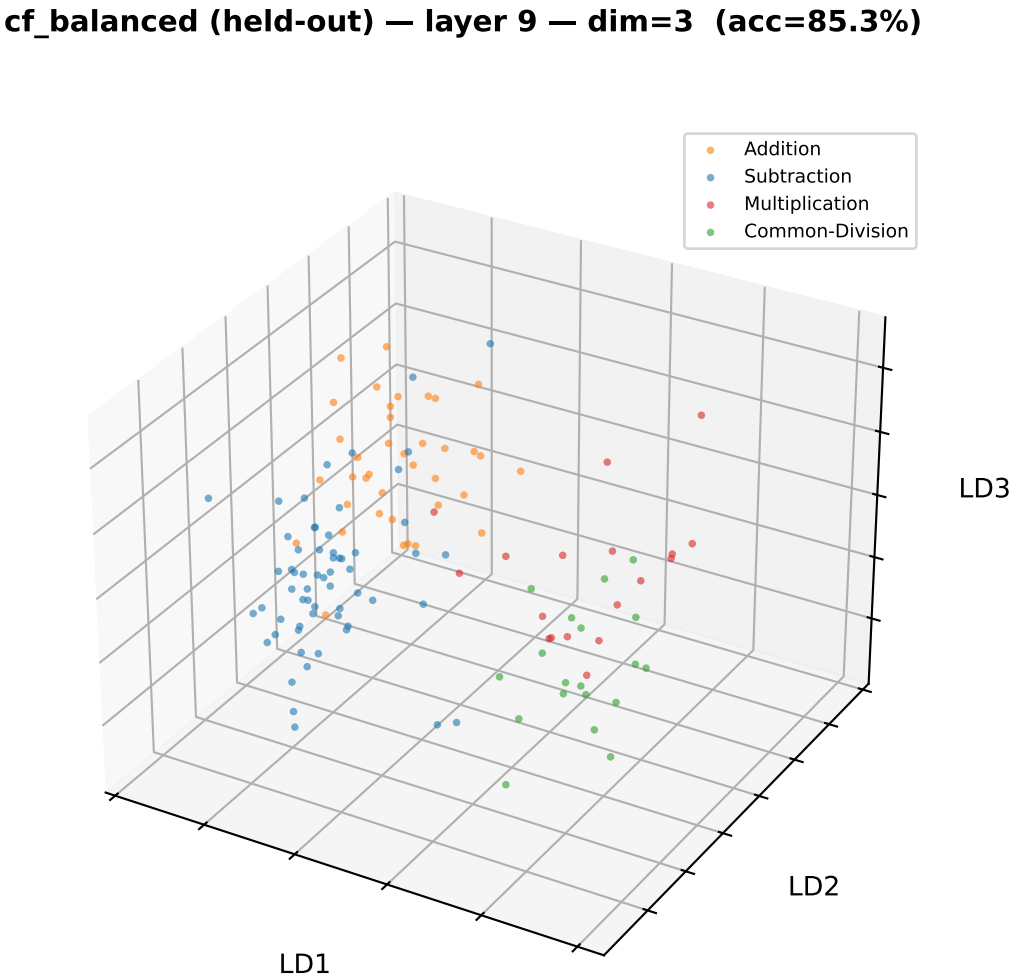

cf_balanced (held-out) — layer 4 — dim=3 (acc=58.1%)
Addition
Subtraction
Multiplication
Common-Division
LD1
LD2
LD3

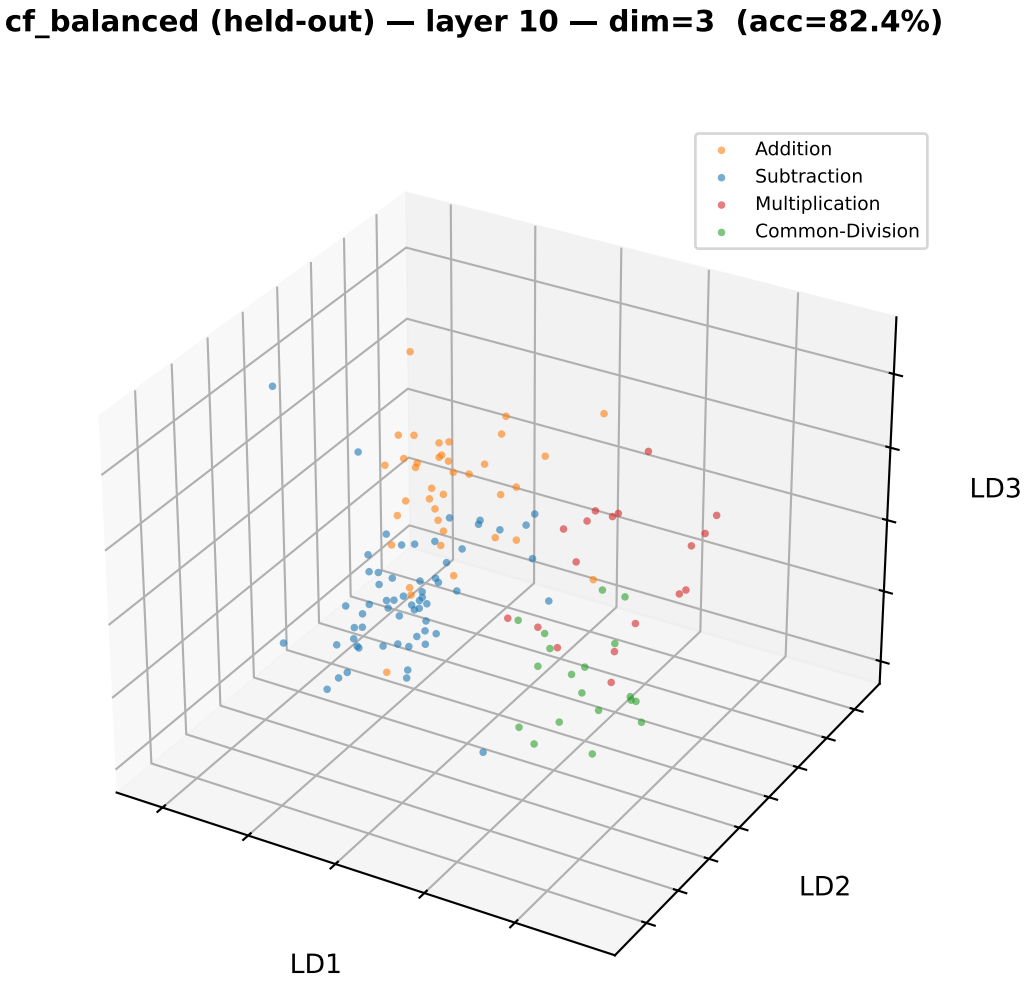

cf_balanced (held-out) — layer 5 — dim=3 (acc=55.1%)
Addition
Subtraction
Multiplication
Common-Division
LD1
LD2
LD3

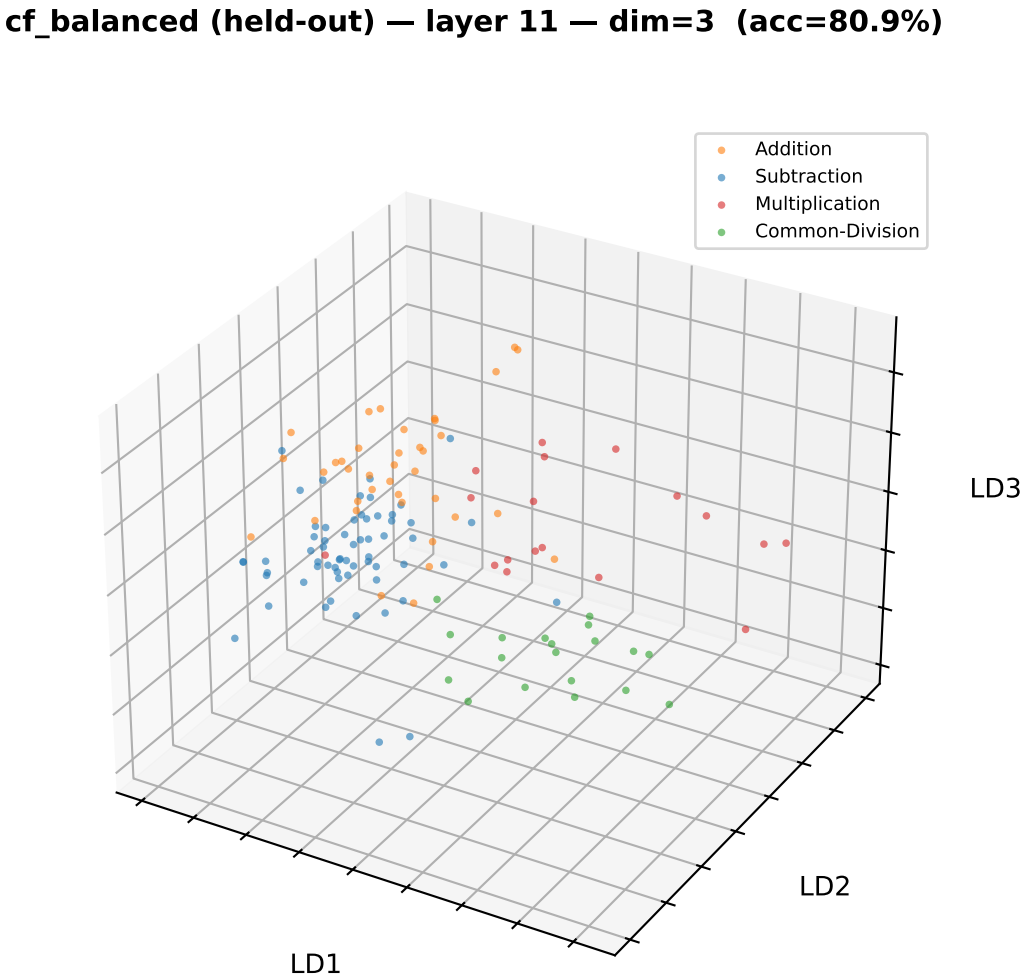

cf_balanced (held-out) — layer 6 — dim=3 (acc=70.6%)
Addition
Subtraction
Multiplication
Common-Division
LD1
LD2
LD3

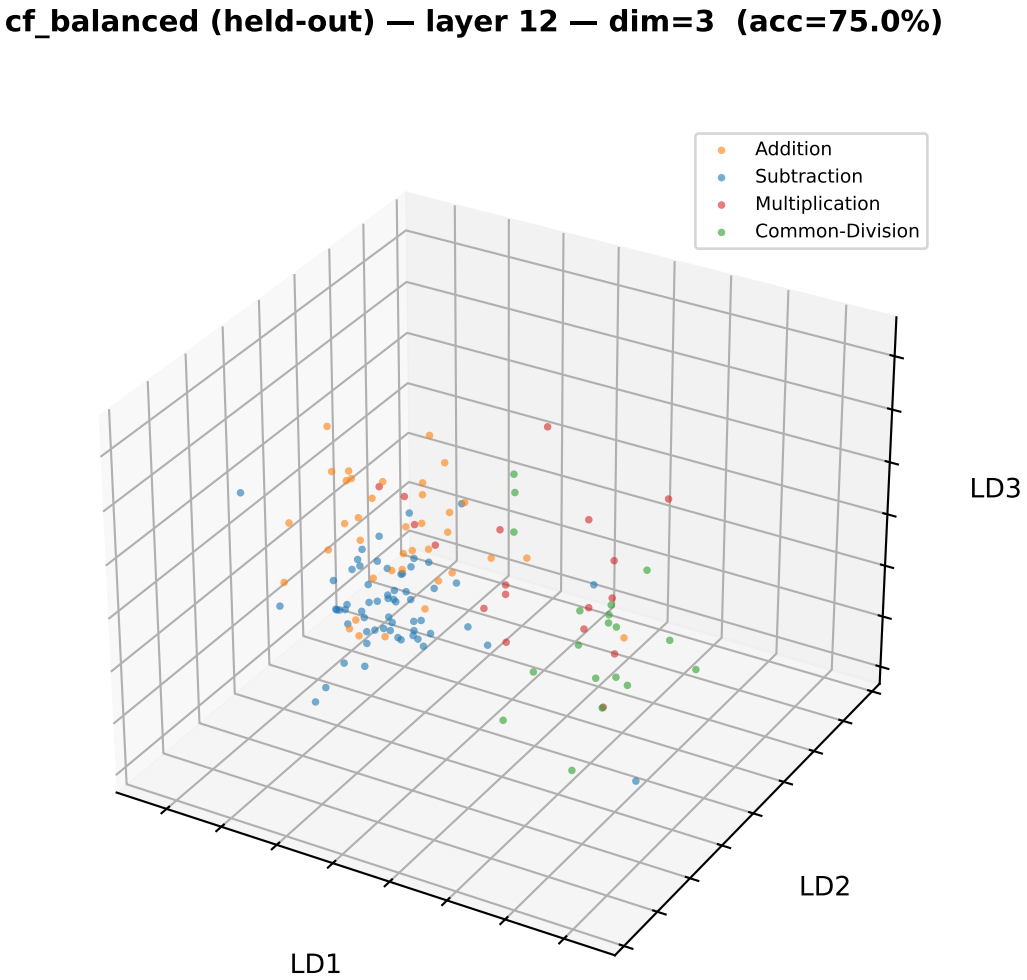

cf_balanced (held-out) — layer 7 — dim=3 (acc=72.8%)
Addition
Subtraction
Multiplication
Common-Division
LD1
LD2
LD3

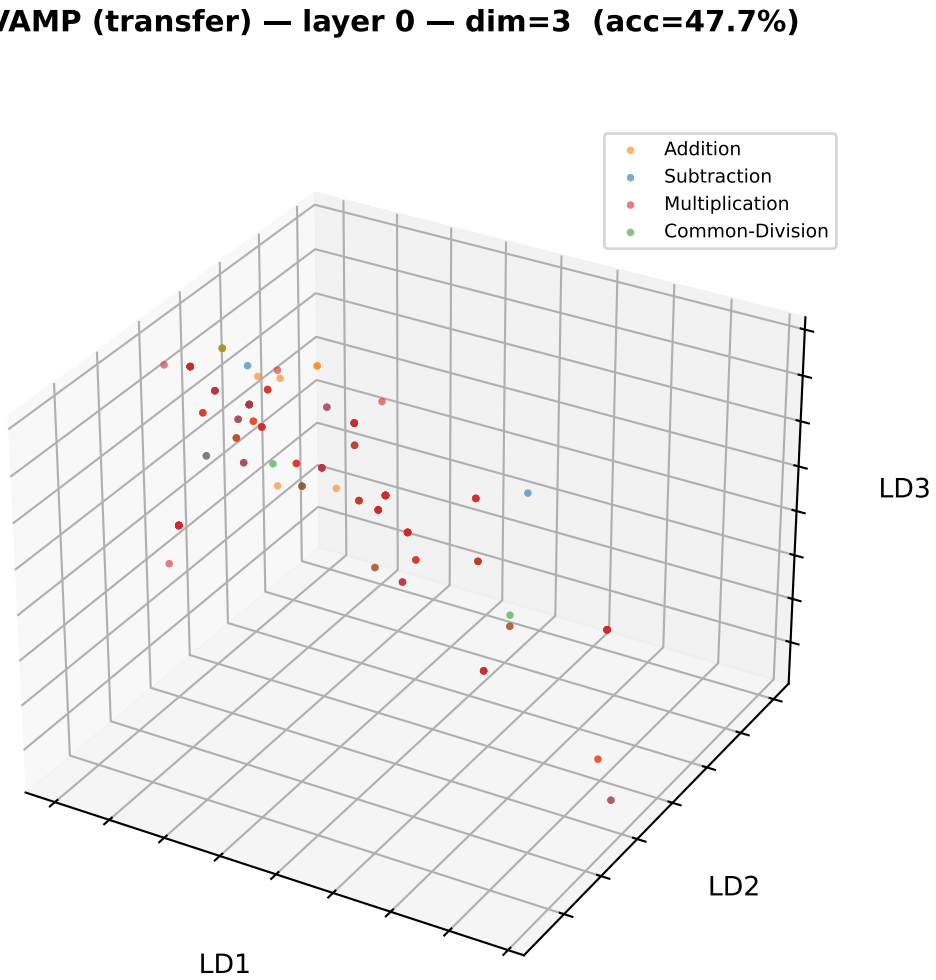

cf_balanced (held-out) — layer 8 — dim=3 (acc=84.6%)
Addition
Subtraction
Multiplication
Common-Division
LD1
LD2
LD3

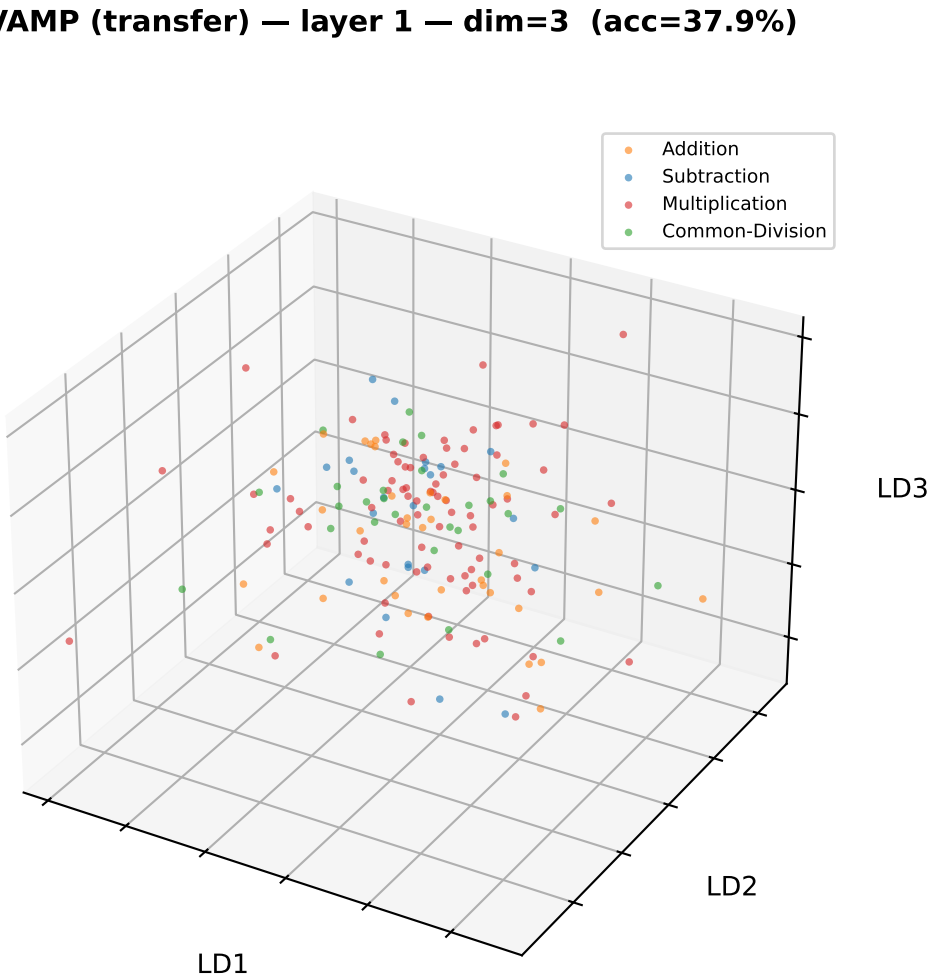

cf_balanced (held-out) — layer 9 — dim=3 (acc=85.3%)
Addition
Subtraction
Multiplication
Common-Division
LD1
LD2
LD3

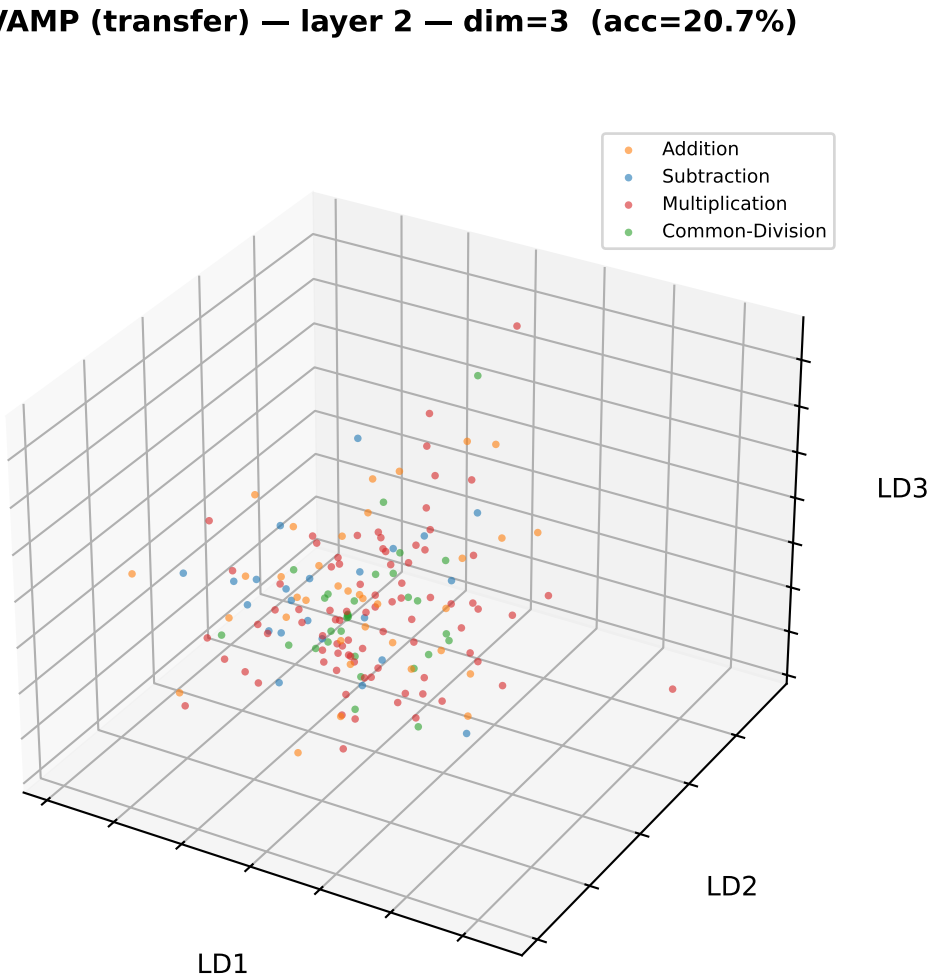

cf_balanced (held-out) — layer 10 — dim=3 (acc=82.4%)
Addition
Subtraction
Multiplication
Common-Division
LD1
LD2
LD3

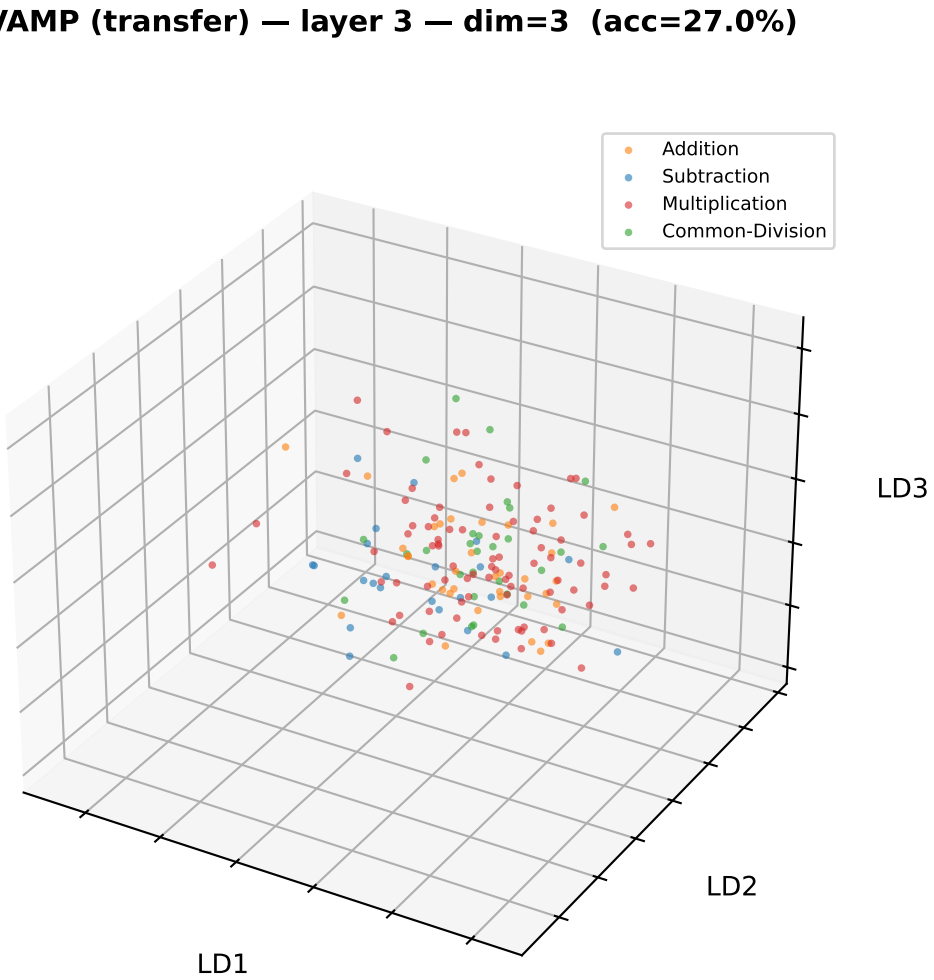

cf_balanced (held-out) — layer 11 — dim=3 (acc=80.9%)
Addition
Subtraction
Multiplication
Common-Division
LD1
LD2
LD3

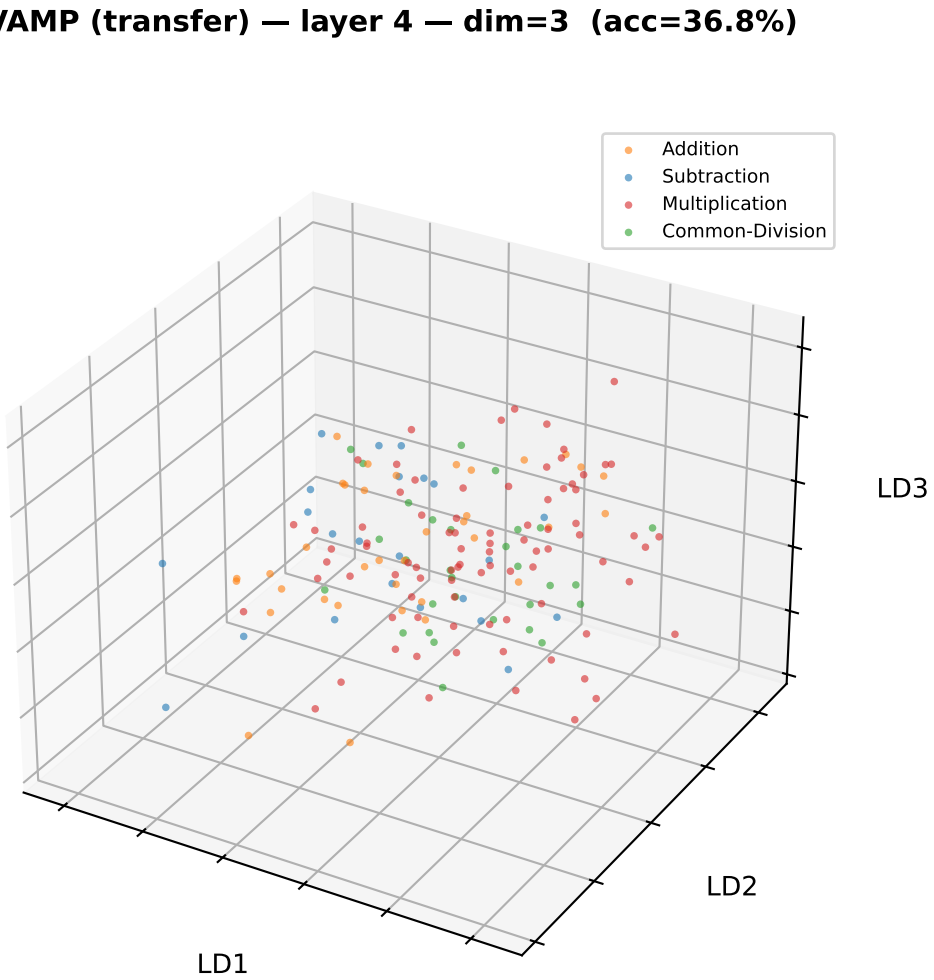

cf_balanced (held-out) — layer 12 — dim=3 (acc=75.0%)
Addition
Subtraction
Multiplication
Common-Division
LD1
LD2
LD3

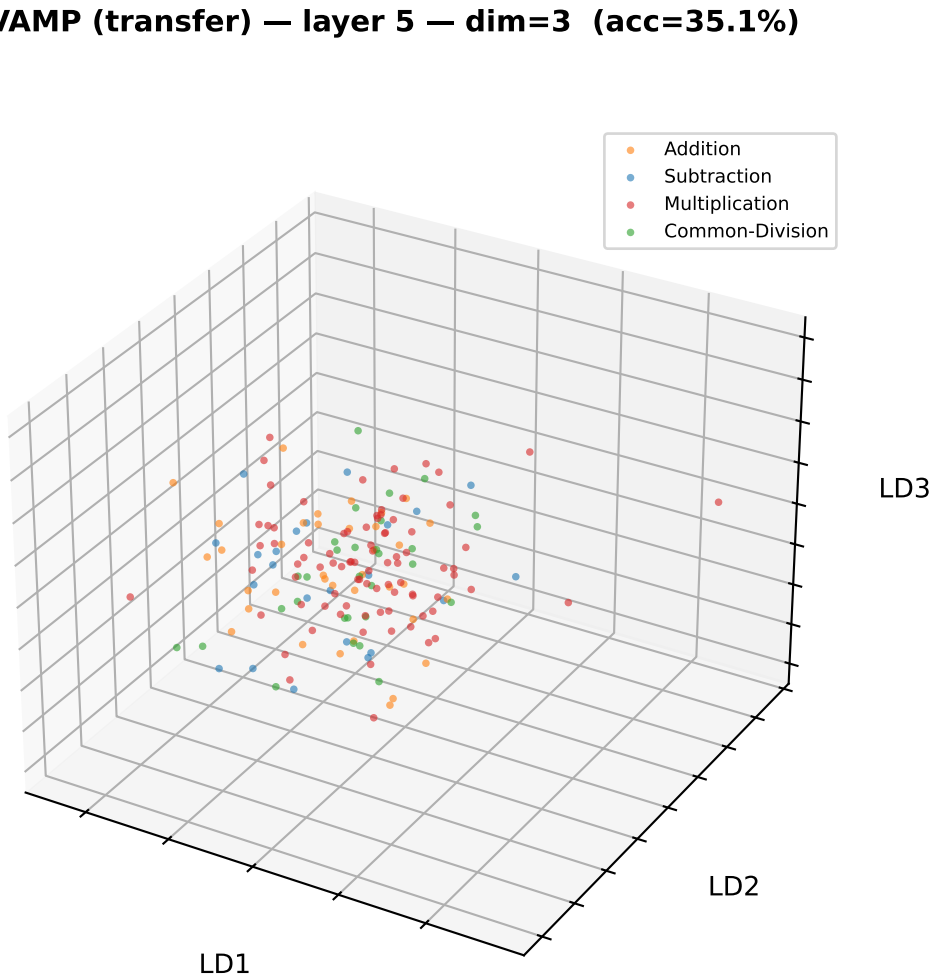

SVAMP (transfer) — layer 0 — dim=3 (acc=47.7%)
Addition
Subtraction
Multiplication
Common-Division
LD1
LD2
LD3

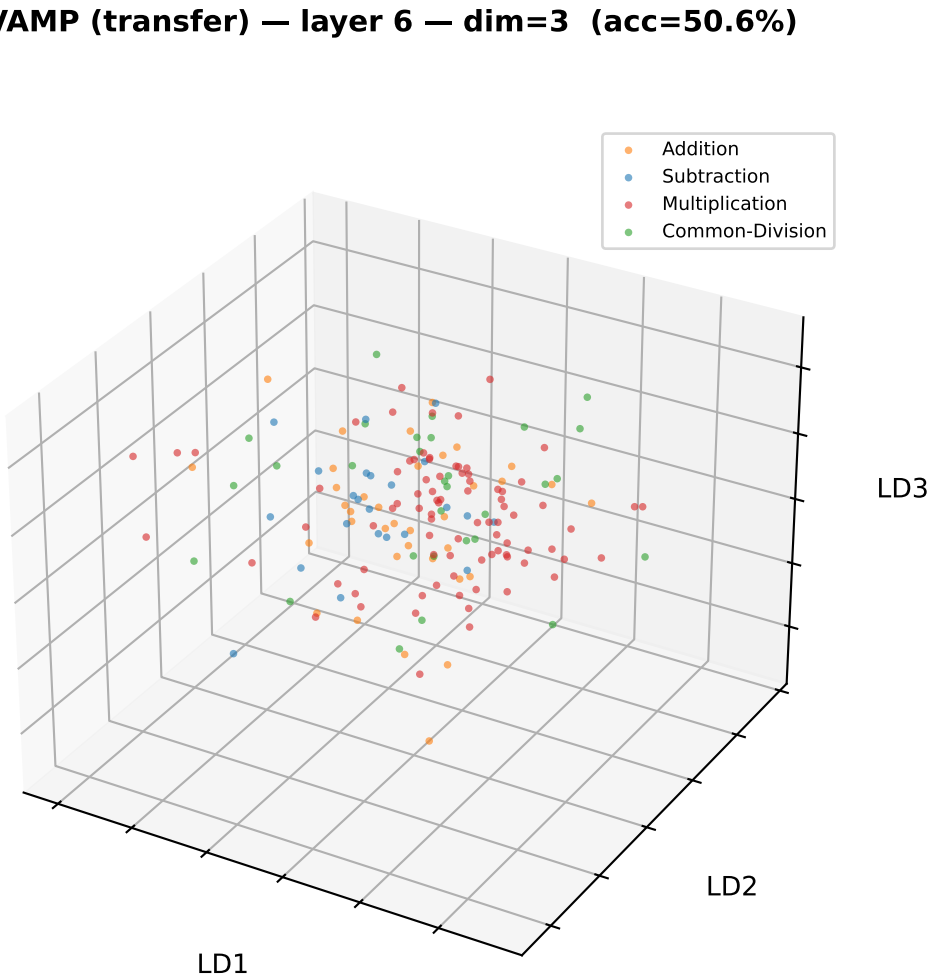

SVAMP (transfer) — layer 1 — dim=3 (acc=37.9%)
Addition
Subtraction
Multiplication
Common-Division
LD1
LD2
LD3

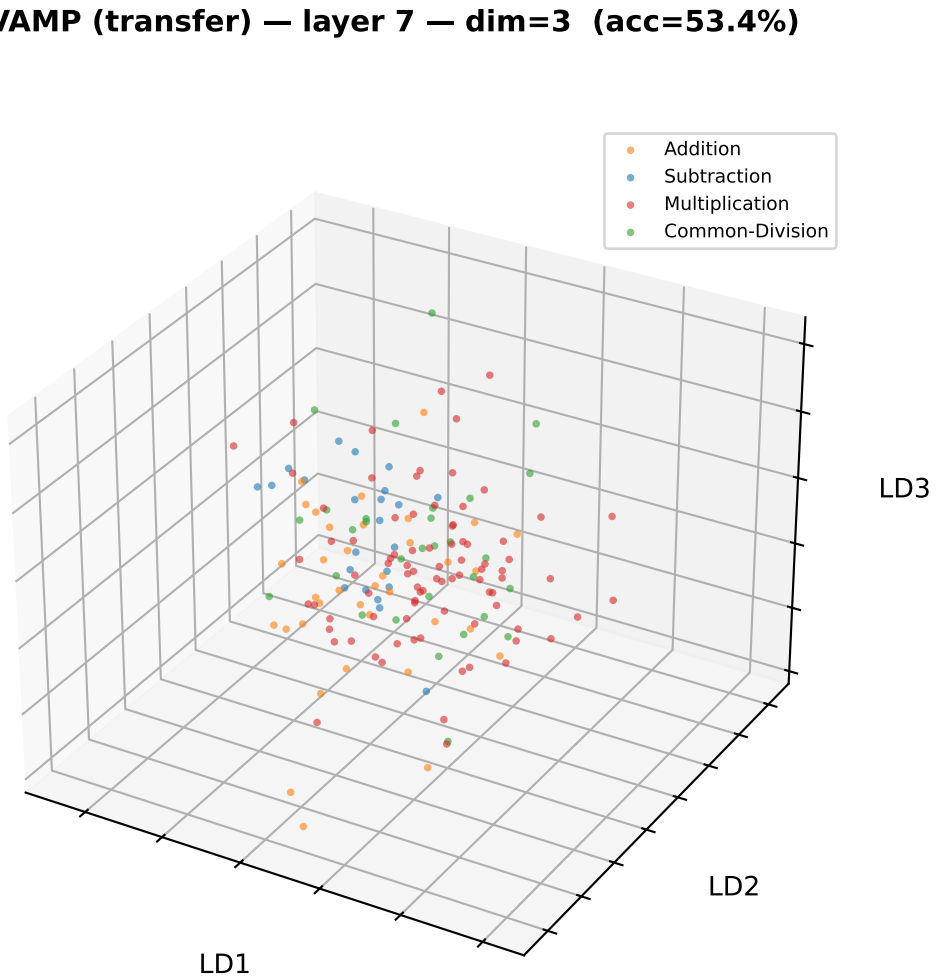

SVAMP (transfer) — layer 2 — dim=3 (acc=20.7%)
Addition
Subtraction
Multiplication
Common-Division
LD1
LD2
LD3

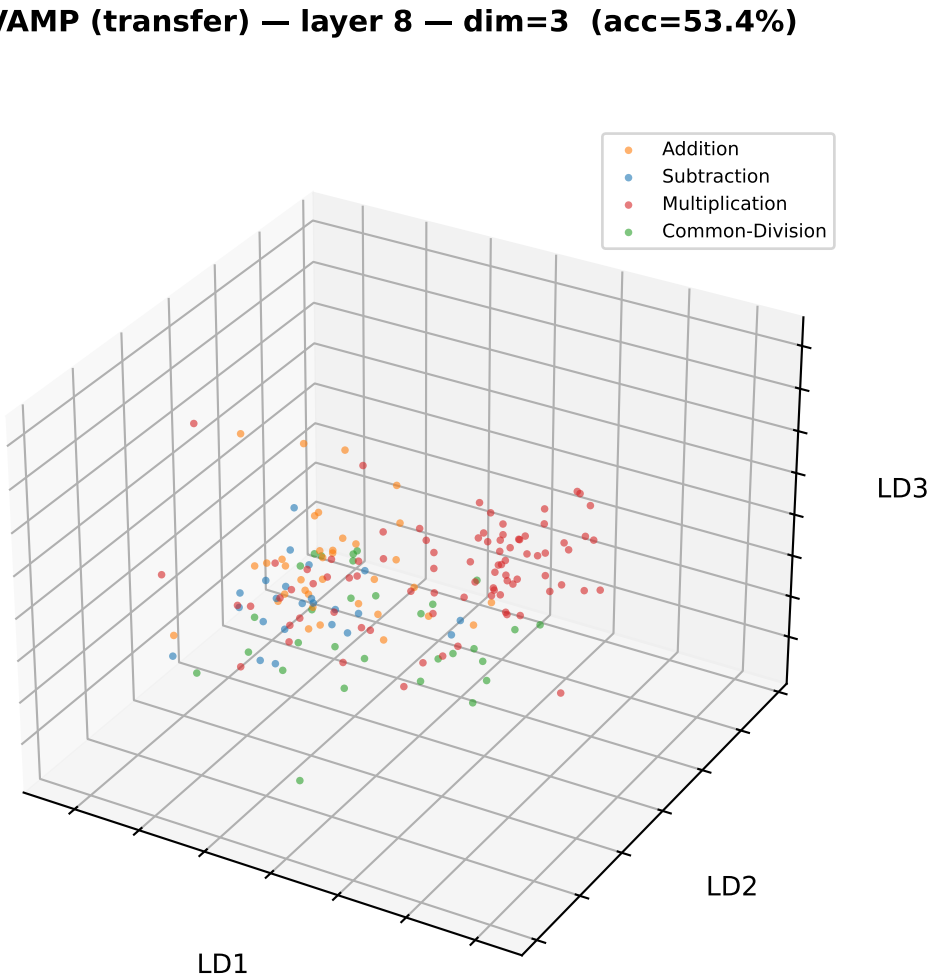

SVAMP (transfer) — layer 3 — dim=3 (acc=27.0%)
Addition
Subtraction
Multiplication
Common-Division
LD1
LD2
LD3

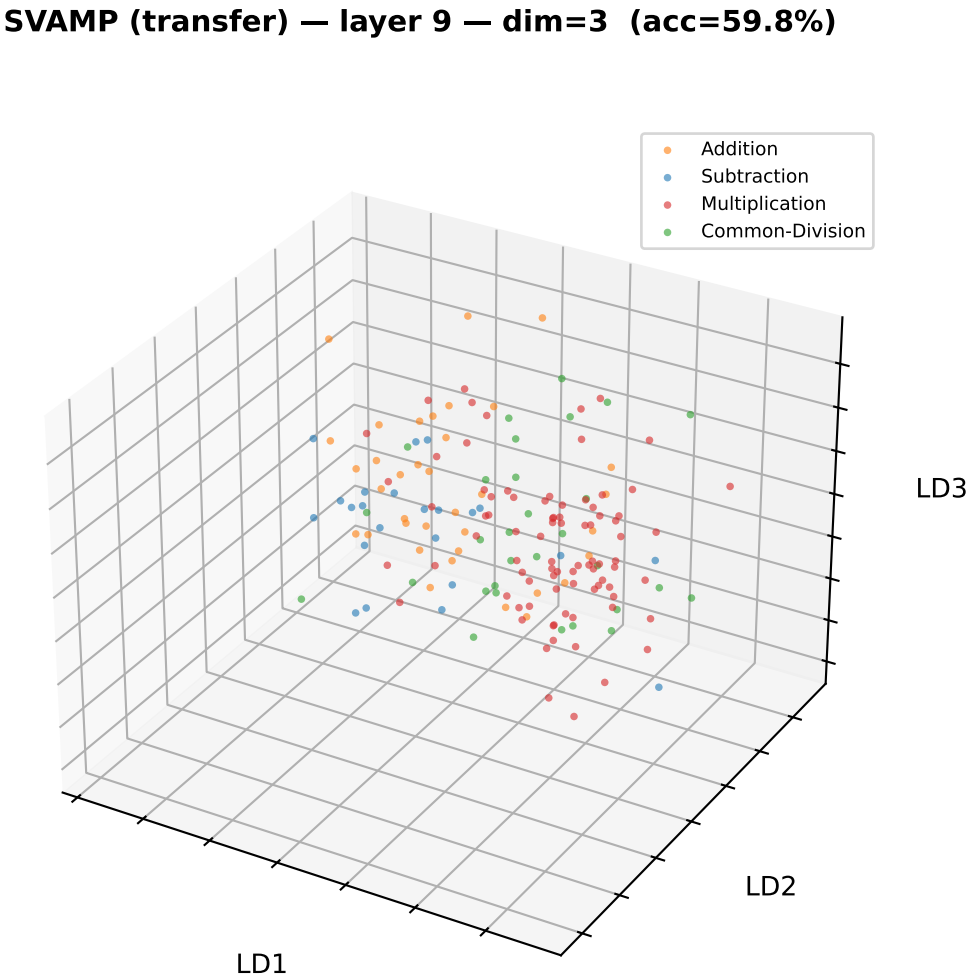

SVAMP (transfer) — layer 4 — dim=3 (acc=36.8%)
Addition
Subtraction
Multiplication
Common-Division
LD1
LD2
LD3

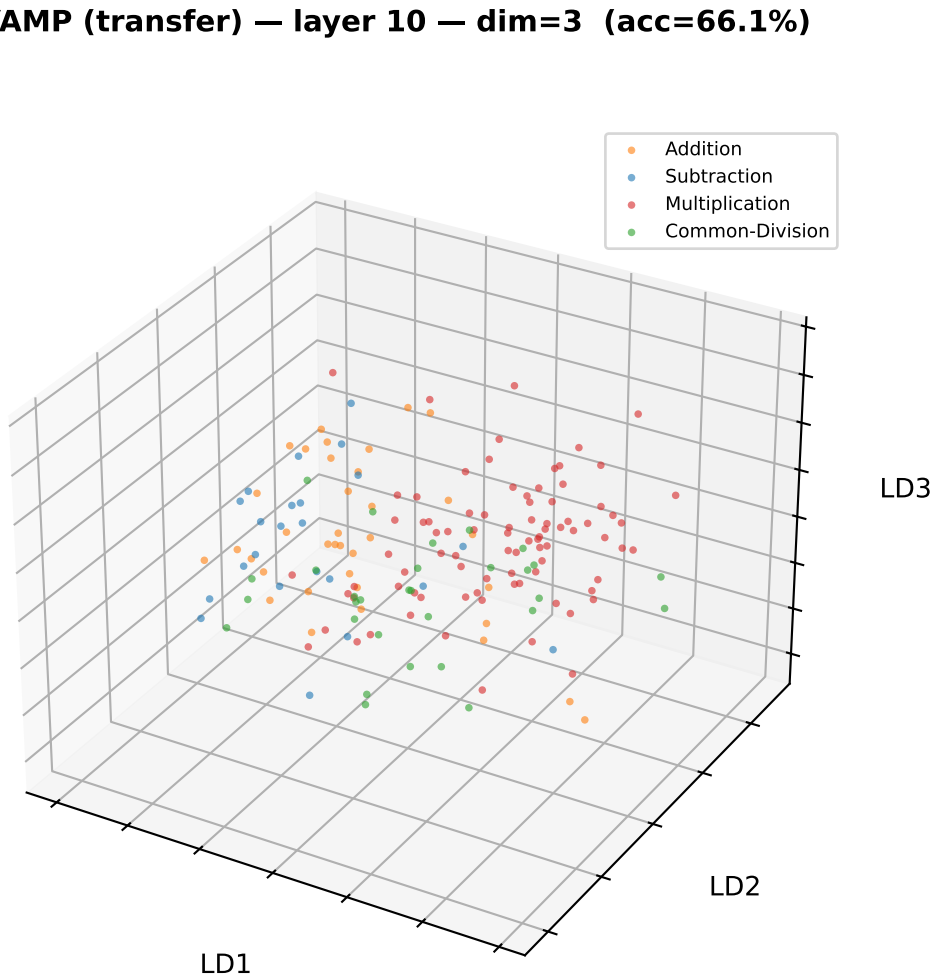

SVAMP (transfer) — layer 5 — dim=3 (acc=35.1%)
Addition
Subtraction
Multiplication
Common-Division
LD1
LD2
LD3

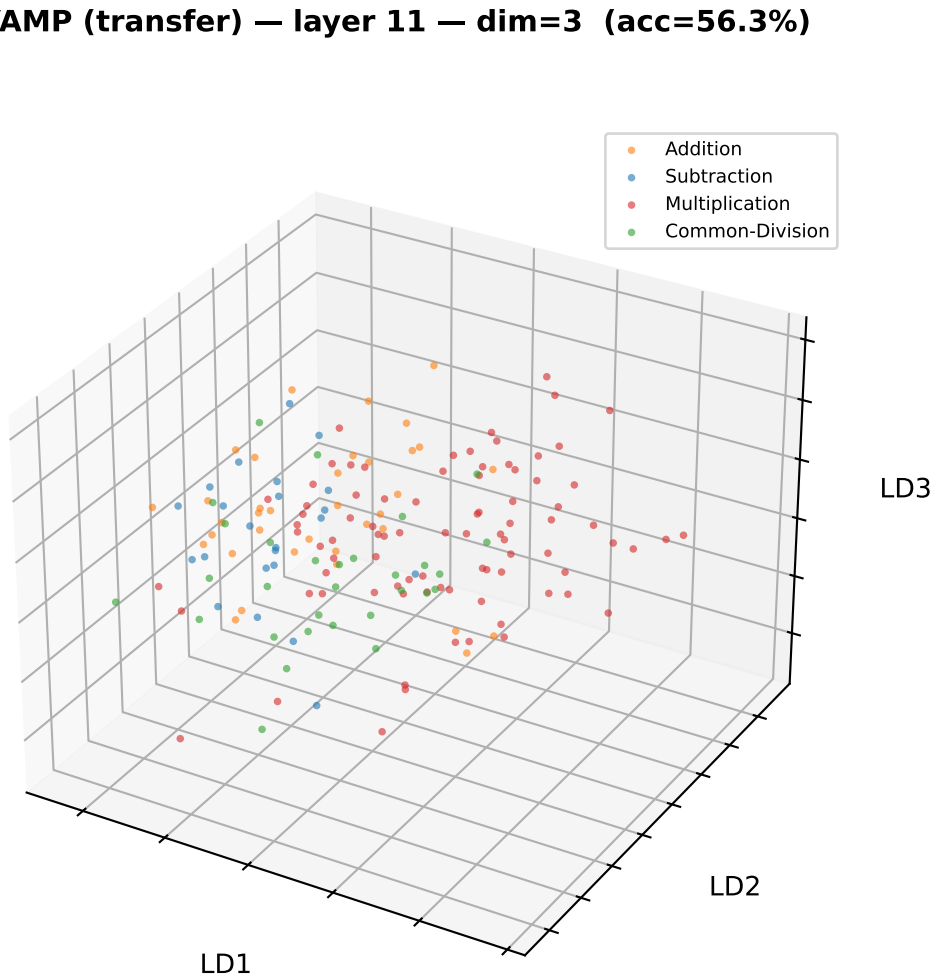

SVAMP (transfer) — layer 6 — dim=3 (acc=50.6%)
Addition
Subtraction
Multiplication
Common-Division
LD1
LD2
LD3

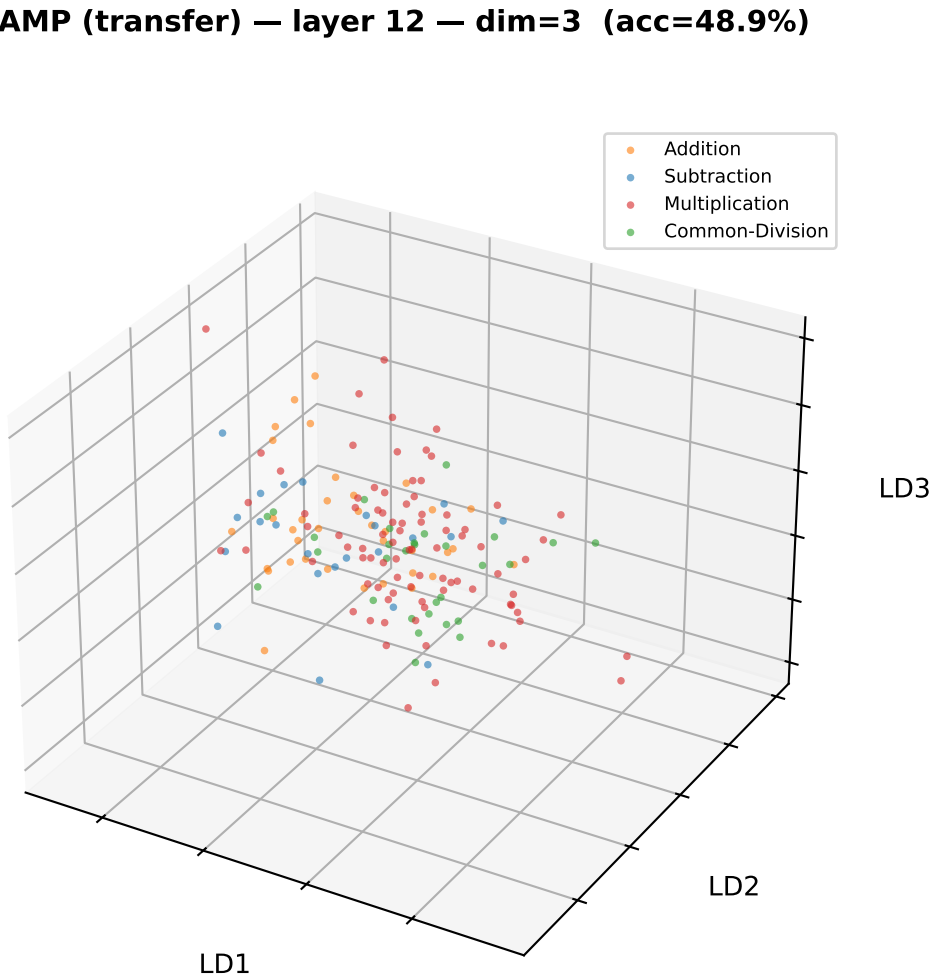

SVAMP (transfer) — layer 7 — dim=3 (acc=53.4%)
Addition
Subtraction
Multiplication
Common-Division
LD1
LD2
LD3


SVAMP (transfer) — layer 8 — dim=3 (acc=53.4%)
Addition
Subtraction
Multiplication
Common-Division
LD1
LD2
LD3


SVAMP (transfer) — layer 9 — dim=3 (acc=59.8%)
Addition
Subtraction
Multiplication
Common-Division
LD1
LD2
LD3


SVAMP (transfer) — layer 10 — dim=3 (acc=66.1%)
Addition
Subtraction
Multiplication
Common-Division
LD1
LD2
LD3


SVAMP (transfer) — layer 11 — dim=3 (acc=56.3%)
Addition
Subtraction
Multiplication
Common-Division
LD1
LD2
LD3


SVAMP (transfer) — layer 12 — dim=3 (acc=48.9%)
Addition
Subtraction
Multiplication
Common-Division
LD1
LD2
LD3